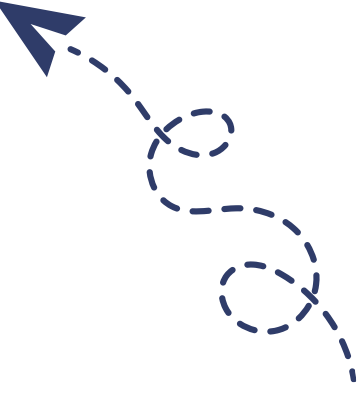




قناة تانية تمريضيانو على التليجرام





Medical-Surgical Nursing Theory Book

Nursing School – Second Year

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In order to support the Ministry of Health and Population in the delivery of education and training for nurses, the University of Sassari and the Italian Agency for Development Cooperation are proud to share the Nursing Curriculum Model for the first level of technical education.

The curriculum, which was developed in 2022, is a benchmark for all educators who are committed to providing students with skills that are up to international standards, in order to deliver high-quality care now and in the future.

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List of content

1. Chapter one: (Foundation for medical surgical nursing)

- Introduction of professional nursing concepts for medical surgical
- Complementary and alternative therapy
- Palliative care and end of life

2. Chapter two: (Perioperative care)

- Care of pre-operative
- Care of intraoperative
- Care of postoperative

3. Chapter three: (Nursing care of respiratory diseases)

- Anatomy and physiology of respiratory system
- Pneumonia
- Bronchial asthma
- Respiratory failure

4. Chapter four: (Nursing care of cardiovascular diseases)

- Anatomy and physiology of cardiovascular system.
- Hypertension
- Ischemic heart disease
- Cardiac surgery

5. Chapter five: (Nursing care of gastrointestinal diseases)

- Anatomy and physiology of gastrointestinal system
- Gastritis
- Peptic ulcer
- Appendicitis
- Cholecystitis
- Viral hepatitis

6. Chapter six: (Nursing care of urinary tract diseases)

- Anatomy and physiology of urinary system
- Urine retention
- Urine incontinence
- Urinary stone
- Renal failure

7. Chapter seven: (Nursing care of neurological disease)

- Anatomy and physiology of nervous system
- Stroke
- Headache
- Epilepsy

8. Chapter eight: (Nursing care of musculoskeletal disease)

- Anatomy and physiology of musculoskeletal system
- Fractures
- Osteoporosis

9. Chapter nine: (Nursing care endocrine of disease)

- Anatomy and physiology of endocrine system
- Diabetes mellitus
- Thyroid disorder

10. Chapter ten: (Nursing care of cancer)

- Definition of cancer.
- Pathophysiology of cancer.
- Types of cancer.
- Stages of cancer.
- Early signs of cancer.
- Management of cancer.
- Common cancer disorders (colon cancer).

11. Chapter eleven (Nursing care with emergency situation)

- Burn
- Trauma
- Coma and Brain Death

Chapter one (Foundation for medical surgical nursing)

Learning objectives

At the end of this lesson the students will be able to:

- Identify the fundamental concepts of nursing and medical-surgical nursing.
- Define the responsibilities of the nursing profession.
- Describe the roles of medical-surgical nurses in various healthcare settings.
- Explain the key influences shaping professional nursing practice.

Introduction

Nursing is a profession centered on providing continuous care to individuals who are ill, injured, disabled, or near the end of life. Nurses also promote the health of people, families, and communities in hospitals and community-based settings. Their responsibilities extend to conducting research, influencing healthcare policy, managing care services, and advocating for patient rights.

Nurses with advanced degrees may independently provide primary and specialized healthcare. Professional nurses collaborate with physicians and supervise practical nurses (LPNs/ENs) and nursing assistants across different care environments.

Definitions of Nursing

Nursing is the protection, promotion, and optimization of health and abilities; prevention of illness and injury; alleviation of suffering clinical care, and advocacy for individuals and populations. **(ANA, 2018).**

Medical-Surgical Nursing

Medical-surgical nursing is broad specialty focused on caring for adolescent to elderly patients across inpatient and outpatient settings. These include hospital medical units, ICUs, rehabilitation centers, clinics, home healthcare, and long-term care facilities. (Academy of Medical-Surgical Nurses [AMSN], 2018; AMSN 2019).

Medical-Surgical Nurse role include:

- Develop and implement individualized patient care plans.
- Administer medications and monitor patients before and after surgery.
- Manage medical devices and interpret health data.
- Provide wound care, start IVs, and insert catheters.
- Collaborate with the healthcare team on discharge planning.
- Educate patients and families about surgical care.
- Maintain accurate health records.
- Follow standardized nursing protocols and safety guidelines.

Influences on Professional Nursing Practice

1. Complex Health Care Environments

- **Technology and Knowledge Growth:** Rapid advances in healthcare tools and information require nurses to continuously adapt.
- **Population Diversity:** Nurses care for patients with varied cultural backgrounds and age groups.
- **Consumer Expectations:** Patients now expect quality care that is efficient and cost-effective.

2. Influences on Health Care Systems

- **Health Financing:** Government and insurance policies impact how nursing care is delivered.

- **National Health Goals:** Initiatives like Healthy People guide nursing education and practice priorities.

3. Supporting Professional Practice

- Organizations like the American Nurses Association (ANA), American Association of Critical-Care Nurses (AACN), and the Oncology Nursing Society (ONS) play major roles in setting practice standards, promoting education, and enhancing care quality across specialties.

Complementary and alternative therapy

Learning objectives

At the end of this lesson the students will be able to:

- Describe three categories of complementary and alternative therapies.
 - Describe the nurse's role related to complementary and alternative therapies.
 - Describe ways that the nurse can use complementary and alternative therapies to provide self-care.
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Complementary and Alternative Therapies

Complementary and alternative therapies (CAT) are a group of diverse medical and health care systems, practices, and products that are not generally considered part of conventional medicine.

NCCAM Categories of Complementary and Alternative Therapies

Category	Description	Examples
Natural products	Practices that use substances found in nature for their impact on health and wellness.	Herbal therapy, dietary supplements, vitamins, minerals, probiotics, aromatherapy
Mind-body medicine	Techniques that enhance mind's ability to affect the physical body.	Meditation, yoga, acupuncture, relaxation breathing, guided imagery, hypnotherapy, prayer, journaling, art therapy
Manipulative and body-based practices	Practices that are based on the manipulation and/or movement of one or more parts of the body.	Massage, yoga

Nurses Can Use Complementary and Alternative Therapies

If you have the training and experience, you can use the following:

- Massage
- Meditation
- Music
- Prayer
- Reiki
- Relaxation breathing
- Therapeutic Touch
- Acupressure
- Animal-assisted therapy
- Aromatherapy
- Healing Touch
- Humor
- Imagery
- Journaling

Palliative Care at End of Life

Learning objectives

At the end of this lesson the students will be able to:

- Discuss the purpose of palliative care.
- Describe the purpose of and services provided by hospice.
- Describe the physical and psychologic manifestations at the end of life.

Palliative Care

Definition:

Palliative care is any form of care or treatment that focuses on reducing the severity of disease symptoms, rather than trying to delay or reverse the progression of the disease itself or provide a cure.

Goals of Palliative Care

- Provide relief from symptoms, including pain.
- Regard dying as a normal process.
- Affirm life and neither hasten nor postpone death.
- Support holistic patient care and enhance quality of life.
- Offer support to patients to live as actively as possible until death.
- Offer support to the family during the patient's illness and in their own bereavement.

Hospice Care

Hospice care is part of palliative care given during the final stages of life. It focuses on showing compassion and giving support to individuals facing terminal illnesses. The goal is to help patients live their remaining days as fully and comfortably as possible.

Hospice services are based on:

- Emotional support for both the patient and family.
 - Physical comfort measures.
 - Respect for the patient's values and dignity.
- **Death** occurs when all vital organs and body systems stop functioning.
 - **Brain Death** is the total and irreversible loss of all brain functions, including the brainstem. A person may still have a heartbeat and be on a ventilator, especially in an intensive care unit (ICU), but cannot recover brain function.
 - **End of life** refers to the final stage of a patient's illness when death is expected soon, the timeline from diagnosis to death varies based on disease progression and individual health status.
 - **End-of-life care** (EOL care) Involves services and support for the dying person and their family.
- **Addresses both:**
- Physical needs like pain management, hygiene, and comfort.
 - Psychosocial needs including emotional support, counseling, and communication.

Physical Manifestations at End of Life	
System	Manifestations
Sensory system	
Hearing	• Usually, last sense to disappear
Touch	• Decreased sensation • Decreased perception of pain and touch
Taste and smell	• Decreased with disease progression
Sight	• Blurring of vision • Sinking and glazing of eyes • Blink reflex absent • Eyelids remain half-open

Cardiovascular system	• Increased heart rate; later slowing and weakening of pulse • Irregular rhythm • Decreased blood pressure • Delayed absorption of drugs administered intramuscularly or subcutaneously
Respiratory system	• Increased respiratory rate • apnea and deep, rapid breathing) • Inability to cough or clear secretions • Irregular breathing, gradually slowing down to gasps
Urinary system	• Gradual decrease in urine output • Incontinence of urine • Inability to urinate
Gastrointestinal system	• Slowing or cessation of GI function • Accumulation of gas • Distention and nausea • Bowel incontinence
Musculoskeletal system	• Gradual loss of ability to move • Sagging of jaw • Difficulty speaking • Swallowing becoming more difficult • Difficulty maintaining body posture and alignment • Loss of gag reflex • Jerking seen in patients on large amounts of opioids
Integumentary system	• Mottling on hands, feet, arms, and legs • Cold, clammy skin • Cyanosis of nose, nail beds, knees • “Waxlike” skin when very near death

Psychosocial Manifestations at End of Life	
• Altered decision making • Anxiety about unfinished business • Decreased socialization • Fear of loneliness • Fear of meaninglessness of one’s life • Fear of pain • Helplessness	• Life review • Peacefulness • Restlessness • Saying goodbyes • Unusual communication • Vision-like experiences • Withdrawal

Students' self-assessment

Using the American Nurses Association's definition of nursing, which activities are within the domain of nursing (*select all that apply*)?

- a. Implementing intake and output for a patient who is vomiting
- b. Establishing and implementing a stress management program for family caregivers of patients with Alzheimer's disease
- c. Explaining the risks associated with the planned surgical procedure when a preoperative patient asks about risks
- d. Developing and performing a study to compare the health status of older patients who live alone with the status of older patients who live with family members
- e. Identifying the effect of an investigational drug on patients' hemoglobin levels
- f. Using a biofeedback machine to teach a patient with cancer how to manage chronic pain
- g. Preventing pneumonia in an immobile patient by implementing frequent turning, coughing, and deep breathing
- h. Determining and giving fluid replacement therapy needed for a patient with serious burns
- i. Testifying to legislative bodies about the effect of health policies on culturally, socially, and economically diverse populations

Palliative care is any form of care or treatment that focuses on reducing the severity of disease symptoms to prevent and relieve suffering and improve quality of life for patients with serious life-threatening illnesses. From the subsequent list, identify the specific goals of palliative care (*select all that apply*).

- a. Regard dying as a normal process.
- b. Minimize the financial burden on the family.
- c. Provide relief from symptoms, including pain.
- d. Affirm life and neither hasten nor postpone death.

- e. Prolong the patient's life with aggressive new therapies.
- f. Support holistic patient care and enhance quality of life.
- g. Offer support to patients to live as actively as possible until death.
- h. Help the patient and family identify and access pastoral care services.
- i. Offer support to the family during the patient's illness and their own bereavement.

من فناء نه نرى ضياعه على التليجرام

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Chapter two perioperative care

Learning objectives

- Define the three phases of perioperative management.
- Explain principle of surgical asepsis
- Demonstrate the nursing tasks required to manage the patient in the preoperative phase.
- Identify roles of operating room team members.
- Identify different patient operative position.
- Discuss post-operative nursing management post-operative complication.

Definitions

Surgery It Is the use of instruments during an operation to treat injuries, diseases, and deformities

Perioperative nursing:

All nursing function associated with the patient surgical experience. Perioperative; incorporates the three phases of the surgical experience which include :- .Preoperative, Intraoperative and Postoperative.

1-Preoperative phase: begins when the decision for surgical intervention is made and ends with the transfer of the patient to operating room table.

2-Intraoperative phase: begins when the patient is admitted or transferred to the operating room and end when he is admitted to the recovery area.

3-Postoperative phase; begins with admission of the patient to the recovery area and end with follow up evaluation in clinical sitting

CLASSIFICATION OF SURGERY

Surgical procedures are classified according to

- Urgency .
- Purpose .
- Extent
- Risk factors are also taken into consideration.

ACCORDING TO URGENCY

- **Emergent:** Patient requires immediate attention to maintain life, disorder may be life threatening. e.g., sever bleeding, gunshot, skull fracture.
- **Urgent/ Imperative:** Patient requires prompt attention within 24 – 30/48 hours. e.g acute gallbladder infection.
- **Required/ Planned:** Patient needs to have surgery; plan within a few weeks or months. e.g., thyroid disorder, cataract.
- **Elective:** Patient should have surgery, failure to have surgery not catastrophic, planned/scheduled with no time requirements. e.g., repaired of scare, simple hernia.
- **Optional:** Decision rests with patient at the preference of patient. e.g., cosmetic surgery.

ACCORDING TO PURPOSE

- **Aesthetic:** Requested by patient for improvement.
- **Diagnostic:** To obtain tissue samples, make an incision, or use a scope to make a diagnosis. e.g., biopsy.
- **Exploratory:** Confirmation or measurement of extent of condition.
- **Preventive:** Removal of tissue before it causes a problem.
- **Curative (Ablative):** Removal of diseased or abnormal tissue. e.g exclusion of tumor or an inflamed appendix.
- **Reconstructive:** Correction of defects of body parts
- **Palliative:** to relieve pain or correct problems.

ACCORDING TO EXTENT

- **Major:** Extensive surgery that involves serious risk and complications, as it involves major organ such as heart.
- **Minor** - Involves minimal complications & blood loss

Risk factors are also taken into consideration.

1- Nutritional and Fluid Status

- Patient have obesity, weight loss, malnutrition, deficiencies in specific nutrients, metabolic abnormalities high risk for surgical complication. Nutritional deficiency should be corrected before surgery.
- Nutrients important for wound healing are: protein, carbohydrates and fats, water, vitamin C, vitamin B complex, vitamin A, vitamin K, magnesium, copper, zinc

2-Hypovolemia, dehydration or electrolyte imbalance.

3-Infection and sepsis

4-Toxic conditions.

5 -immunological abnormalities.

6-Drug or Alcohol Use

7 -Age: very young people and elderly patient.

8 -Presence of disease such as pulmonary disease, cardiovascular disease (hypertension) endocrine (DM) and Pregnancy.

9-Concurrent or prior pharmacotherapy: Medication history for Example, should be stop aspirin 7-10 days before surgery.

preoperative Phase

Pre-Operative Nursing Management:

a. Assessment:

1.Psychosocial Assessment and Care because the patient has fears from anesthesia, pain, death, fear of disturbance on Body image. etc.

2 .Physiologic nursing assessment: Personal data such as (age, sex), presence of pain, nutritional status (obesity or malnutrition), fluid and electrolyte balance , presence of infection and any disease.

Nursing diagnosis:

- Anxiety related to fear of the unknown or surgical experience (anesthesia, pain) and outcome of surgery.
- Knowledge deficit regarding preoperative procedure.

Patient outcome:

- Relieve anxiety and increase knowledge regarding procedure and include in surgical regimen.

Nursing preparation:

- Explain procedure to the patient about surgery.
- Explain all nursing care and any possible discomfort to promote faster recovery
- Allow the patient ask any question about surgery and answer it.
- Introduce the person who undergoing the major surgery to the patient who have successfully recovered from these operations.

Legal aspect:

- Any one undergoing surgery must sign an operative permit to protect the patient from unwanted surgical procedure and protect the health care staff from legal action.
- Defined as written informed consent (permission obtained from a patient to perform a specific test or procedure) should be taken before surgery

Physiologic aspect:

Before the date of surgery:

- Correct any dietary deficiencies.
- Reduce an obese person weight
- Correct fluid and electrolyte imbalance.
- Restore adequate blood volume with blood transfusion.
- Treat chronic disease

- Cure infectious disease
- Ensuring that necessary tests have been or will be performed such as liver function, blood glucose level, ECG, X-ray, Blood grouping and matching, PT, PTT etc.

Pre-operative teaching

- Instructing and demonstrating exercises that will benefit the person during postoperative period such as deep breathing, coughing to promote optimal lung expansion and respiratory function.
- turning and moving, foot and leg exercise to improve circulation and prevent venous stasis.
- Planning for discharge and any projected changes in lifestyle due to surgery

Preparing the Person Before Surgery

Preparing the skin :

- Have full bath to reduce microorganisms in the skin without injury the skin and shaved hair from area of surgery.

Preparing the GI tract

- NPO after midnight; cleansing enema as required if the patient undergoing abdominal or pelvis surgery .

Preparing for anesthesia

- Visit done by anesthetist complete respiratory, neurological and cardiovascular examination
- Avoid alcohol and cigarette smoking for at least 24 hours before surgery.

Promoting rest and sleep

Measures to reduce preoperative sleeplessness and restlessness include:

- A well – ventilated room.
- Comfortable, clean bed.
- Back rub.

- Warm fluid (if the fluid is not contraindication)
- Administer sedatives as ordered on the night before surgery.

Preparing the Person on the Day of Surgery

- Early morning care
- Awaken one hour before preoperative medications
- Morning bath, mouth wash
- Provide clean gown
- Remove hairpins, link long hairs, cover hair with cap.
- Remove dentures, foreign materials (chewing gum), colored nail polish, hearing aid, contact lens
- Take baseline vital signs before preoperative medication
- Check ID band and skin preparation
- Check for special orders – enema, GI tube insertion, IV line
- Check NPO
- Have client void before preoperative medication.
- Continue to support emotionally.
- Accomplish “preoperative care checklist.”

Pre operation checklist day of surgery

- Preoperative Medications/ Pre anesthetic Drugs such as:
- Narcotics such as: Morphine sulfate and Meperidine (Demerol)
- Analgesia; enhancement of postoperative pain relief
- Antianxiety and sedative hypnotics such as Diazepam (Valium) is Sedation and anxiety reduction.
- Antiemetic: such as (Zofran): to Control nausea and vomiting; may be effective into the postoperative period
- H2 antagonist: such as Ranitidine (Zantac)
- And Famotidine (Pepcid)
- To Reduction of acidic gastric secretions in case aspiration occurs.

- Antibiotic: such as Cefazolin and Ampicillin. to prevent the postoperative infection.
- Anticholinergic: such as Atropine sulfate
- lead to Secretion reduction

Intraoperative Phase

Begins when the client is transferred onto the OR table and ends with admission to the recovery room.

Nursing activities during intraoperative include :

- Providing safety and maintaining an aseptic environment
- Ensure proper functioning of equipment
- Providing the surgeon with specific instruments and supplies for the surgical field.
- Proper documentation

The Surgical Team

- Patient
- Surgeon
- assistance surgeon
- scrub nurse
- circulatory nurse
- anesthesiologist

The circulating nurse: protects the patient's safety and health by

1. Monitoring the activities of the surgical team.
2. Checking the OR conditions
3. Continually assessing the patient for signs of injury and implementing appropriate interventions.
4. Monitors aseptic practices to avoid breaks in technique.

5. verifying consent, coordinating the team, and ensuring cleanliness, proper temperature, humidity, lighting, safe function of equipment, and the availability of supplies and materials.
6. Arranges sterile and non-sterile equipment and open sterile equipment to scrub nurse.
7. Sends for client at proper time.
8. Confirms the client allergies.
9. Checks medical record for completeness.
10. Position the patient on the room table in accordance with type of operation.
11. Counts sponge, needle and instruments with scrub nurse before surgery

Role Of the Scrub Nurse:

1. Performs a surgical hand scrub and wear sterile gown and gloves.
2. Setting up the sterile tables and maintain sterile field
3. Prepares sutures, ligatures, and special equipment.
4. Assist with surgical draping of the client.
5. Observe progress of surgical procedure
6. Hands surgeon instrument, sponges and necessary supplies during procedure.
7. The scrub nurse and the circulatory nurse count all needles, sponges and instruments to ensure not retained as foreign body in the patient.

Important positions During Surgery

- **Dorsal Recumbent:** for hernia repair, mastectomy, bowel resection
- **Trendelenburg:** for lower abdomen, pelvic surgeries.
- **Lithotomy:** for vaginal repairs, D and C, rectal surgery
- **Prone:** for spinal surgeries, laminectomy
- **Lateral:** for kidney, chest, hip surgeries

Potential adverse effect of surgery and anesthesia to the patient

1. Allergic reaction.
2. Cardiac dysrhythmia from adverse effect of anesthetic agents.
3. Stress response is elicited
4. Central nervous system agitation, seizures and respiratory arrest.
5. Defense against infection is lowered
6. Organ functions are disturbed
7. Body image may be disturbed
8. Lifestyles may change

Principles Of Surgical Asepsis

1. Moisture causes contamination because microorganism travel more easily through moist environment.
2. Never assume that an object is sterile should be ensure that it is labeled as sterile.
3. Always face the sterile field because objects that are out of the line of vision may be accidentally contaminated.
4. Sterile articles may touch only sterile articles or surfaces because anything considered unsterile transfer microorganisms to the sterile object it touches
5. Sterile equipment or areas must be kept above the waist and on top of the sterile field .
6. Prevent unnecessary traffic and air currents around the sterile area.
7. Open, unused sterile articles are no longer sterile after the procedure.
8. A person who is considered sterile who becomes contaminated must reestablish sterility.
9. surgical technique is a team effort

Anesthesia

is a state of narcosis (sever central nervous system depression produced by pharmacological agent)

Types of Anesthesia

General, Regional, Epidural, Local.

Potential intraoperative complication

- Nausea and vomiting.
- Anaphylaxis: occurs when the patient contact with foreign substance.
- Hypoxia and other respiratory complication.
- Hypothermia; during anesthesia, the patient temperature may fall. Glucose metabolism is reduced and as a result, metabolic acidosis.
- Malignant hyperthermia: is rare inherited muscle disorder that is chemically induced by anesthetic agent.

Postoperative Phase

Extends from the time the client is admitted to the recovery room, to the time he is transported back into the surgical unit, discharged from the hospital, until the follow-up care.

Goals:

- Maintain adequate body system functions
- Restore homeostasis
- Alleviate pain and discomfort
- Prevent postoperative complications
- Ensure adequate discharge planning and teaching

Admission to post anesthesia care unit (PACU)

Goal:

- To promote safe recovery from anesthesia
- Administer oxygen by nasal cannula or mask as ordered

- Continuous monitoring is done for ECG, pulse oximetry, and bp measurements
- Assess surgical site and dressing
- Check for patency of catheter, drains and tubes
- Measure vital signs every 5 to 15 minutes
- Provide warming blanket.

Phases of post anesthesia care unite (PACU)

- **In phase I PACU: used during immediate recovery phase, intensive nursing care is provided.**
- **In phase II PACU: the patient is prepared for self-care or care in the hospital or an extended care setting.**
- **In phase III PACU: the patient prepared for discharged.**
- **Post anesthetic care (phase I)**

The nurse who admits patient to the post anesthetic care unit (PACU)

Nursing management in the post anesthesia care unit

Systematic assessment is essential during the post-operative period to detect quickly any complications and to individualize nursing care that promote optimal recovery from the surgery.

Assessment in the immediate postoperative period (phase I PACU): -

1-Respiratory assessment- :

- Check airway potency and monitor respiratory rate and depth.
- Auscultate breath sound. inspect skin color.
- Observe chest expansion.

2-Cardiac assessment- :

- Monitor blood pressure and heart rate and rhythm at least every 15 min. the first hour, every 30 min. the next 2 hours, every hour for 4 hrs, and finally, every 4 hrs for the first 24 hrs assess EC.G.

3 -Neuralgic assessment- :

- Check papillary response.
- Monitor muscle strength to determine muscle relaxant reversal.

4 -surgical site assessment- :

- Monitor for drainage (color, amount, and consistency)
- Observe for hemorrhage or hematoma formation.

5-Pain management- :

- -Assess for both subjective and objective manifestations of pain.
- Administer analgesics as prescribed.

6-Renal function assessment- :

- Monitor amounts of urinary output for clients with indwelling catheter (at least 30 ml/hr)
- For clients without a urinary catheter, palpate and percuss for bladder distension or scan with portable bladder ultrasound.

7-Intravenous fluids- :

- Assess type, amount of solution, flow rate, security and potency of tubing, and infusion site.

Phase II: Nursing intervention the immediate post-operative period: -

1-Ensure maintenance of patent airway and adequate respiratory function through:

- Lateral position with neck extended (recovery position)
- Keep airway in place until fully awake
- Suction secretions
- Encourage deep breathing
- Administer humidified oxygen as ordered

2 .Maintaining cardiovascular stability through:

Manifestations

- Apprehension, restlessness, thirst, cold, moist, pale skin
- Deep rapid respiration, low body temperature
- Low blood pressure, low hemoglobin
- Progressive weakness.

Management: Check vital signs

- Administering I V fluids, blood and blood products and medication that elevate blood pressure.
- Administer oxygen therapy.
- Administer Vitamin K as ordered
- Pressure dressings

3 .Relieving pain and anxiety :

- Monitor physiological status .
- Manage pain and discomfort by administer opioid administer by IV.
- Provides psychological support to relieve the patient fears and anxiety.

4 .Controlling nausea and vomiting.

- a. The nurse should administer antiemetic drugs at the patient first report of nausea to control the problems
- Determined readiness for discharge from PACU to surgical unit according to

Parameters for Discharge from Recovery Room to surgical ward which include:

- **Activity:** able to move spontaneously or on commands.
- **Respiration:** ability to breathe deeply and cough.
- **Circulation:** BP is within +/-20 mmHg of the preop level.
- **Consciousness:** fully awake and responsive.
- **O2 saturation:** Able to maintain o2 sat more than 95 % on room air.

Phases III: Preparing the postoperative patient for direct discharge:

2-3 days after surgery (discharge planning/teaching) about:

- Self-care activities
- Activity limitation or intolerance
- Diet and medications
- Complications
- Referrals, follow-up check up

Postoperative Complications

1 -Femoral phlebitis/ deep thrombophlebitis

Manifestations

- Pain, Redness, Swelling, Heat/warmth, Positive Homan's sign.

Nursing Interventions (prevention)

- Hydrate adequately to prevent hemoconcentration
- Encourage leg exercises and ambulate early
- Avoid any restricting devices that can constrict and impair circulation

Nursing Interventions (Active)

- Bed rest, elevate the affected leg with pillow support
- Wear ant embolic support hose from the toes to the groin
- Avoid massage on the calf of the leg
- Initiate anticoagulant therapy as ordered

2-Pulmonary complications such as atelectasis, bronchitis, bronchopneumonia and lobar pneumonia.

Nursing Interventions

- Reinforce deep breathing, coughing, and turning exercises to enhance the respiratory function.
- Encourage early ambulation
- use Incentive spirometry

3-Intestinal obstruction

Manifestations

- Intermittent, sharp, colicky abdominal pains
- Nausea and vomiting
- Abdominal distention
- Diarrhea (incomplete obstruction), no bowel movement (complete)
- Return flow of enema is clear.

Nursing Interventions

- NGT insertion
- Administer electrolyte/ IV as ordered
- Prepare for possible surgical intervention

4- Wound infections

Causes

- Staphylococcus aureus, E- coli, Anaerobic bacteria.

Clinical manifestations

- Redness, swelling, pain, warmth
- Pus or other discharge on the wound
- Foul smell from the wound
- Elevated temperature; chills and Tender lymph nodes
- Rule of thumb to determined infection:
- Fever within first 24 hours – pulmonary infection
- Within 48 hours – urinary tract infection
- Within 72 hours – wound infection

Preventive interventions

- Strict aseptic technique
- Wound care
- Keep unit clean
- Maintain hand washing.

- Antibiotic therapy as ordered

Nursing interventions

- Apply abdominal binders can provide support and guard against dehiscence.
- Encourage proper nutrition (high protein, vitamin C)
- Stay with client, have someone call for the doctor
- Keep in bed rest
- Supine or Semi-Fowler's position, bend knees to decrease pressure.
- Cover exposed intestine with sterile, moist saline dressing
- Reassure, keep him/her quiet and relaxed
- Prepare for surgery and repair of wound

Students' self-assessment

➤ **multiple-choice questions (MCQs) about the phases of operative care:**

1- Which phase includes the actual surgical procedure?

- a) Preadmission testing
- b) Intraoperative phase
- c) Postoperative phase
- d) None of the above

2- What is the primary purpose of maintaining NPO for 6 to 8 hours before surgery?

- a) To prevent malnutrition
- b) To prevent electrolyte imbalance
- c) To prevent aspiration pneumonia
- d) To prevent intestinal obstruction

3- Which position is commonly used for patients undergoing abdominal surgery?

- a) Supine
- b) Trendelenburg
- c) Fowler's position
- d) None of the above

4- Which factor ensures the validity of informed written consent?

- a) The patient is of legal age with a proper mental disposition
- b) If the patient is a child, secure consent from the parents or legal guardian
- c) The consent is secured before administration of preoperative medications
- d) If the patient is unable to write, the nurse signs the consent for the patient

5- Which phase involves the actual surgical procedure?

- a) Preadmission testing
- b) Intraoperative phase
- c) Postoperative phase
- d) None of the above

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Chapter three

(Nursing care of respiratory diseases)

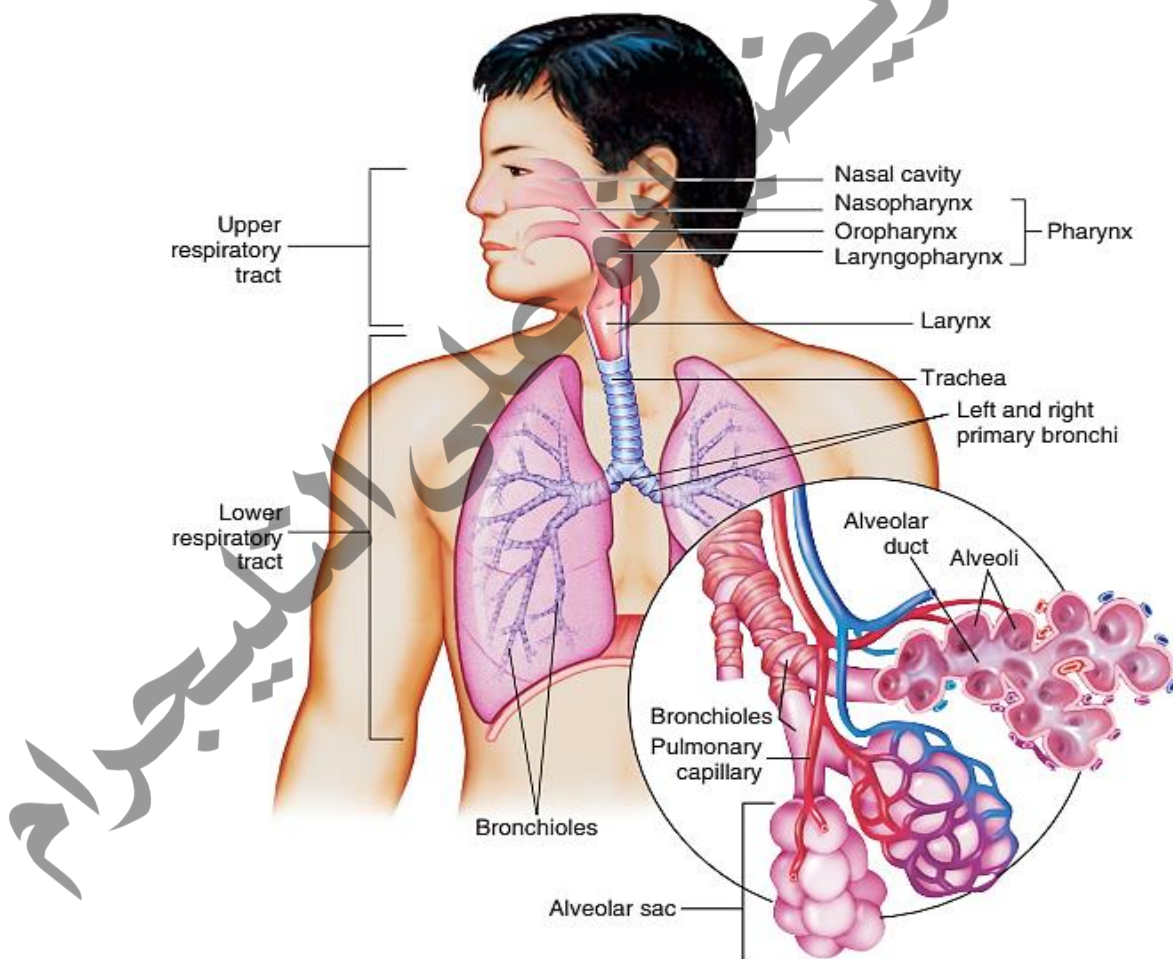
Learning objectives

- Anatomy of respiratory system
- Physiology of respiratory system
- Discuss common respiratory disorders

Introduction

The function of the respiratory system is to exchange gases (oxygen and carbon dioxide) with the external environment, the respiratory system maintains the level of these gases within a narrow range, regardless of the demand for oxygen.

Anatomy of respiratory system



Structural plan of the respiratory organs, showing the pharynx, trachea, bronchi, and lungs. The *inset* shows the grapelike alveolar sacs where the interchange of oxygen and carbon dioxide takes place through the thin walls of the alveoli. Capillaries surround the alveoli. (From Patton KT, Thibodeau GA: *Anatomy and physiology*, ed 8, St. Louis, 2013, Mosby.)

Physiology of respiratory system

Respiratory Passages:

- Nose or mouth
- Pharynx
- Larynx and associated structures
- Trachea
- Right and Left main bronchi
- Bronchi and bronchioles
- Lung alveoli

The function of respiration is to:

- Supply oxygen, remove carbon dioxide
- Regulate the pH
- Excretion of certain anesthetic drugs.

Mechanics of respiration

- Each respiratory cycle is composed of:

A- Inspiration

B- Expiration

Respiratory cycle

- Consists of inspiratory and expiratory phases with a pause in between
- Normal resting respiratory rate (cycles) in adults:

12-16 cycle/min., it is double in children

Common respiratory disorders

Pneumonia

Definition:

- Pneumonia is an acute lung infection with inflammation accompanied by accumulation of exudate in the alveoli
- The risk of pneumonia increases with aspiration, central nervous system depression, chronic illness, COPD, dehydration, existence of a tracheostomy opening, immobility, immunosuppression, intubation, pain in the thoracic cavity, and use of general anesthesia

Causes

- It may result from a bacterial, fungal, or viral infection or from exposure to a chemical irritant, through aspiration or gas inhalation
- Pneumonia remains a major cause of morbidity and mortality among elderly and chronically ill people

Signs and symptoms

- Dyspnea or tachypnea, fatigue, and fever commonly occur
- Signs of cyanosis or hypoxia may occur in those with advanced disease
- Irritability or restlessness may signal cerebral hypoxia
- Sputum may vary in amount, color, and consistency, depending on the causative agent
- Lung auscultation may reveal crackles, rhonchi, bronchial breath sounds over areas of consolidation, or wheezes over the affected areas; breath sounds may be decreased in those with advanced disease

Diagnosis

- Chest X-ray, ABG studies, sputum culture (for bacterial infection), and serologic testing (for viral infection) may be prescribed
- Needle or open biopsy may obtain lung tissue specimens (for fungal infection), and cold agglutinins may reveal antibodies associated with *Mycoplasma pneumonia* infection

Treatment

- ◆ Pneumonia is treated with antibiotics to eradicate the infecting organism
- ◆ A bronchodilator is used to open narrowed airways

Nursing interventions

- ◆ Teach effective coughing and deep breathing to improve airway clearance
- ◆ Provide adequate hydration to liquefy secretions; thin secretions are easier to expectorate
- ◆ Implement chest physiotherapy; postural drainage uses gravity to clear secretions, and percussion and vibration loosen secretions, making them easier to cough up
- ◆ Administer oxygen to aid ventilation, improve Pao₂, and preserve oxygenation
- ◆ Suction the patient's airway as needed to maintain patency and to clear secretions
- ◆ Advise the patient to limit activity and to rest for long periods to decrease oxygen consumption
- ◆ Provide pain medication as needed to allow effective coughing and deep breathing
- ◆ Maintain adequate nutrition to offset the increased use of calories secondary to infection
- ◆ Teach the patient how to contain secretions to reduce the risk of spreading infection

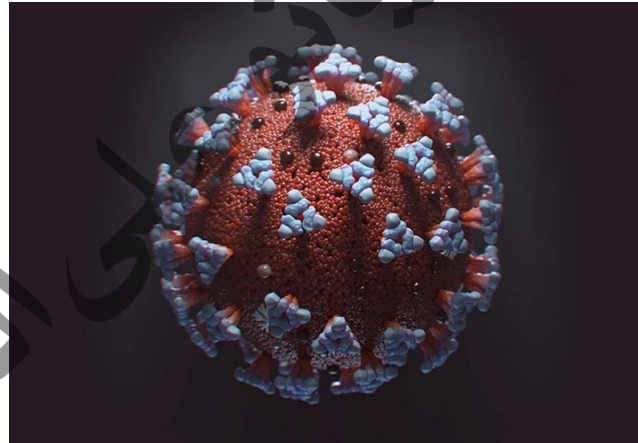
- ◆Practice good hand hygiene techniques to reduce the risk of spreading infection
- ◆Teach relaxation and stress-reduction techniques; anxiety can compromise the immune system and increase the risk of infection

Coronavirus Disease 2019 (COVID-19)

Introduction

Coronaviruses comprise a vast family of viruses, seven of which are known to cause disease in humans. Some coronaviruses that typically infect animals have evolved to infect humans. [Severe acute respiratory syndrome](#) (SARS) and [Middle East respiratory syndrome](#) (MERS) also are caused by coronaviruses that “jumped” from animals to humans.

More than 8000 individuals developed SARS, nearly 800 of whom died of the illness (mortality rate, approximately 10%), before it was controlled in 2003. MERS continues to resurface in sporadic cases.



Definition

Coronavirus disease 2019 (COVID-19) is defined as illness caused by a novel coronavirus called severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2; formerly called 2019-nCoV), which was first identified amid an outbreak of respiratory illness cases in Wuhan City, Hubei Province, China.

Pathophysiology

- SARS-CoV-2, the virus that causes COVID-19, enters patient body through patient mouth, nose or eyes (directly from the airborne droplets or from the transfer of the virus from patient hands to patient face).
- It then travels to the back of patient nasal passages and mucous membrane in the back of patient throat.
- It attaches to cells there, begins to multiply and moves into lung tissue. From there, the virus can spread to other body tissues.

Transmission

Coronavirus is likely spread:

- The virus travels in respiratory droplets released into the air when an infected person coughs, sneezes, talks, sings or breathes near you. person may be infected if he inhales these droplets.
- You can also get coronavirus from close contact (touching, shaking hands) with an infected person and then touching your face.

Risk factors

- Live in or have recently traveled to any area with ongoing active spread.
- Have had close contact with a person who has a laboratory-confirmed or a suspected case of the COVID-19 virus. (Close contact is defined as being within 6 feet of an infected person for a cumulative total of 15 minutes or more over a 24-hour period).
- Are over the age of 60 with pre-existing medical conditions or a weakened immune system.

Symptoms and Causes

The time between becoming infected and showing symptoms (incubation period) can range from two to 14 days. The average time before experiencing symptoms is five days. Symptoms can range in severity from very mild to severe.

COVID-19 symptoms vary from person to person. In fact, some infected people don't develop any symptoms (asymptomatic). In general, people with COVID-19 report some of the following symptoms:

- ✓ Fever or chills.
- ✓ Cough.
- ✓ Shortness of breath or difficulty breathing.
- ✓ Tiredness.
- ✓ Muscle or body aches.
- ✓ Headaches.
- ✓ New loss of taste or smell.
- ✓ Sore throat.
- ✓ Congestion or runny nose.
- ✓ Nausea or vomiting.
- ✓ Diarrhea.

Diagnosis and tests

COVID-19 is diagnosed with a laboratory test. healthcare provider may collect a sample of patient saliva or swab patient nose or throat to send for testing.

According to current Centers for Disease Control and Prevention (CDC) recommendations, patient should self-isolate until patient has met both of the following criteria:

- It's been five days since patient symptoms first appeared and are improving.
- Patient has not had a fever for 24 hours and has not used fever-lowering medications during this time.

Management and treatment

Treatments for COVID-19 vary depending on the severity of patient symptoms. If patient is not in the hospital or doesn't need supplemental oxygen, no specific antiviral or immunotherapy is recommended.

Depending on the severity of patient COVID symptoms, patient may need:

- Supplemental oxygen (given through tubing inserted into patient's nostrils).
- Some people may benefit from an infusion of monoclonal antibodies.
- Antiviral medications may reduce the risk of hospitalization and death in certain patients with COVID-19.
- Mechanical ventilation (oxygen through a tube inserted down patient's trachea). Patients are given medications to keep patient comfortable and sleepy as long as patient's receiving oxygen through a ventilator.
- Extracorporeal membrane oxygenation (ECMO). Patients continue to receive treatment while a machine pumps patient's blood outside patient's body. It takes over the function of patient's body's lungs and heart.

Nursing Interventions

Based on assessment data, nursing interventions for COVID-19 should focus on monitoring vital signs, maintaining respiratory function, managing hyperthermia, and reducing transmission.

- **Monitor vital signs** – particularly temperature and respiratory rate, as fever and dyspnea are common symptoms of COVID-19.
- **Monitor O2 saturation** – normal O2 saturation as measured with pulse oximeter should be 94 or higher; patients with severe COVID-19 symptoms can develop hypoxia, with values dropping low enough to warrant supplemental oxygen.
- **Manage fever** – use appropriate therapy for hyperthermia, including adjusting room temperature, eliminating excess clothing and covers, using cooling mattresses, applying cold packs to major blood vessels, starting or increasing intravenous (IV) fluids as allowed, administering antipyretic medications as prescribed, and readying oxygen therapy in the event of respiratory problems resulting from the metabolic demands for oxygen during a fever.

- **Maintain respiratory isolation** – isolation rooms should be well-marked with limited access; all who enter the restricted-access room should use personal protective equipment, such as masks and gowns.
- **Enforce strict hand hygiene** – to reduce or prevent transmission of coronavirus, patients should wash hands after coughing, as should all who enter or leave the room.
- **Provide information** – educate the patient and patient's family members of the transmission of COVID-19, the tests to diagnose the disease, disease process, possible complications, and ways to protect oneself and one's family from coronavirus.

Prevention

The best defense to prevent getting COVID-19 is to get vaccinated. You should also follow the same steps you would take to prevent getting other viruses, such as the common cold or the flu.

- Wash your hands for at least 20 seconds — especially before eating and preparing food, after using the bathroom, after wiping your nose, and after coming in contact with someone who has a cold.
- Wear a multilayered cloth facemask that fits snugly on your face and covers your mouth, nose and chin as recommended by the CDC.
- Avoid touching your eyes, nose and mouth to prevent the spread of viruses from your hands.
- Cover your mouth and nose with a tissue when sneezing and coughing or sneeze and cough into your sleeve. Throw the tissue in the trash. Wash your hands afterward. Never cough or sneeze into your hands!
- Avoid close contact (within 6 feet) with those who have coughs, colds or are sick. Stay home if you're sick.

- If you're prone to sickness or have a weakened immune system, stay away from large crowds of people. Follow the directions of your healthcare authorities, especially during outbreaks.
- Clean frequently used surfaces (such as doorknobs and countertops) with a virus-killing disinfectant.
- Use hand sanitizers that contain at least 60% alcohol if soap and water are not available.
- Greet people with a friendly gesture instead of shaking hands.
- Get enough sleep, eat a healthy diet, drink plenty of liquids and exercise if you are able. These steps will strengthen your immune system and help you fight off infections more easily.

Asthma

Definition:

Asthma is a chronic reactive airway disorder that involves episodic, reversible airway obstruction resulting from bronchospasms, increased mucus secretions, and mucosal edema

It's **characterized by** airway inflammation, intermittent airflow obstruction, and bronchial hyper responsiveness

Predisposing factors

- ▶ exposure to tobacco or wood smoke
- ▶ inhaling other respiratory irritants, such as perfumes or cleaning products
- ▶ exposure to airway irritants at the workplace
- ▶ breathing in allergy-causing substances (allergens), such as molds, dust, or animal dander
- ▶ upper respiratory tract infection, such as a cold, influenza, sinusitis, or bronchitis
- ▶ exposure to cold or dry weather
- ▶ emotional excitement or stress

- ▶physical exertion or exercise (exercise-induced asthma)
- ▶reflux of stomach acid (gastro esophageal reflux disease, or GERD)

Signs and symptoms

(Cough, dyspnea, and wheezing are the three most common signs and symptoms of asthma)

- ◆Tachycardia, tachypnea, mild systolic hypertension
- ◆cyanosis, confusion, and lethargy indicate the onset of life-threatening status asthmatics and respiratory failure

Diagnosis

- ◆ABG analysis
- ◆Radio allegro sorbent testing
- ◆A complete blood count (CBC)
- ◆Chest X-rays
- ◆Pulmonary function tests
- ◆Skin testing may identify specific allergens
- ◆Pulse oximetry measurements may show decreased oxygen saturation

Nursing management

- ◆Give prescribed inhalers and asthma medications
- ◆Place the patient in high Fowler's position
- ◆Encourage pursed-lip and diaphragmatic breathing
- ◆Administer prescribed humidified oxygen
- ◆Adjust oxygen according to the patient's vital signs and ABG values
- ◆Assist with intubation and mechanical ventilation, if appropriate
- ◆Perform postural drainage and chest percussion, if tolerated
- ◆If the patient is intubated, suction as needed
- ◆Treat the patient's dehydration with I.V. or oral fluids as tolerated
- ◆Keep the room temperature comfortable
- ◆Use an air conditioner or a fan in hot, humid weather

◆Monitor the patient's vital signs, intake and output, response to treatment, signs and symptoms of theophylline toxicity, breath sounds, ABG results, pulmonary function test results, pulse oximetry, complications of corticosteroid treatment, and anxiety level

Acute respiratory failure

Description

◆In acute respiratory failure, the lungs' inability to adequately maintain arterial oxygenation or eliminate carbon dioxide results in inadequate ventilation

Causes

accumulated secretions secondary to cough suppression; airway irritants; any condition

bronchospasm.

Signs and symptoms

◆changes in mental status, such as confusion, and cyanosis of the oral mucosa, lips, and nail beds

◆absent breath sounds, wheezes, rhonchi, and crackles in the lung fields

◆use of accessory muscles, pursed-lip breathing, nasal flaring, ashen skin, and tachypnea

Diagnosis

◆ABG analysis.

◆Complete blood count (CBC).

◆Chest X-rays

◆Electrocardiography (ECG) may show arrhythmias, Pulse oximetry may show decreased arterial oxygen saturation

Nursing interventions

◆Monitor the patient's vital signs, pulse oximetry readings, and ECG

◆Administer prescribed drugs as indicated (via inhalation, the oral route, nebulization, or I.V., as appropriate)

- ◆Orient the patient frequently
- ◆Administer humidified, supplemental oxygen as ordered
- ◆Maintain a patent airway
 - ◆Help clear the patient's secretions with postural drainage and chest physiotherapy; also suction as necessary
- ◆Encourage pursed-lip breathing
- ◆Encourage the use of an incentive spirometer to promote lung expansion
- ◆Reposition the patient every 1 to 2 hours to help mobilize secretions
- ◆ Perform or assist with oral hygiene
 - ◆Position the patient for comfort, with the head of the bed elevated, if appropriate, to promote optimal gas exchange
- ◆Maintain a normal body temperature with antipyretics
- ◆Cluster nursing activities and schedule care to provide frequent rest periods
- ◆Prepare for intubation and mechanical ventilation if needed
- ◆Provide sedation as necessary
- ◆ Provide education and emotional support to the patient and family

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Students' self-assessment

Choose the correct answer for each question:

1. Which of the following is a known cause of pneumonia?
 - A) Hypertension
 - B) Chemical gas inhalation
 - C) Iron deficiency
 - D) Thyroid dysfunction
2. What is the most common symptom of pneumonia?
 - A) Loss of appetite
 - B) Skin rash
 - C) Dyspnea and fever
 - D) Swollen limbs
3. Which diagnostic test is most appropriate for identifying bacterial pneumonia?
 - A) Urinalysis
 - B) ECG
 - C) Sputum culture
 - D) Skin test
4. Which symptom is most specific to COVID-19?
 - A) Dry cough
 - B) Loss of taste or smell
 - C) Chest pain
 - D) Weight gain
5. What is a correct preventive measure against COVID-19 transmission?
 - A) Handshaking
 - B) Using hand sanitizer with at least 60% alcohol
 - C) Sharing personal items
 - D) Touching your face after contact with surfaces
6. Which of the following may trigger an asthma attack?
 - A) Eating sugar
 - B) Exposure to cold dry air
 - C) Calcium deficiency
 - D) High blood pressure

7. What is the recommended position for a patient during an asthma attack?
- A) Lying flat
 - B) Trendelenburg position
 - C) High Fowler's position
 - D) Prone position
8. Which symptom may indicate the onset of acute respiratory failure?
- A) Increased appetite
 - B) Mental confusion
 - C) Elevated systolic blood pressure
 - D) Excessive sweating
9. Which test is used to evaluate oxygen and carbon dioxide levels in the blood?
- A) Urinalysis
 - B) CBC
 - C) Arterial Blood Gas (ABG) analysis
 - D) ECG
10. Which nursing intervention helps mobilize respiratory secretions?
- A) Giving antipyretics
 - B) Restricting fluid intake
 - C) Postural drainage and chest physiotherapy
 - D) Administering sedatives

Chapter four

(Nursing care of cardiovascular diseases)

Learning objectives

- Describe the anatomy of cardiovascular system.
- Discuss the physiology of cardiovascular system.

Cardiovascular system

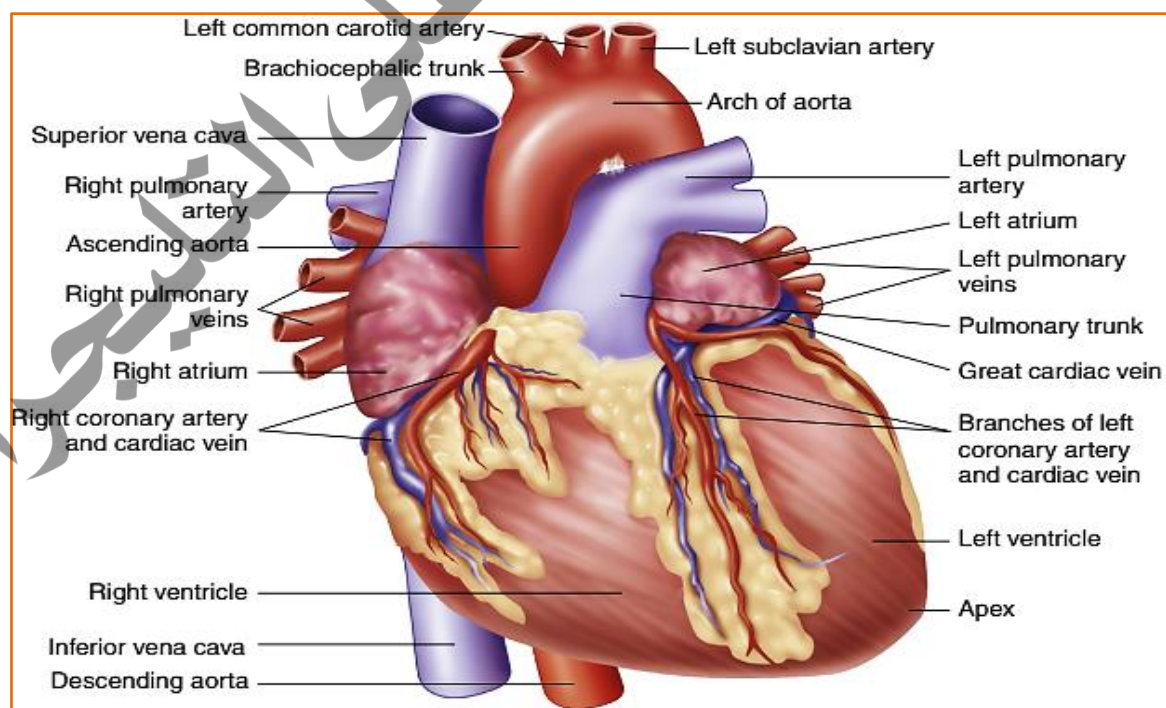
Introduction

The cardiovascular system is a complex system of distribution of nutrients, gases, electrolytes, removal of waste products of metabolism and other substances.

Component of cardiovascular system

1. Heart.
2. Blood vessels.

Anatomy and physiology of cardiovascular system



The heart

It is located in the thoracic cavity (chest) in the mediastinum (between the lungs), behind and to the left of the sternum (breast bone). The heart rests on the diaphragm in the thoracic cavity.

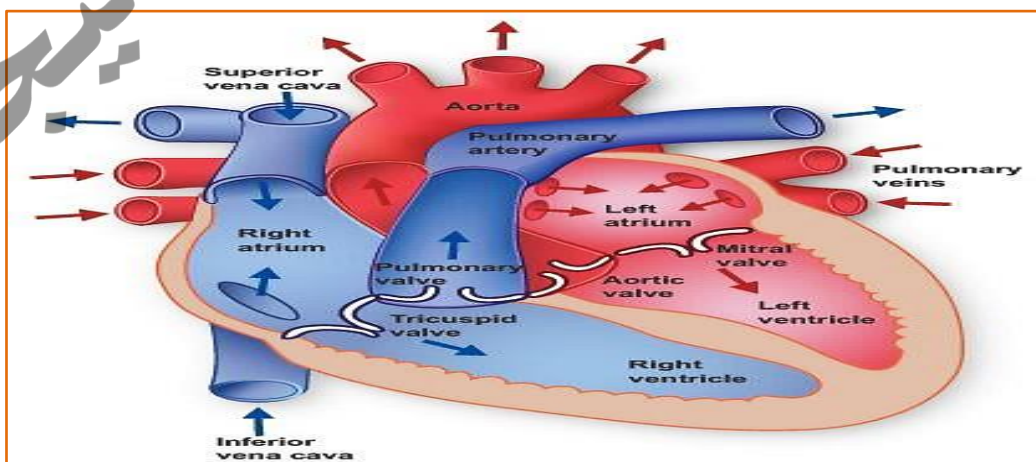
Walls of the heart

1. **Pericardium** is a double-layered, serous membrane that covers the entire structure. Between the two thin membranes is a serious fluid that allows friction-free movement of the heart.
2. **Myocardium** forms the bulk of the heart wall and is the thickest and strongest layer of the heart. It is composed of cardiac muscle tissue.
3. **Endocardium** (innermost layer) is composed of a layer of endothelial cells and a layer of subendocardial connective tissue.

Chambers of the heart

The heart has **four chambers**, two atria and two ventricles (left and right).

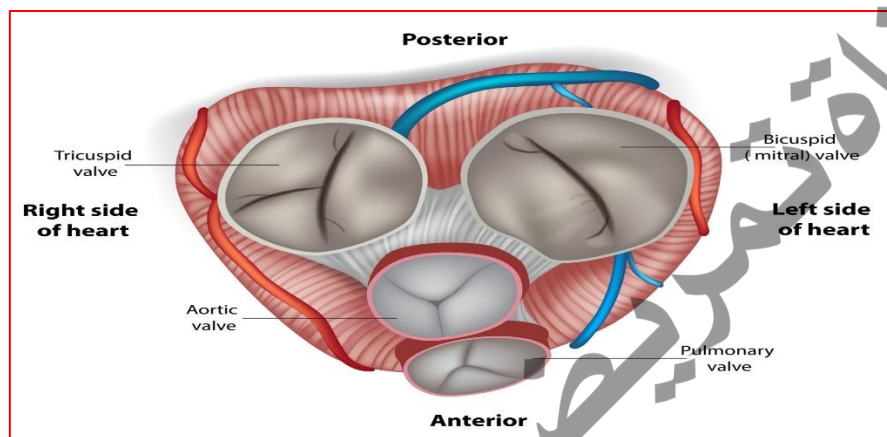
- The right heart pump receives deoxygenated blood (blood that has given up some of its oxygen to the cells) from the tissues and pumps it out into the pulmonary circulation (the lungs).
- The left heart pump receives oxygenated blood from the pulmonary circulation and pumps it out to the rest of the body (the systemic circulation).



Valves of the heart

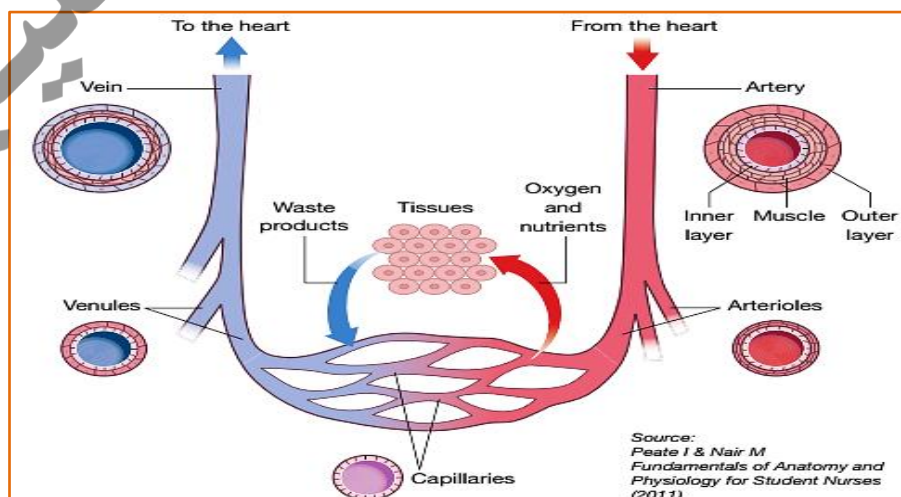
Between the atria and the ventricles (**the atrioventricular valves**).

1. **The tricuspid valve** lies between the right atrium and the right ventricle.
2. **The bicuspid (mitral) valve** lies between the left atrium and the left ventricle.
3. **Pulmonic valve** lies between the right ventricle and pulmonary artery.
4. **Aortic valve** lies between the left ventricle and aorta.



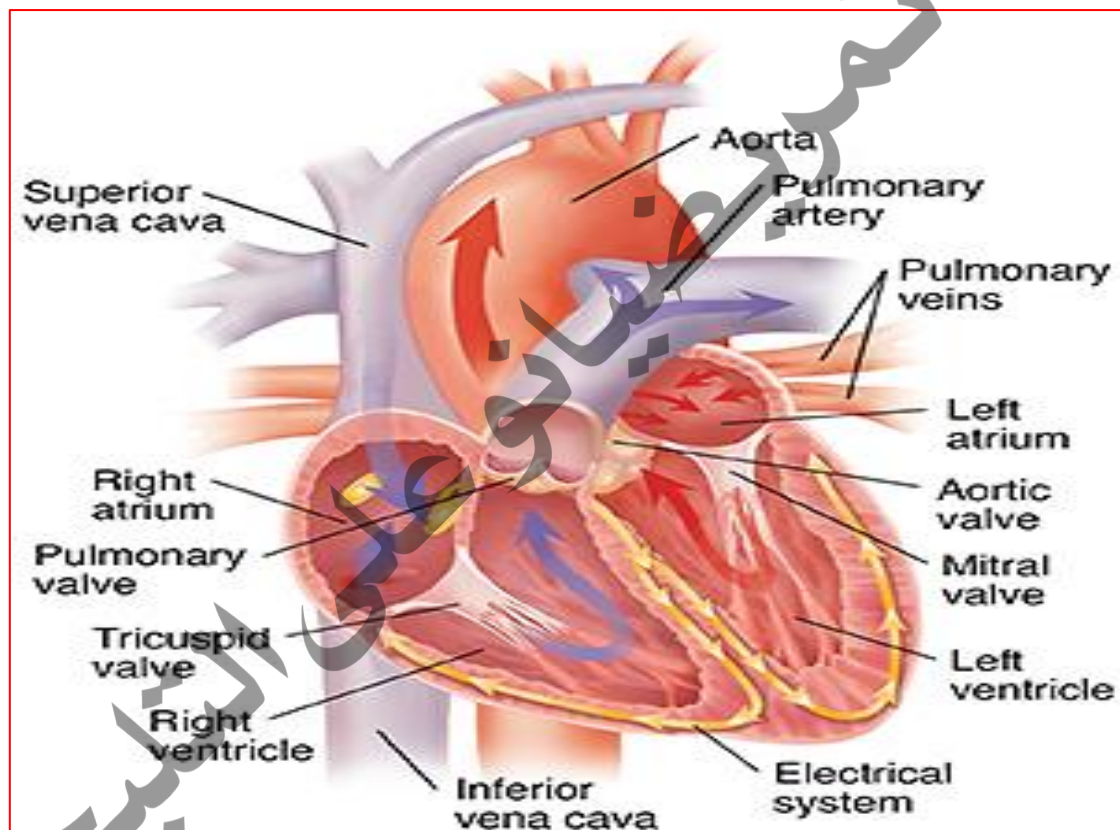
Blood vessels: Blood vessels are part of the circulatory system that transports blood throughout the body. There are three major types of blood vessels:

- **The arteries**, which carry the blood away from the heart.
- **The veins**, which carry blood from the capillaries back towards the heart
- **The capillaries**, which enable the actual exchange of water, nutrients and chemicals between the blood and the tissues.



Blood vessels of the heart

1. **The aorta** carries and distributes oxygen-rich blood to all arteries.
2. **The coronary arteries** supply oxygenated and nutrient filled blood to the heart muscle.
3. **The pulmonary arteries** carry de-oxygenated blood to the lungs.
4. **Four pulmonary veins** carry oxygenated blood from lungs to the heart.
5. **The venae cava** (superior and inferior) carry de-oxygenated blood from various regions of the body to the right atrium of the heart.



Hypertension

Learning objectives

- Define hypertension
 - Explain the pathophysiology of hypertension.
 - Identify factors affecting blood pressure.
 - Differentiate the types of hypertension.
 - list the signs and symptoms of hypertension.
 - Mention diagnostic evaluation of hypertension.
 - Explain medical and nursing management of hypertension.
 - Enumerate complication of hypertension.
-

Introduction

Hypertension Known as the “silent killer” disease, hypertension affects many Americans. Factors such as lifestyle, hereditary, comorbidities, sleep deprivation, high sodium intake in diets, and atherosclerotic disease are significant contributors to the development of increased pressures.

Definition of hypertension

Hypertension refers to an intermittent or sustained elevation in diastolic or systolic blood pressure.

Pathophysiology of hypertension

- Changes in the arteriolar wall that cause increased resistance
- Abnormally increased tone in the sensory nervous system, causing increased peripheral vascular resistance
- Increased blood volume resulting from renal or hormonal dysfunction
- Increased arteriolar thickening caused by genetic factors, leading to increased peripheral vascular resistance

- Abnormal renin release, resulting in the formation of angiotensin II, which constricts the arterioles and increases blood volume.

Factors affecting blood pressure

This factor affecting blood pressure and may lead to hypertension that include:

- Sleep apnea.
- Advancing age.
- Ethnicity/race.
- Increased sodium intake.
- Diabetes mellitus.
- Smoking.
- Lack of physical exercise.
- Nervous system regulation.
- Arterial baroreceptors and chemoreceptors.
- The renin-angiotensin-aldosterone mechanism.
- The balance of body fluids.
- Heredity.
- Obesity.

Types of hypertension

1. **Essential (also called primary or idiopathic) hypertension** is chronic elevation of BP from an unknown cause.
2. **Secondary hypertension**, which results from renal disease or another identifiable cause such as a tumor of the adrenal gland, or a congenital defect of the aorta.
3. **Malignant hypertension:** a severe, fulminant form and a medical emergency; commonly arises from both type

Classification of hypertension

Blood Pressure Categories			
			
BLOOD PRESSURE CATEGORY	SYSTOLIC mm Hg (upper number)		DIASTOLIC mm Hg (lower number)
NORMAL	LESS THAN 120	and	LESS THAN 80
ELEVATED	120 – 129	and	LESS THAN 80
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1	130 – 139	or	80 – 89
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2	140 OR HIGHER	or	90 OR HIGHER
HYPERTENSIVE CRISIS (consult your doctor immediately)	HIGHER THAN 180	and/or	HIGHER THAN 120

Signs and symptoms of hypertension

- Blood pressure measurements of more than 140/90 mm Hg on two or more readings taken at two or more visits after an initial screening
- Early morning headaches.
- Dull aching headaches throughout the day.
- Orthostatic hyper- or hypotensive change
- Severe anxiety.
- Fatigue.
- Proteinuria.
- Visual disturbances.
- Bloody nose.
- Shortness of breath.

Diagnostic evaluation for hypertension

- Laboratory tests: comprehensive panel, lipids, hepatic levels, erythrocyte, sedimentation rate (ESR), C-reactive protein, renal function tests, blood glucose level.

- Radiographic studies: magnetic resonance imaging (MRI), computed tomography (CT), x-ray, and ultrasonography (brain, kidney, heart).
- Ophthalmoscopic examination for hypertensive retinopathy.
- ECG.

Medical management for hypertension

- 1- **Diuretics:** Increase urine output by inhibiting sodium and water reabsorption by the kidney
- 2- **Sympatholytic as:** Beta Blockers
- 3- **Angiotensin-Converting Enzyme (ACE) Inhibitors**
- 4- **Calcium Channel Blockers**
- 5- **Direct Vasodilators**

Nursing management for hypertension

- 1- Assess patient's baseline blood pressure. Two to three readings in a quiet and calm environment will give the nurse a good estimation of blood pressure values.
- 2- Assess the patient general condition to identify causes leading to hypertension.
- 3- Assess for presence of complication related to hypertension.
- 4- Assess the patient life style (smoking, obesity, activity per day and exercise).
- 5- Assess for signs of edema.
- 6- Determine what risk factors are present in the patient. Consider race/ethnicity, diet, weight, comorbidities, exercise levels, levels of stress, other medications, and psychological status.

- 7- Monitor intake and output for patient administer diuretics and monitor potassium level.
- 8- Monitor the patient for presence of medication side effect.

1- Education

1- Teach patient how to made life style modification:

- L**—Limit salt, caffeine, and alcohol.
- I**—Include daily potassium and calcium.
- F**—Fight fat and cholesterol.
- E**—Exercise regularly (walking).
- S**—Stay on your BP regimen.
- T**—Try to quit smoking.
- Y**—Your medications are to be taken daily.
- L**—Lose weight.
- E**—End-stage complications will be avoided!

- 2- Teach patients to take their prescribed medications and awareness that although blood pressure levels will be more manageable with blood pressure medications, stopping the medications will revert to increased pressure again.
- 3- Teach patients the importance of monitoring side effects of hypertensive medications: hypotension. They should not fail to report dizziness, syncope, or falls to their health care provider. Remind the patient to shift positions slowly to avoid orthostatic changes.

- 4- Teach patients the importance of follow-up appointments with cardiology, ophthalmology, endocrinology (if DM present), and nephrology (if renal impairment noted).

من فضاء نهضة مصر
مركز تطوير وتنمية
المهارات
على التليجرام

Ischemic heart disease

Learning objectives

- Define ischemic heart disease.
 - Understand pathophysiology of ischemic heart disease.
 - List causes and risk factors of ischemic heart disease.
 - Differentiate between types of ischemic heart disease.
 - Recognize signs and symptoms of ischemic heart disease.
 - list diagnostic evaluation of ischemic heart disease.
 - Discuss medical and nursing management of ischemic heart disease.
 - enumerate complication of ischemic heart disease.
-

Introduction

The arterial wall serves multiple functions and can actively regulate blood flow and pressure, tissue perfusion or ischemia, and coagulation, while also participating in atherogenesis formation and progression. Atherosclerotic plaque constitutes the underlying entity of atherosclerotic diseases including coronary artery disease (CAD).

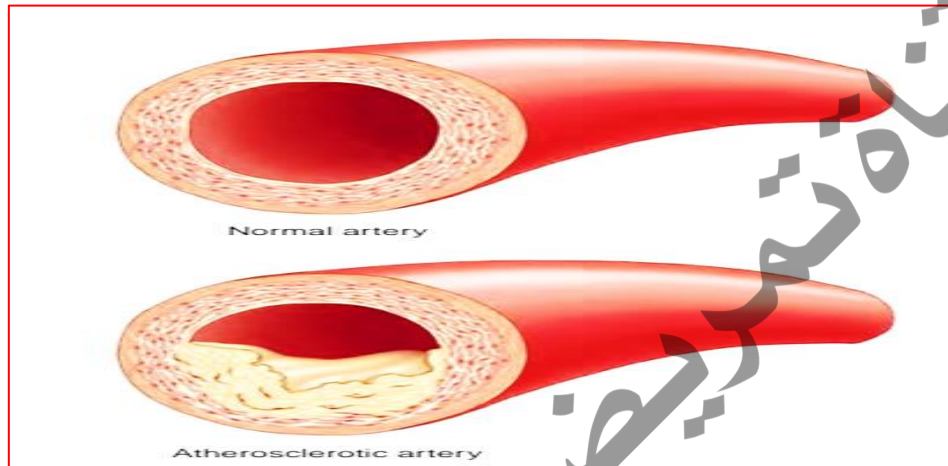
Definition of ischemic heart disease

- Heart disease that results from narrowing of coronary arteries over time as a result of atherosclerosis
- Primary effect: loss of oxygen and nutrients to myocardial tissue because of diminished coronary blood flow

Pathophysiology of ischemic heart disease

- Fatty, fibrous plaques progressively occlude the coronary arteries, reducing the volume of blood that can flow through them, leading to myocardial ischemia

- As atherosclerosis progresses, luminal narrowing and vascular changes impair the diseased vessel's ability to dilate. This causes a precarious balance between myocardial oxygen supply and demand, threatening the myocardium beyond the lesion.
- When oxygen demand exceeds what the diseased vessels can supply, localized myocardial ischemia results



Causes and risk factors of ischemic heart disease

Modifiable Risk Factors	Nonmodifiable Risk Factors
Blood pressure	Age
Cigarette/tobacco use	Gender
Serum cholesterol/lipid levels	Family history
Diet	Race
Lifestyle and physical activity	

Source: Adapted from National Institute of Health (nih.gov); Urden, L. D., Stacy, K. M., & Lough, M. E. (2020). *Priorities in critical care nursing* (8th ed.). St. Louis, MO: Elsevier

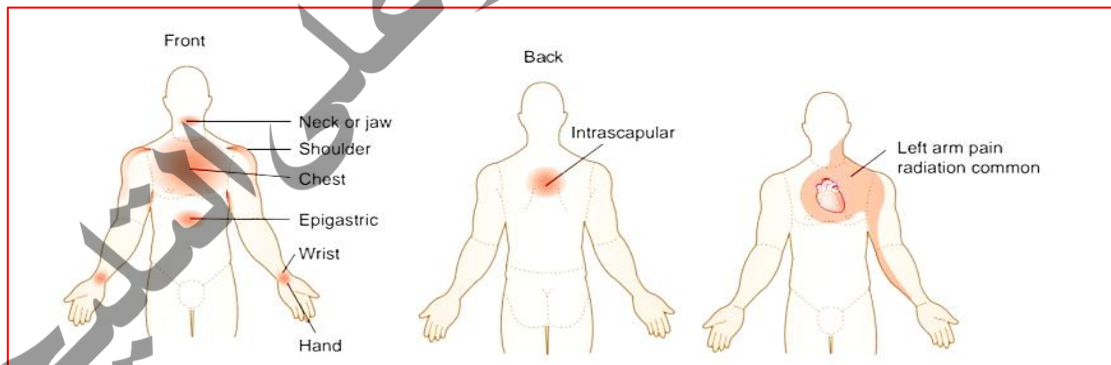
Types of ischemic heart disease

- **Stable angina:** Chest pain that is predictable and caused by a similar pattern, activity, or event. Typically, patients achieve pain control within 5 minutes of onset with rest and the use of sublingual nitroglycerin.

- **Unstable angina:** The result of limited or insufficient blood flow through the coronary artery. USA is a change in the previously established stable pattern of angina and requires medical intervention.
- Cardiac biomarkers are not elevated.
- **NSTEMI:** Partial blockage of a coronary artery producing ST-segment depression. Moderate damage may be caused to the left ventricle.
- **STEMI:** Total blockage of a coronary artery producing ST-segment elevation or new left bundle branch block (LBBB). Cardiac biomarkers are significantly elevated, and there is potential for major damage to the left ventricle.

Signs and symptoms of ischemic heart disease

- **Stable** — pain that's predictable in frequency and duration and relieved with nitrates and rest
- **Unstable** — increased pain that's easily induced
- **Prinzmetal's or variant** — pain that results from unpredictable coronary artery spasm

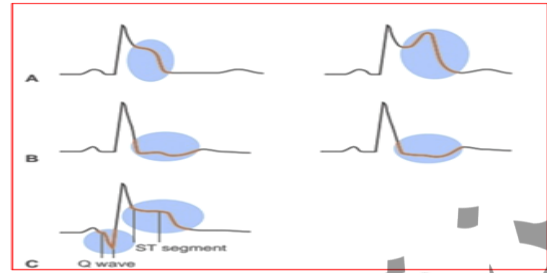


Other signs and symptoms of CAD include:

- Nausea
- Vomiting
- Weakness
- Diaphoresis
- Cool extremities.

Diagnostic evaluation for ischemic heart disease

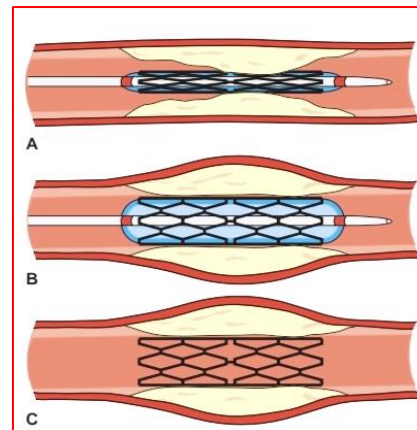
- 1- ECG.
- 2- Serum myoglobin.
- 3- Serum CK-MB.
- 4- CBC.
- 5- Serum magnesium and potassium.
- 6- Vital signs, oxygen saturation, intake and output.
- 7- Echocardiogram.
- 8- Possible coronary interventions: angioplasty, stents, or cardiac bypass.



Management for patient with ischemic heart disease

Medical management

- 1- Oxygen.
- 2- Morphine sulfate.
- 3- Nitrates. Vasodilate to reduce preload and afterload. Reduce oxygen consumption of myocardium.
- 4- Fab Four cardiac medications: aspirin, statin, ACEI, beta Blocker.
- 5- Platelet aggregation inhibitors.
- 6- Thrombolytics.
- 7- Anticoagulants.
- 8- Antidysrhythmic.
- 9- Vasodilators.
- 10- Percutaneous coronary interventions and stents.
- 11- Myocardial revascularization–CABG.



Nursing management

1. Ensure vascular access is established. IV access may be necessary to use to administer drugs for pain relief.
2. Intervention Monitor location, duration, intensity, and radiation of pain.
3. Monitor the vital signs.
4. Administer oxygen as ordered via nasal cannula to increase oxygen availability to myocardium.
5. Obtain a 12-lead ECG as ordered to determine ischemia or injury of the myocardium with evaluation of the ST segment.
6. Remain with patient and reassess pain in 5 minutes after.
7. Sit the patient upright, preferably at a 45-degree angle.

Patient Education

- Teach patient risk factors for IHD
- avoidance of activities that precipitate episodes of pain
- effective coping mechanisms to deal with stress
- need to follow the prescribed drug regimen
- low-sodium, low fat and low-calorie diet
- importance of regular, moderate exercise.

Complication of Ischemic Heart Disease

- 1- Dysrhythmias.
- 2- Cardiogenic shock.
- 3- Heart failure/ pulmonary edema.
- 4- Emboli.
- 5- Rupture of muscles or valves of the heart, septal rupture.
- 6- Pericarditis (inflammation of the heart muscle).

Cardiac surgery

Learning objectives

- Identify Indications for cardiac surgery.
 - List Types of cardiac surgery.
 - Discuss nursing interventions before for cardiac surgery.
 - Discuss nursing interventions after for cardiac surgery.
 - Mention Complications of cardiac surgery.
-

Introduction

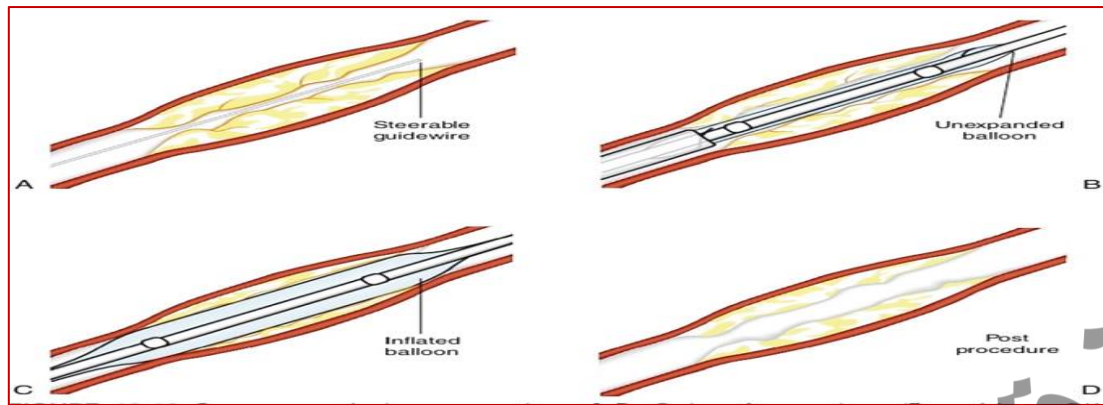
An explosion in new technology has facilitated these advancements and allowed surgeons to perform coronary artery bypass graft (**CABG**) surgery, valve surgery, ablation for atrial fibrillation procedures, and even proximal aortic operations without median sternotomy, which has been the traditional approach for access to the heart.

Indications for cardiac surgery

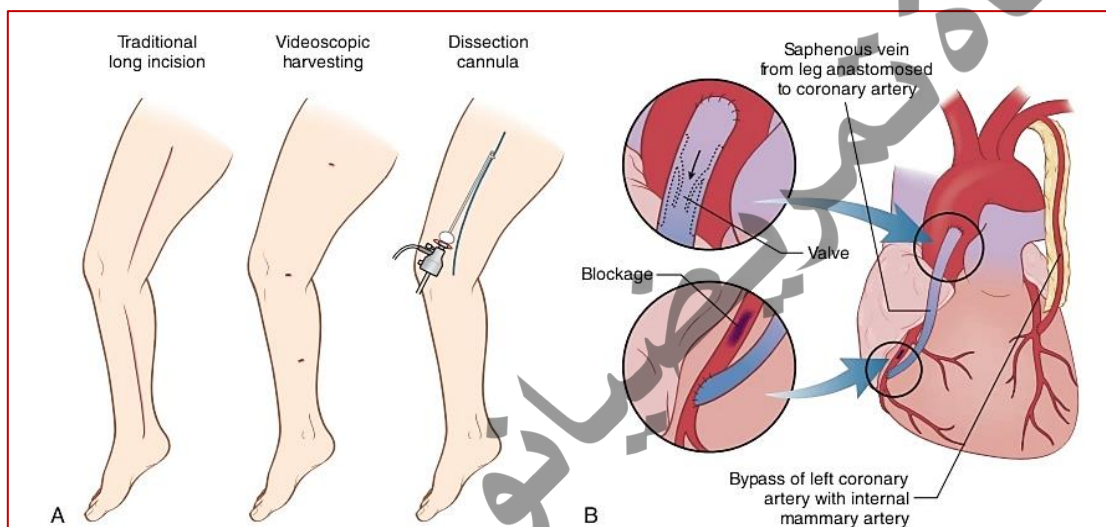
- Asymptomatic or Mild Angina.
- Stable Angina.
- MI.
- Unstable Angina
- Presence of Poor LV Function.
- Failed PCI.
- Previous CABG.

Types of cardiac surgery:

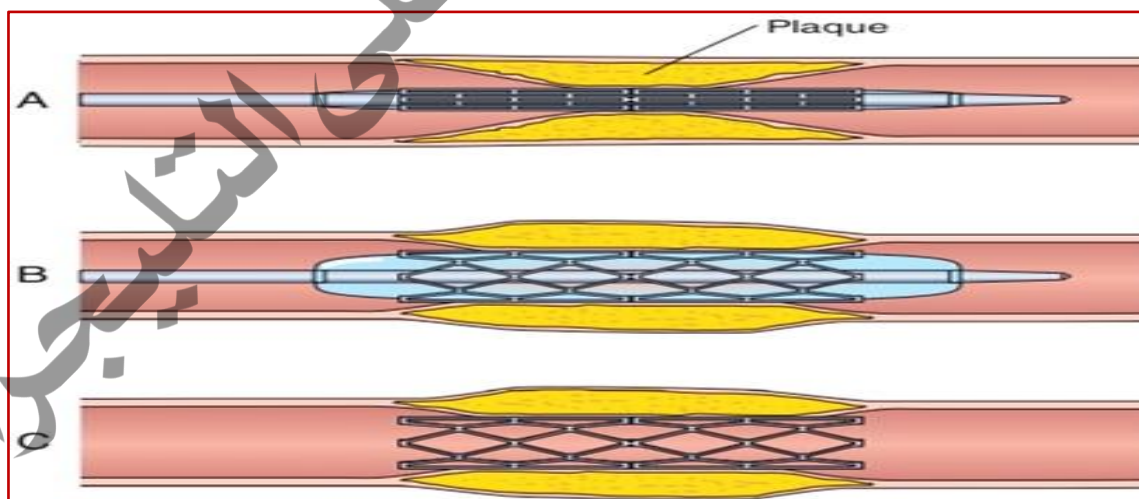
1. Percutaneous coronary intervention (PCI):
2. Balloon angioplasty:



3. Coronary Artery Bypass Graft (CABG):



4. Coronary Artery Stent



5. Percutaneous Trans-luminal Coronary Angioplasty (PTCA)

6. Minimally Invasive Coronary Artery.

Preparation for Surgery (preoperative care)

- A nursing assessment is important to provide baseline data that can be used for postoperative comparison and early discharge planning.
- In addition to routine admission testing, patients with chronic obstructive pulmonary disease (COPD) may have pulmonary function tests and baseline arterial blood gases (ABGs) done.
- Patients with carotid bruits have carotid studies to determine the amount of occlusion in the carotid artery.
- Drugs that increase bleeding include aspirin, often stopped 3 to 7 days preoperatively; warfarin (Coumadin), often stopped 4 to 5 days preoperatively; and heparin, usually stopped 4 hours preoperatively.
- The patient usually takes nothing by mouth (NPO) 8 to 12 hours before surgery.
- Explanations of expected procedures and care including pain management, endotracheal tube, methods of communicating, ventilator, chest tubes, coughing and deep breathing exercises (IV lines, urinary catheter, incision care, and various equipment alarms) are provided to the patient and family.
- It should be emphasized that patients are not able to talk while the endotracheal tube is in place.

Nursing care after cardiac catheterization and angiography:

- Maintain the patient on bed rest (time varies depending on the size of the catheter used and the method for preventing arterial bleeding).
- Keep the extremity used for catheter insertion immobile.
- Mark the hematoma with a marker around outer perimeter, to aid in assessing for an increase in bleeding.

- Assess for bruits.
- Maintain head-of-bed elevation no higher than 30 degrees.
- Monitor peripheral pulses, color, and sensation of the extremity distal to insertion site (every 15 minutes 3 4; every 30 minutes 3 4; every 1 hour).
- In addition, monitor the opposite extremity pulse to assess for presence of equal pulses and color, and sensation bilaterally.
- Observe cardiac rhythm.
- Encourage fluid intake if not contraindicated.
- Monitor intake and output.
- Observe for an adverse reaction to dye (angiography).
- Assess for chest pain, back pain, and shortness of breath and notify health care provider.
- Monitor for hypotension; administer fluids and vasopressors as ordered or based on protocol.
- Rewarm the patient gradually (if applicable).
- Monitor and treat fluid and electrolytes, hemoglobin, hematocrit, renal function, and coagulation studies.
- Provide pain relief.
- Assess wounds and provide incisional care per hospital protocol.
- Gradually increase the patient's activity.

Complications of Cardiac Surgery:

Patients who have had cardiac surgery should be closely monitored for complications such as:

- Dysrhythmias (atrial fibrillation, atrial flutter, ventricular tachycardia, ventricular fibrillation)

Student self-assessment

1. What early manifestation(s) is the patient with primary hypertension likely to report?

- a. No symptoms
- b. Cardiac palpitations
- c. Dyspnea on exertion
- d. Dizziness and vertigo

2. What accurately describes the pathophysiology of CAD?

- a. Partial or total occlusion of the coronary artery occurs during the stage of raised fibrous plaque.
- b. Endothelial changes may be caused by chemical irritants, such as hyperlipidemia or by tobacco use.
- b. Collateral circulation in the coronary circulation is more likely to be present in the young patient with CAD.
- c. The leading theory of atherogenesis proposes that infection and fatty dietary intake are the basic underlying causes of atherosclerosis.

What are manifestations of acute coronary syndrome (ACS) (select all that apply)?

- a. Dysrhythmia
- b. Stable angina
- c. Unstable angina
- d. ST-segment-elevation myocardial infarction (STEMI)
- e. Non-ST-segment-elevation myocardial infarction (NSTEMI)

Which characteristics describe unstable angina (select all that apply)?

- a. Usually precipitated by exertion
- b. New-onset angina with minimal exertion
- c. Occurs only when the person is recumbent
- d. Characterized by increased duration or severity
- e. Usually occurs in response to coronary artery spasm

A patient is scheduled to have CABG surgery. What should the nurse explain is involved with the procedure?

- a. A synthetic graft will be used as a tube for blood flow from the aorta to a coronary artery distal to an obstruction.

- b. A stenosed coronary artery will be resected, and a synthetic arterial tube graft will be inserted to replace the diseased artery.
- c. The internal mammary artery will be detached from the chest wall and attached to a coronary artery distal to the stenosis.
- d. Reversed segments of a saphenous artery from the aorta will be anastomosed to the coronary artery distal to an obstruction

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Chapter five

(Nursing care of gastrointestinal diseases)

Learning objectives

At the end of this chapter student will be able to:

- Describe the anatomy and physiology of digestive system.
 - Discuss the most common digestive system.
-

The digestive system

Introduction

Food is essential to life, which is necessary for the chemical reactions that take place in every body cell. Most of the foods we eat are just too large to pass through the plasma membranes of the cells. The process of breaking down food molecules for the body's cells to use is called digestion, and the organs which work together to perform this function are termed the digestive system.

Anatomy and physiology of digestive system

The digestive organs are commonly divided into two main groups: the gastrointestinal (GI) tract (also called the alimentary canal) and the accessory structures.

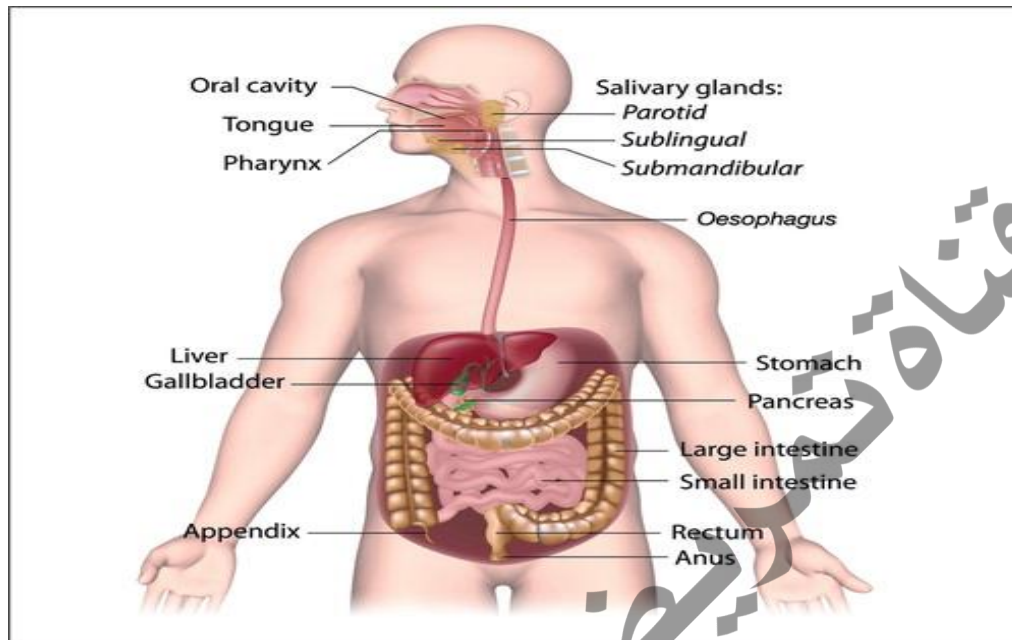
1- The gastrointestinal (GI) tract:

The gastrointestinal tract is a continuous tube which extends from the mouth to the anus and which runs through the ventral body cavity. From the time food is eaten until it is digested and eliminated, it is in the gastrointestinal tract.

The organs which make up the gastrointestinal tract are the **mouth, pharynx, esophagus, stomach, small intestine, and large intestine**. These organs are sometimes referred to as **the primary organs of the digestive system**.

2- Accessory structures:

These structures include the **teeth, tongue, salivary glands, liver, gallbladder, and pancreas.**

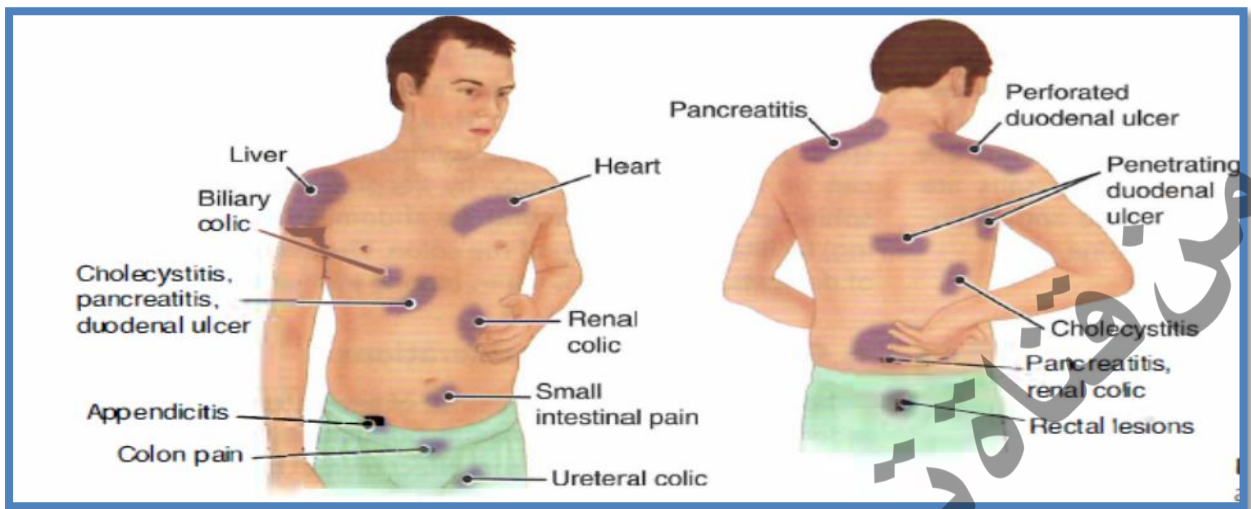


The digestive system

Functions and stages of digestive process:

- (1) **Ingestion:** Taking food, drink, or medicines into the body by mouth.
- (2) **Peristalsis:** Alternating contraction and relaxation of the walls of a tubular structure by which food is moved along the digestive tract.
- (3) **Digestion:** The processes by which food is broken down **chemically** and **mechanically** for the body's use.
- (4) **Absorption:** the passage of substances (water, salts, vitamins, carbohydrates, proteins, and fats) through the intestinal mucosa of the villi into the blood or lymph.
- (5) **Defecation or elimination:** The discharge of indigestible substances from the body.

Common Gastrointestinal disorders



Common sites of referred abdominal pain

Gastritis

Learning objectives:

At the end student will be able to:

- Define the gastritis.
- Discuss pathophysiology of gastritis.
- Mention causes of gastritis.
- Enumerates signs & symptoms of gastritis.
- List diagnostic procedure used in gastritis diagnosis.
- Explain management of gastritis.

➤ Definition

Gastritis is a disease which results from an inflammation of the gastric mucosa. It is characterized by pain, swelling, and irritation of the mucosal membrane of the stomach.

Gastritis is either an acute or a chronic depending on how long the signs and symptoms persist.

- **Acute gastritis** is an inflammation of the stomach lining that occurs suddenly and lasts shortly within one or two days and even less than a month.
- **Chronic gastritis** is an inflammation of the gastric mucosa that occurs gradually and persists for more than a month and even for some years.

➤ **Pathophysiology:**

» **Acute gastritis**

Acute gastritis also referred to as reactive gastritis occurs as a result of the trigger by factors such as NSAIDs, stress, bile reflux, radiation, alcohol abuse, cocaine addiction, and ischemic damage. The outcome of these triggers may result in the protective mucosal layer being altered, and acid secretion produces mucosal reddening, edema, ulcers, hemorrhage, and erosion of the gastric mucosa.

» **Chronic gastritis**

In the majority of cases, the *H. pylori* infection persists leading to accumulation of large number of chronic inflammatory cells leading to active chronic gastritis.

➤ **Causes of gastritis:**

- **Helicobacter pylori (Hp):** is the main cause of gastritis.
- **Autoimmune response.**
- **Chemical irritants;** for example, alcohol, salicylates.
- **Bacterial infections or toxins;** for example, staphylococcal food poisoning, scarlet fever, pneumonia.
- **Viral infections;** "viral gastroenteritis," measles, hepatitis, influenza.
- **Allergy;** for example, to shellfish.
- **Drugs & Ingestion of poisons.**
- **Dietary indiscretion;** excess of tea or coffee, excess of heavily spiced food.

➤ Signs and symptoms:

- Anorexia, loss of appetite.
- Epigastric fullness (full stomach) and pressure in that area.
- Nausea and vomiting.
- Headache and dizziness.
- Examination indicates mild stomach tenderness.
- Hemorrhage present if cause is chemical irritants.
- Diarrhea, colic, malaise, fever, chills, headache, and muscle cramps if toxins or infections are the cause.

➤ Diagnosis:

★ First investigations to order:

- ✓ Helicobacter pylori urea breath test.
- ✓ H pylori fecal antigen test.
- ✓ CBC.

★ Investigations to consider:

- ✓ Endoscopy.
- ✓ H pylori rapid urease test.
- ✓ Gastric mucosal histology
- ✓ Serum vitamin B12.

★ Emerging tests:

- ✓ H pylori culture/polymerase chain reaction (PCR).

➤ Management:

■ Medical and pharmacological Management:

Treatment depends on the cause of the illness and the symptoms of the patient. Generally, remove the offending agent, and treat the specific infections.

- 1- Elimination of causes (such as treatment of H-Pylori by antibiotics, discontinuation of NSAIDs, smoking cessation).
- 2- If the patient has become dehydrated, correct the water and electrolyte disturbances. Intravenous fluids may be necessary to correct dehydration and electrolyte.
- 3- Drugs to control nausea, vomiting, and diarrhea may be prescribed. Tigran (trimethobenzamide) is a medicine that can be given to combat nausea and vomiting.
- 4- Antibiotics, antacids.
- 5- Histamine-2 (H2) receptor antagonists, such as cimetidine.
- 6- For massive bleeding: blood transfusion.

CAUTION: Do NOT use aspirin because aspirin can be very irritating to the stomach lining.

■ **Nursing Management:**

- 1- Do not give the patient anything by mouth until he has stopped vomiting. Then, begin by giving him clear liquid and, later, progress to a diet of soft foods as the patient can tolerate them.
- 2- Assess the patient's symptoms and administer the prescribed symptomatic relief medications such as antacids and antiemetics.
- 3- Monitor intake and output closely.
- 4- Administer and monitor IV therapy when ordered to replace lost fluids.
- 5- Weigh daily to monitor weight loss and encourage the prescribed diet to maintain nutrition.
- 6- Instruct the patient to:
 - Eliminate irritating foods, such as caffeine and spicy foods.
 - Reduce anxiety.

➤ **Complications:**

- Hemorrhage, anemia.
- Dehydration.
- Obstruction.
- Perforation "Peritonitis.
- Gastric cancer.
- Peptic ulcer.
- Atrophic gastritis.

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Appendicitis

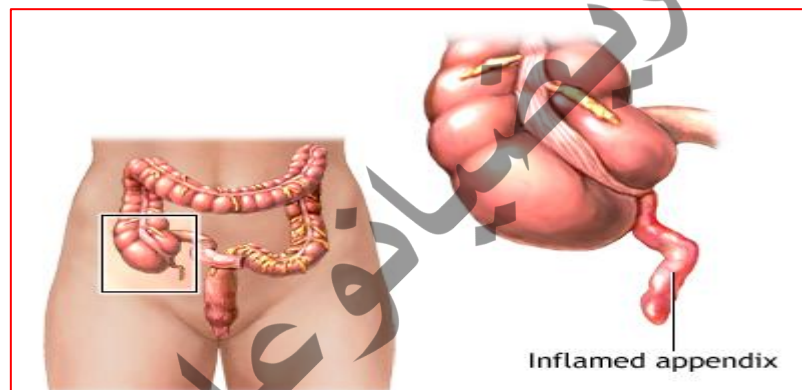
Learning objectives:

At the end student will be able to:

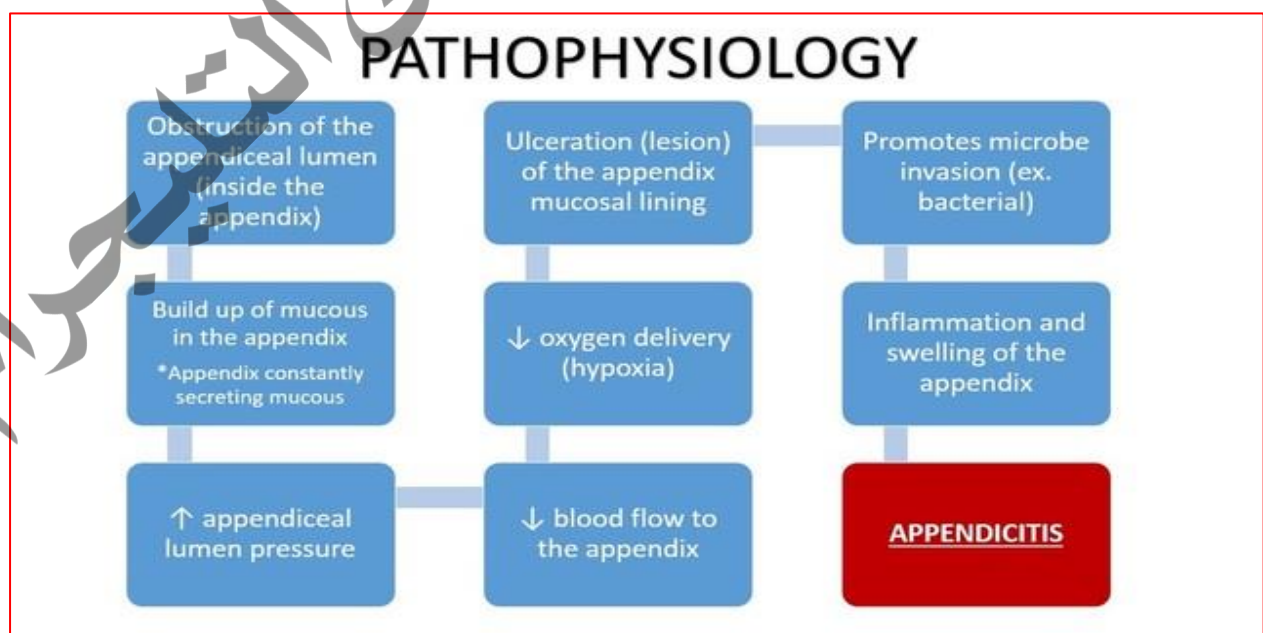
- Define the appendicitis.
- Discuss pathophysiology of appendicitis.
- Mention causes of appendicitis.
- Enumerates signs & symptoms appendicitis.
- Discuss diagnostic procedure used in appendicitis diagnosis.
- Explain management of appendicitis.

➤ Definition

Appendicitis is the inflammation of the vermiform appendix.



➤ Pathophysiology:



➤ Causes of appendicitis:

An obstruction, or blockage, of the appendiceal lumen causes appendicitis.

Sources of blockage include:

- Stool, parasites, or growths that clog the appendiceal lumen.
- Enlarged lymph tissue in the wall of the appendix, caused by an infection in the GI tract or elsewhere in the body.
- Inflammatory bowel disease (IBD), which includes ulcerative colitis.
- Long-lasting disorders that cause irritation and ulcers in the GI tract.
- Trauma to the abdomen.

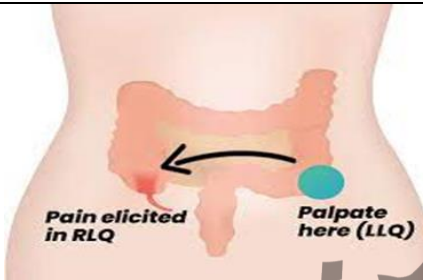
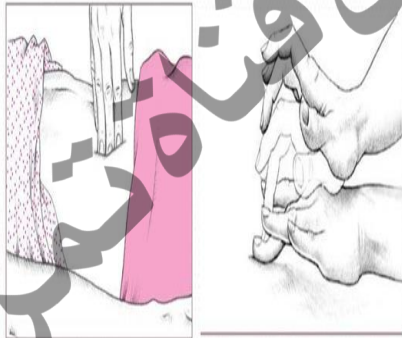
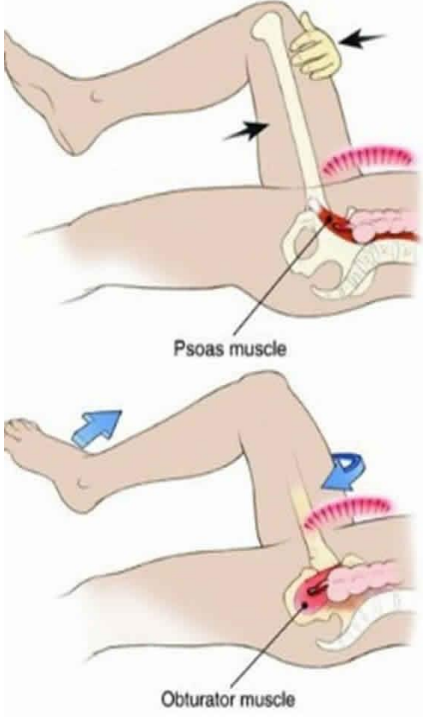
➤ Signs and symptoms:

- Generalized abdominal pain that localizes in the right lower quadrant (the major symptom of pancreatitis).
- Anorexia.
- Nausea and vomiting.
- Abdominal rigidity or guarding.
- Rebound tenderness.
- Fever.
- Elevated white blood cell count.

➤ Diagnosis:

★ Physical Exam

Details about the person's abdominal pain are key to diagnosing appendicitis. The health care provider will assess the pain by touching or applying pressure to specific areas of the abdomen.

Sign	Description	
Rovsing's sign	-Pressure lower left side produces pain on your right side.	
Blumberg's sign (also known as rebound tenderness).	-Pressure applied to the sore area feels worse when it's released	
Dunphy's sign	-Coughing makes the pain worse.	
Psoas sign	If your appendix is behind your colon rather than in front, appendicitis might irritate your psoas muscle. You might find yourself flexing your right hip to shorten the muscle, which relieves pain. A healthcare provider might try extending your right hip or rotating it outward. If this hurts, it's called the psoas sign. If rotating it inward hurts, it's called the obturator sign .	

★ Laboratory Tests:

- Blood tests: Blood tests can show signs of infection, such as a high white blood cell count, also may show dehydration or fluid and electrolyte imbalances.
- C-reactive protein count (CRP).
- Urinalysis: Urinalysis is used to rule out a urinary tract infection or a kidney stone.
- Pregnancy test: may be performed for women of childbearing age to rule out ectopic pregnancy and before x-rays are obtained.

★ Imaging Tests:

- ✓ Abdominal ultrasound
- ✓ CT scan.

➤ Management:

▪ Medical management:

- **IV fluids:** To correct fluid and electrolyte imbalance and dehydration, IV fluids are administered prior to surgery.
- **Antibiotic therapy:** To prevent sepsis, antibiotics are administered until the surgery is performed.
- **Drainage:** When perforation of the appendix occurs, an abscess may form and the patient is initially treated with antibiotics and the surgeon may place a drain in the abscess.

▪ Surgical management:

Surgery to remove the appendix is called an appendectomy. A surgeon performs the surgery using one of the following methods:

- **Laparotomy:** Laparotomy removes the appendix through a single incision in the lower right area of the abdomen.

- **Laparoscopic surgery:** Laparoscopic surgery uses several smaller incisions and special surgical tools fed through the incisions to remove the appendix.
- **Nursing Management:**
 - Administer IV fluids as ordered to maintain hydration.
 - Keep the patient NPO until symptoms subside and/or surgery is ruled out.
 - **Position:** the patient in Fowler's or semi-Fowler's position. This position relaxes the abdominal muscles and reduces pain.
 - Never apply heat to the abdomen, as this may cause the appendix to rupture.
 - Analgesics are normally withheld since they mask symptoms.
 - **Prevent infection:**
 - Maintain a clean environment.
 - Provide wound care to the postoperative patient, and assess incision frequently for signs of infection.
 - Monitor patient temperature and heart rate for signs of potential infection.
 - Administer antibiotics as prescribed by the provider.
 - Reduce patient anxiety by keeping the patient informed of the plan of care and ensure the patient is aware of diagnosis and treatment options.
 - Encourage patients to walk as able/ permitted to maintain circulation.
 - Monitor for adequate bowel movements. Encourage adequate water intake and use of a stool softener.

Discharge and Home Care Guidelines:

- **Removal of sutures:** The nurse instructs the patient to make an appointment with the surgeon to remove the sutures between the 5th and 7th days after surgery.
- **Activities:** Heavy lifting is to be avoided postoperatively; however, normal activity can be resumed within 2 to 4 weeks.
- **Home care:** A home care nurse may be needed to assist with incision care and to monitor the patient for complications and wound healing.

➤ **Complication:**

- A burst appendix.
- Peritonitis.
- Appendiceal abscess.

Cholecystitis

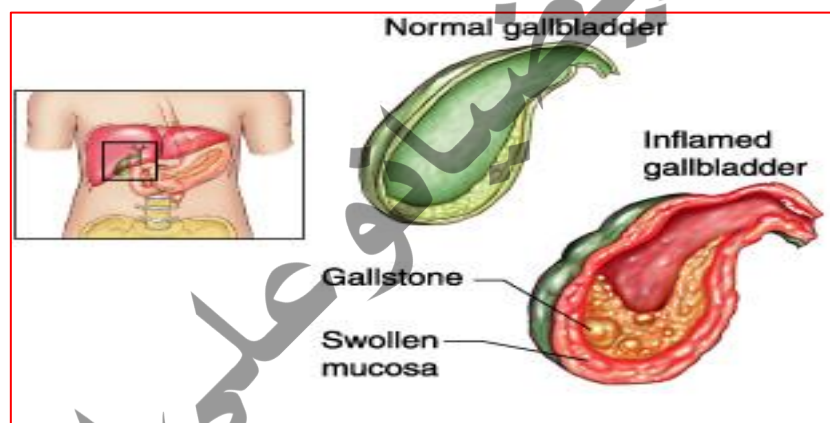
Learning objectives:

At the end student will be able to:

- Define the cholecystitis.
- Discuss pathophysiology of cholecystitis.
- Mention causes of cholecystitis.
- Enumerate signs & symptoms cholecystitis.
- Interpret diagnostic procedure used in cholecystitis diagnosis.
- Explain management of cholecystitis.

➤ Definition

Cholecystitis is inflammation of the gallbladder. In ninety percent of the cases, this condition is associated with gallstones.



➤ Pathophysiology:

- **Obstruction:** Calculous cholecystitis occurs when a gallbladder stone obstructs the bile outflow.
- **Chemical reaction:** Bile remaining in the gallbladder initiates a chemical reaction; autolysis and edema occur.
- **Compression:** Blood vessels in the gallbladder compressed, compromising its vascular supply.

➤ Causes of cholecystitis:

- Gallstones lodge in the neck of the gallbladder.

- Bile duct blockage.
- Bile stasis or the lack of gallbladder contraction.
- Infection.

➤ **Signs and symptoms:**

- Intolerance of fatty foods or heavy meals.
- A low-grade fever, chills, and a light increase in the white blood count when inflammation of the gallbladder is severe.
- Pain in the right upper quadrant of the abdomen or pain radiating to the scapula and right shoulder (this pain called referral pain).
- Rigid abdominal muscles, shallow respirations, and increased pulse rate when pain is severe.
- Prolonged gallbladder attack may cause liver damage with these symptoms: jaundice, pruritus, dark urine, and clay-colored stools.

➤ **Diagnosis:**

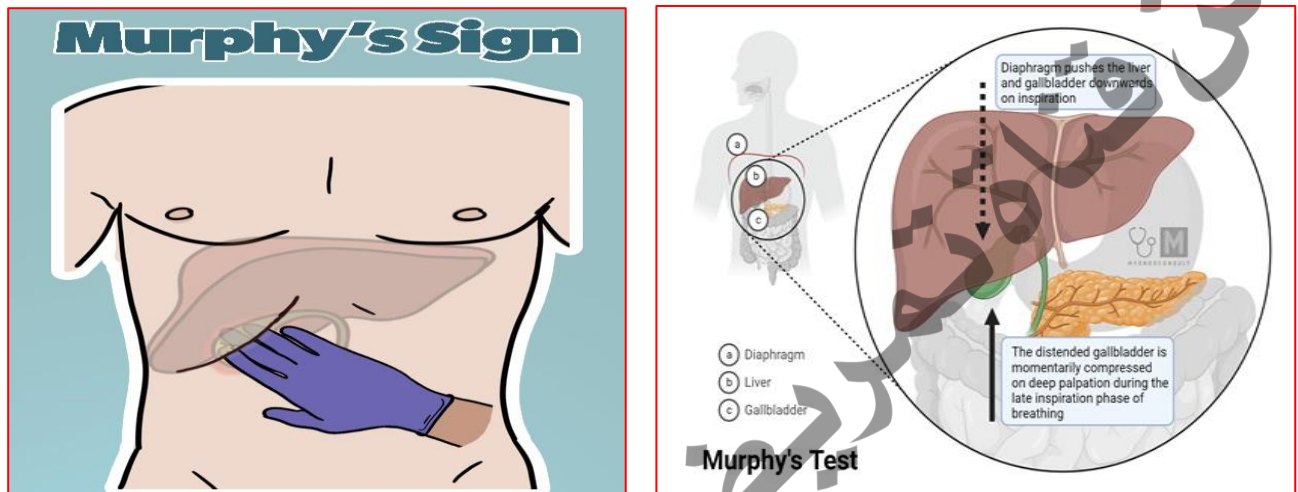
★ **Physical Exam**

Murphy's sign test:

- Have the patient lie supine on the exam table.
- Position examiner left hand, with fingers pointing toward the midline, on the patient's lowermost right anterior rib cage. Your index finger should rest on the most inferior rib.
- Extend examiner left thumb and push it into the patient's abdomen without leaning on the rib cage.
- Instruct the patient to take a deep breath, feeling the rib cage move toward you during inspiration.
- Observe the patient's breathing and assess the degree of tenderness.
- Repeat the test as a placebo maneuver by placing your hand in the same position but without pushing your thumb inward. Note if the patient can complete a full inspiration.

A positive Murphy's sign: when the patient experiences pain or tenderness severe enough to cause an abrupt halt in inspiration, usually toward the end of the breath. This suggests acute cholecystitis.

A negative Murphy's sign: when the patient can complete a full inspiration without significant pain or tenderness.



★ Laboratory Tests:

- Complete blood count (CBC): CBC: Moderate leukocytosis (acute).
- Serum bilirubin and amylase: Elevated.
- Serum liver enzymes—AST; ALT; ALP; LDH: Slight elevation.
- Prothrombin levels: Reduced when obstruction to the flow of bile into the intestine decreases' absorption of vitamin K.

★ Imaging Tests:

- **Ultrasound (also called sonography):** It is used to see the liver and gallbladder and check blood flow through different vessels.
- **Belly X-ray:** This test makes pictures of internal tissues, bones, and organs using invisible electromagnetic energy beams.
- **CT scan.**
- **RCP (endoscopic retrograde cholangiopancreatography):** This is used to find and treat problems in your liver, gallbladder, bile ducts, and pancreas.

Diagnostic criteria for acute cholecystitis:

A. Local signs of inflammation:

(1) Murphy's sign, (2) RUQ mass/pain/tenderness.

B. Systemic signs of inflammation:

(1) Fever, (2) elevated CRP, (3) elevated WBC count.

C. Imaging findings: imaging findings characteristic of acute cholecystitis.

» Definite diagnosis:

(1) One item in **A** and one item in **B** are positive.

(2) **C** confirms the diagnosis when acute cholecystitis is suspected clinically.

➤ Management:

▪ Medical management:

- **Fasting:** The patient may not be allowed to drink or eat at first in order to take the stress off the inflamed gallbladder; IV fluids are prescribed to provide temporary food for the cells.
- **Supportive medical care:** This may include restoration of hemodynamic stability and giving antibiotics

▪ Pharmacologic Therapy

- **Antibiotic therapy:** Levofloxacin and Metronidazole for prophylactic antibiotic coverage against the most common organisms.
- **Promethazine** may control nausea and prevent fluid and electrolyte disorders.
- **Acetaminophen** may control inflammatory signs and symptoms and reduce pain.

▪ Surgical management:

- **Cholecystectomy:** Cholecystectomy is most commonly performed by using a laparoscope and removing the gallbladder.

Student self-assessment

1. A positive Rovsing's sign is indicative of appendicitis. The nurse knows to assess for this indicator by palpating the:

- A. Right lower quadrant.
- B. Left lower quadrant.
- C. Right upper quadrant.
- D. Left upper quadrant.

2. Which clinical manifestation would the nurse expect a client diagnosed with gastritis to exhibit?

- A. Anorexia, loss of appetite.
- B. Epigastric fullness (full stomach) and pressure in that area.
- C. Headache and dizziness.
- D. All the above.

3. Ali, a 60-year-old woman, is diagnosed with cholecystitis. Which diet, when selected by the client, indicates that the nurse's teaching has been successful?

- A. 4-6 small meals of low-carbohydrate foods daily.
- B. High-fat, high-carbohydrate meals.
- C. Low-fat, high-carbohydrate meals.
- D. High-fat, low protein meals.

4. A major symptom of pancreatitis that brings patient to medical care is:

- A. Severe abdominal pain.
- B. Fever.
- C. Jaundice.
- D. Mental agitation.

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Chapter six (Nursing care of urinary tract diseases)

Learning objectives

At the end of this chapter student will be able to:

- Describe the anatomy and physiology of urinary system.
- Discuss the most common urinary system.

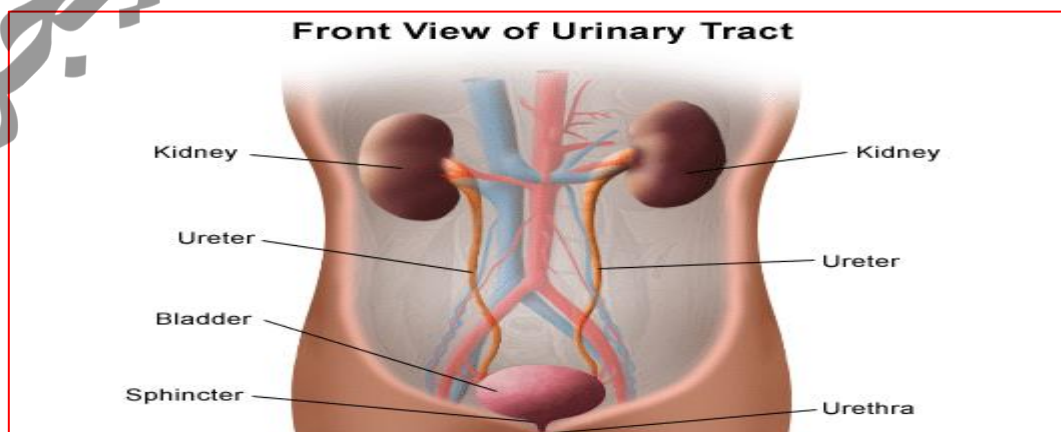
The urinary system

Introduction

Metabolism of nutrients by the body produces wastes that must be removed from the body. Although excretory processes involve several organ systems (e.g., lungs excrete carbon dioxide), it is mainly the urinary system that removes nitrogenous wastes from the body.

Anatomy and physiology of urinary system

The primary organs in the urinary system are the **paired kidneys**. To properly do their job, the kidneys act first as blood “filters”, and then as blood “processors”. In addition to the kidneys, the urinary system also includes the **ureters**, which transport the urine from the paired kidneys to the **urinary bladder** where it is collected and stored. Once the bladder is full, the urine exits the body via the **urethra**.



Functions of the urinary system:

- **Excretion:** removal of waste material from the blood plasma and the disposal of this waste in the urine.
- **Elimination:** removal of waste from other organ systems. From digestive system – undigested food, water, salt, ions drugs. From respiratory system – CO₂, H⁺, water, toxins. From skin – water, NaCl, nitrogenous wastes (urea, uric acid, ammonia, creatinine).
- **Water balance:** kidney tubules regulate water reassertion and urine concentration.
- **Regulation of pH:** volume, and composition of body fluids.
- **Production of erythropoietin:** for hematopoiesis, and renin for blood pressure regulation.

Common Urinary tract disorders

Urinary retention

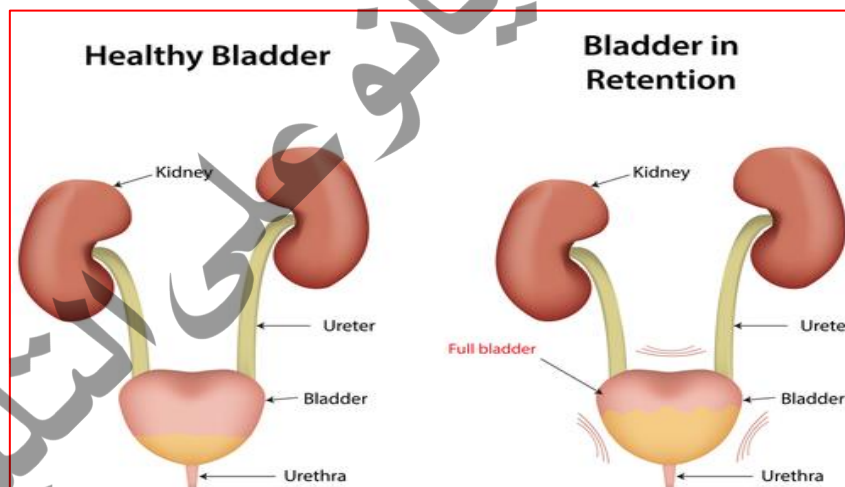
Learning objectives:

At the end student will be able to:

- Define the urinary retention.
- Discuss pathophysiology of urinary retention.
- Mention causes of urinary retention.
- Enumerates signs & symptoms of urinary retention.
- Discuss diagnostic procedure used in urinary retention diagnosis.
- Explain management of urinary retention.

➤ Definition:

Urinary retention is the inability to empty the bladder completely; Urinary retention can be acute or chronic.



➤ Pathophysiology:

Disease Progression → Unable to produce an effective voiding contraction



Incomplete emptying and/or within a usual time span



Urine retention

➤ **Causes of urinary retention:**

- **Obstruction of the urethra.**
- **Nerve problems:** (brain or spinal cord infections or injuries).
- **Medications:** (antihistamines, anticholinergics/antispasmodics)
- **Weakened bladder muscles** (with aging).
- **Medical conditions:** like disk surgery, or hysterectomy.

➤ **Signs and symptoms:**

» **The symptoms of acute urinary retention:**

- Inability to urinate.
- Painful, urgent need to urinate.
- Pain or discomfort in the lower abdomen.
- Bloating of the lower abdomen

» **The symptoms of chronic urinary retention:**

- Urinary frequency.
- Urination eight or more times a day.
- Trouble beginning a urine stream.
- A weak or an interrupted urine stream.
- An urgent need to urinate with little success when trying to urinate.
- Feeling the need to urinate after finishing urination.
- Mild and constant discomfort in the lower abdomen and urinary tract.

➤ **Diagnosis:**

★ **Physical Exam**

The healthcare provider may be able to feel a distended bladder by lightly tapping on the lower abdomen.

★ **Postvoid Residual Measurement:**

This test measures the amount of urine left in the bladder after urination. The health care provider inserts the catheter through the urethra into the bladder, a procedure called catheterization, to drain and measure the amount of remaining

urine. A postvoid residual of 100 mL or more indicates the bladder does not empty completely.

★ **Imaging Tests:**

- **Cystoscopy:** to look inside the urethra and bladder.
- **Computerized tomography (CT) scans.**
- **Urodynamic tests:** look at how well the bladder and urethra store and release urine.
- **Electromyography:** uses special sensors to measure the electrical activity of the muscles and nerves in and around the bladder and sphincters.

➤ **Management:**

▪ **Medical management:**

- **Bladder drainage:** Involves catheterization to drain urine.
- **Urethral dilation:** Treats urethral stricture by inserting increasingly wider tubes into the urethra to widen the stricture.
- **Prostate Medications:** Dutasteride, finasteride, doxazosin.

▪ **Surgical management:**

- **Urethral Stents:** Another treatment for urethral stricture involves inserting an artificial tube, called a stent, into the urethra to the area of the stricture.
- **Prostate surgery:** To treat urinary retention caused by benign prostatic hyperplasia.
- **Internal urethrotomy:** An urologist can repair a urethral stricture by performing an internal urethrotomy.
- **Cystocele or rectocele repair:** Women may need surgery to lift a fallen bladder or rectum into its normal position.

■ **Nursing Management:**

1. Obtain focused urinary history emphasizing the character and duration of urinary retention symptoms.
2. Perform a focused physical assessment or review the results of a recent physical including palpation of the lower abdomen for obvious bladder distension; neurological examination including perineal skin sensation.
3. Determine the urinary residual volume by catheterizing the patient immediately after urination.
4. Complete a bladder log, including patterns of urine elimination, patterns of urine loss (if present), nocturia, and volume and type of fluids consumed for a period of 3 to 7 days.
5. Insert an indwelling catheter (followed infection control precaution) to provide continuous drainage of urine.

Patient Education:

- Teach the patient with urinary retention and infrequent voiding to urinate by the clock (Timed or scheduled voiding may reduce urinary retention by preventing bladder over distension).
- Teach the client who is unable to void specific strategies to manage this potential medical emergency including:
 - Drinking a cup of hot tea or coffee, a warm cup of coffee or tea stimulates the bladder and may promote voiding.
 - Attempting urination in complete privacy.
 - Placing the feet solidly on the floor to relax the pelvic muscles and may encourage voiding.
 - If unable to void using these strategies, taking a warm bath or shower and voiding (if possible) while still in the tub or the shower. Warm water also stimulates the bladder and may produce voiding.

- If unable to void within 6 hours, or if bladder distension is producing significant pain, seeking urgent or emergency care.

➤ **Complication:**

- Urinary tract infections (UTIs).
- Bladder damage.
- Kidney damage.
- Urinary incontinence after prostate, tumor, or cancer surgery.

Urinary Incontinence

Learning objectives:

At the end student will be able to:

- Define the urinary incontinence.
 - Discuss pathophysiology of urinary incontinence.
 - Differentiate between types of incontinence.
 - Mention causes of urinary incontinence.
 - Enumerate signs & symptoms urinary incontinence.
 - Interpret diagnostic procedure used in urinary incontinence diagnosis.
 - Explain management of urinary incontinence.
-

➤ Definition

Urinary incontinence, also known as **overactive bladder**, is the involuntary loss of urine due to difficulties controlling the bladder, frequently seen in older individuals, particularly women.

➤ Pathophysiology:

Urinary incontinence occurs when there is dysfunction in either the storage function or occasionally, in the emptying function of the lower urinary tract. Urethral sphincter dysfunction and bladder dysfunction can co-exist and various components of the continence mechanism may compensate one another. For example, women may experience anatomical or neuromuscular injury during childbirth but remain asymptomatic until there is a loss of urethral sphincter function due to aging.

- **Stress incontinence** is due to pelvic floor weakness or prolapse and/or loss of the normal urethra vesical angle.
- **Urge incontinence** is uninhibited bladder contractions caused by irritation or loss of neurologic control of bladder contractions.

➤ Types of urinary incontinence:

There are 3 major types of urinary incontinence:

- **Stress urinary incontinence:** involuntary leakage on effort, activities, or exertion.
- **Urgency urinary incontinence:** involuntary leakage accompanied or preceded by urgency (compelling desire to void which is difficult to defer).
- **Mixed urinary incontinence:** a mixture of urgency and stress incontinence.

Other forms of incontinence include:

- **Overflow incontinence:** occurs when small amounts of urine leak from a bladder that is always full. This condition tends to occur in males with enlarged prostates that prevent the complete emptying of the bladder.
- **Functional incontinence:** occurs when cognitive or physical impairments prevent the patient from voiding independently and appropriately.
- **Continuous incontinence:** the complaint of continuous urinary leakage.
- **Situational incontinence:** e.g., giggle.

➤ Causes of urinary incontinence:

Temporary or short-term causes of incontinence can include:

- **Urinary tract infections (UTIs).**
- **Pregnancy:** uterus places extra pressure on the bladder as it expands. Most women who experience incontinence during pregnancy.
- **Medications:** Incontinence can be a side effect of certain medications, including diuretics and antidepressants.
- **Constipation:** Chronic constipation can cause to have bladder control issues.

Chronic or long-term causes of incontinence can include:

- Pelvic floor disorders.
- Stroke.
- Diabetes.
- Menopause.
- Multiple sclerosis (MS).
- Enlarged prostate.
- After prostate cancer surgery.

➤ Signs and symptoms:

The main symptom of incontinence is a leakage of urine.

- Leaking urine during everyday activities, such as lifting, bending, coughing, or exercising.
- Being unable to hold in urine after feeling a sudden, strong urge to urinate.
- Leaking urine without any warning or urge.
- Being unable to reach a toilet in time.
- Wetting your bed during sleep.
- Leaking during sexual activity.

➤ Diagnosis:

★ Physical Exam

Look for any physical reason that could be causing incontinence. This includes: a pelvic exam, checking the size of a man's prostate.

- **Stress test:** During this test, the examiner asks the patient to cough to see if any urine leaks from this action. If the patient noticed leakage during other activities, like running or jumping, the examiner may ask the patient to repeat those actions to see if a leakage issue has.

- **Pad test:** give the patient a pad to wear, which will catch any leaked urine. At the end of the test, this pad will be checked to see how much urine you lost.

★ **Laboratory Tests:**

- Urinalysis: to test for infections or blood in urine.

★ **Imaging Tests:**

- **An ultrasound of bladder.**
- **Cystoscopy.**
- **Urodynamic testing:** to check how much bladder can hold and how urethral sphincter muscle is working.

➤ **Management:**

- **Medical management:**
- **Pelvic floor exercise:** Pelvic floor muscle training (PFMT) involves strengthening the pelvic floor muscles. It needs to be continued for 3–4 months before determining its success. It should be done with three sets of 8–12 slow maximal contractions sustained for 6–8 seconds and repeated 3–4 times per week.
- **Bladder training:** Bladder training is the initial treatment for urinary incontinence, being noninvasive, inexpensive and easy. This includes PFMT, a scheduled voiding program with gradual increases in the duration between voids, and urge suppression techniques with distraction or relaxation.

▪ **Pharmacologic Therapy**

- **Anticholinergic medications:** form the basis of pharmacological treatment in urinary incontinence.
- **Imipramine, a tricyclic antidepressant:** may reduce detrusor contractility and increases outlet resistance.

▪ **Surgical management:**

- **Intravesical botulinum toxin A injections or neuromodulation:** For detrusor overactivity refractory to oral medications.
- **Myomectomy or bladder augmentation:** This reduces the efficacy of detrusor contraction and thus improves continence.
- **Ileal conduit urinary diversion may be performed.**

▪ **Nursing Management:**

- Perform a focused history of the incontinence including duration, frequency and severity of leakage episodes, and alleviating and aggravating factors.
- Assess client for potentially reversible causes of acute/transient urinary / chronic incontinence (e.g., urinary tract infection [UTI]).
- Assess the home, acute care, or long-term care environment for accessibility to toileting facilities, paying particular attention to the following:
 - Distance of toilet from bed, chair, living quarters.
 - Characteristics of the pathway to the toilet, including barriers such as stairs, loose rugs on the floor, and inadequate lighting.
 - Characteristics of the bathroom, including patterns of use; lighting; height of toilet from floor; presence of hand rails to assist transfers to toilet; and breadth of door and its accessibility for wheelchair, walker, or another assistive device.
 - Assess client for mobility, including ability to rise from chair and bed; ability to transfer to toilet and ambulate; and need for physical assistive devices such as a cane, walker, or wheel chair.
 - Protective products may be needed to protect the skin from breakdown and prevent leakage onto clothing. Incontinence underwear has a

waterproof liner and built-in cloth pad to absorb large amounts of urine to protect skin from moisture and control odor.

- Begin a prompted voiding program or patterned urge response toileting program:
 - Determine the frequency of current urination using an alarm system or check and change device.
 - Record urinary elimination and incontinent patterns on a bladder log to use as a baseline for assessment and evaluation of treatment efficacy.
 - Begin a prompted toileting program based on the results of this program; toileting frequency may vary from every 1.5 to 2 hours, to every 4 hours.

Patient Education:

Lifestyle changes can help with incontinence. Losing weight, drinking less caffeine (found in coffee, tea, and many sodas), preventing constipation, and avoiding lifting heavy objects may help with incontinence. Limiting fluid intake before bedtime and scheduling prescribed diuretic medication in the morning or early afternoon are also helpful.

Teaching Pelvic Floor Exercises: Kegel exercises are designed to make pelvic floor muscles stronger.

- Start by finding the right muscles. There are two easy ways to do this: stop the stream of urine as are urinating or imagine that are trying to stop the passage of gas. Squeeze the muscles you would use to do both.
- Lie on the floor. Pull in the pelvic muscles and hold for a count of 3. Then relax for a count of 3. Work up to 10 to 15 repeats each time you exercise.
- Complete pelvic exercises at least three times a day. Try to use three different positions while performing the exercises: lying down, sitting, and standing.

- Be patient. Most people notice an improvement after a few weeks, but the maximum effect may take up to 3-6 weeks.

➤ **Complication:**

- Urinary tract infection.
- Sleep disorders.
- Skin problem.
- Depression.

Student self-assessment

1. A patient is experiencing which type of incontinence if she experiences leaking urine when she coughs, sneezes, or lifts heavy objects?

- A. Overflow.
- B. Functional.
- C. Stress.
- D. Urge.

2. A postvoid residual measurement of ----- indicates the bladder does not empty completely.

- A. 75 ml.
- B. 100ml.
- C. 50ml.
- D. 25ml.

3. -----is the inability to voluntarily empty the bladder.

- A. Urinary retention
- B. Prostatic hyperplasia
- C. Post void residual measurement
- D. Catheterization.

4. A major symptom of urinary incontinence is:

- A. Skin problem.
- B. Fever.
- C. Leakage of urine.
- D. Depression.

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Chapter seven (Nursing care of neurological disease)

Learning objectives

- Explain the anatomy of the nervous system.
- Understand the physiology of the nervous system.

Neurological system

Introduction

The nervous system is the “executive suite” of the human body. It directs all other systems and provides the unique ability for thought, emotion, understanding of complex information, and integration of numerous stimuli. As the recipient of all sensory information for analysis, the nervous system generates intellectual and motor responses aimed at maintaining the integrity of life structures.

Definition

The nervous system is an organ system that contains a network of specialized cells called neurons. This is the master controlling and communicating system of the body. It coordinates the action of an animal and transmits signals between the different parts of the body. Every thought, movement and emotions reflect the activity of the nervous system.

Anatomy and physiology of neurological system

The nervous system is divided into two parts:

- **Central nervous system (CNS)**

The CNS consists of the brain and spinal cord, which occupy the dorsal body cavity and act as the integrating and command centers of the nervous system.

- **Peripheral nervous system (PNS)**

- Nerves connect PNS with CNS
- Includes 12 pairs of cranial nerves, 31 pairs of spinal nerves
- It divided into **somatic** (voluntary). **autonomic** (involuntary) nervous systems. **Autonomic nervous system** comprised of **sympathetic**, **parasympathetic** nervous systems

Organization of the nervous system

- Network of brain, spinal cords, nerves
- Sensory/afferent, integrative, motor/efferent functions

Sensory/Afferent

- Receptors monitor external, internal environment
- Conscious stimuli (e.g., vision, hearing, touch)
- Unconscious stimuli (e.g., pH, blood pressure)

Integrative

Sensory/afferent input received by central nervous system →→information processed → interpreted → response initiated

Motor/Efferent

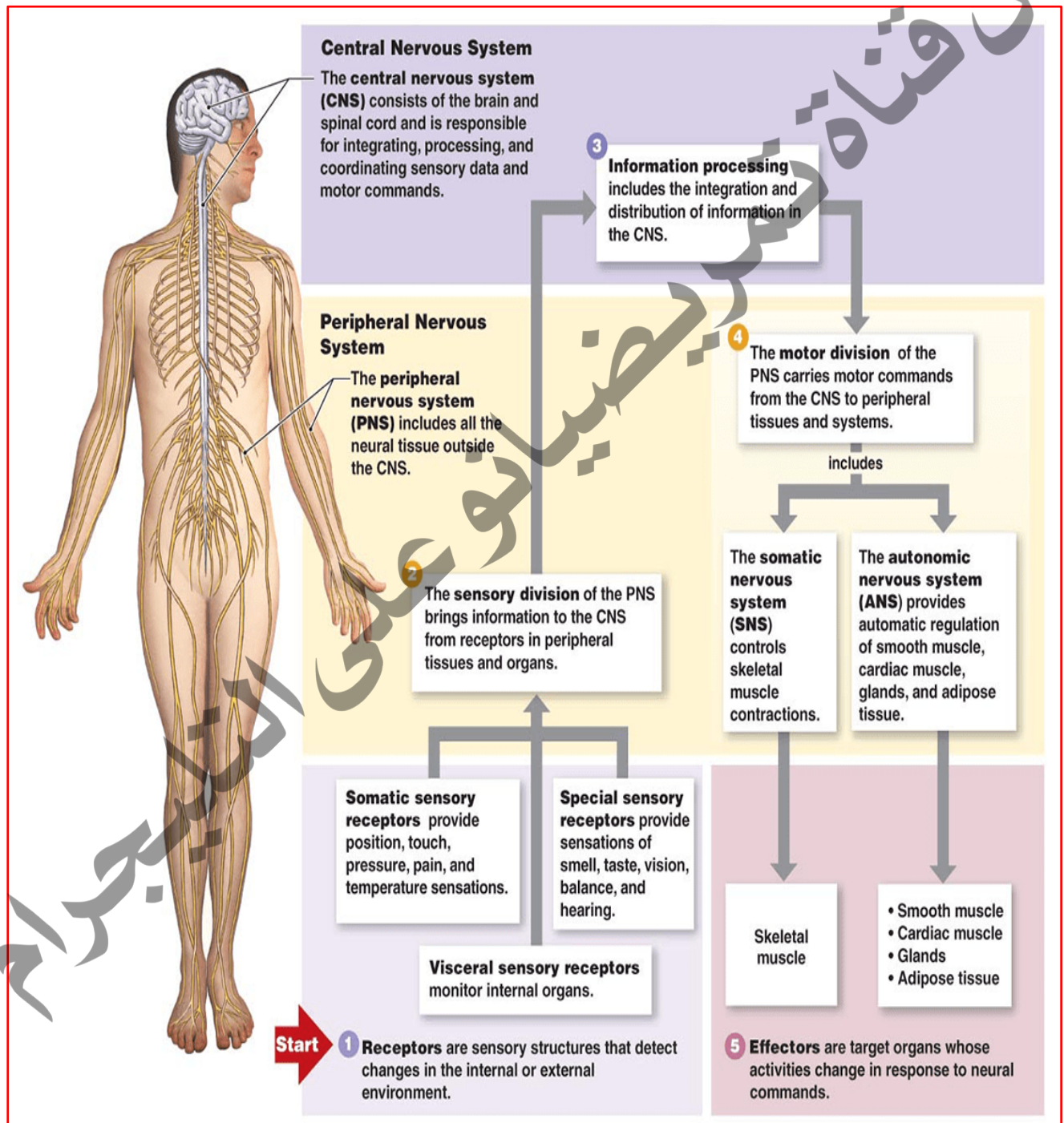
- Brings motor information from central nervous system to periphery
- Controls actions of effector organs (e.g., muscles, glands)

Functions of the nervous system

1. **To monitor changes that takes place inside and outside the body.** The nervous system utilizes the million sensory receptors to carry out this function. Any changes or stimuli occurring are noted by the nervous system and the gathered data is now called a sensory input.
2. Another important function of the nervous system is to **process and interpret the sensory input or gathered data.** It is the working of this

system to make decision about what should be done at each moment. This is the process known as **integration**.

3. As the nervous system has reached a decision of what response and appropriate action to be done in response to the stimuli, it then **effects a response by activating muscles or glands through motor output**.



Headache

Learning objectives

- Define the headache.
 - Discuss pathophysiology of headache.
 - Enumerate causes and of headache.
 - Recognize signs, symptoms & characteristics of headache.
 - Identify types of headaches.
 - Differentiate the condition relieving and exacerbating headache.
 - Discuss the diagnostic evaluation of headache.
 - Explain medical and nursing management of headache.
-

Introduction

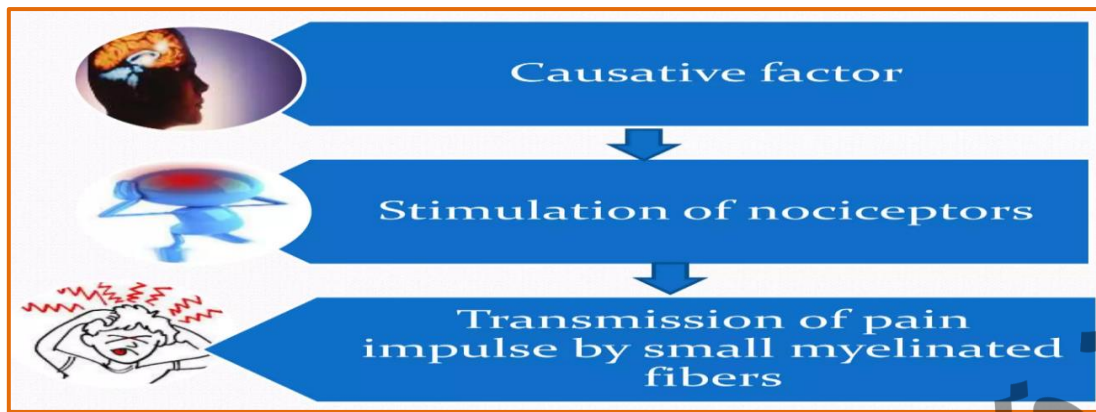
One of the most common of all human physical complaints. Headache is actually a symptom rather than a disease a stress response, vasodilation (migraine), skeletal muscle tension (tension headache), or a combination of factors. headache is a common symptom of neurologic disorders. However, most headaches are transient events and do not indicate a serious pathological condition.

Definition of headache

A headache is a pain or discomfort in the head, scalp, or neck

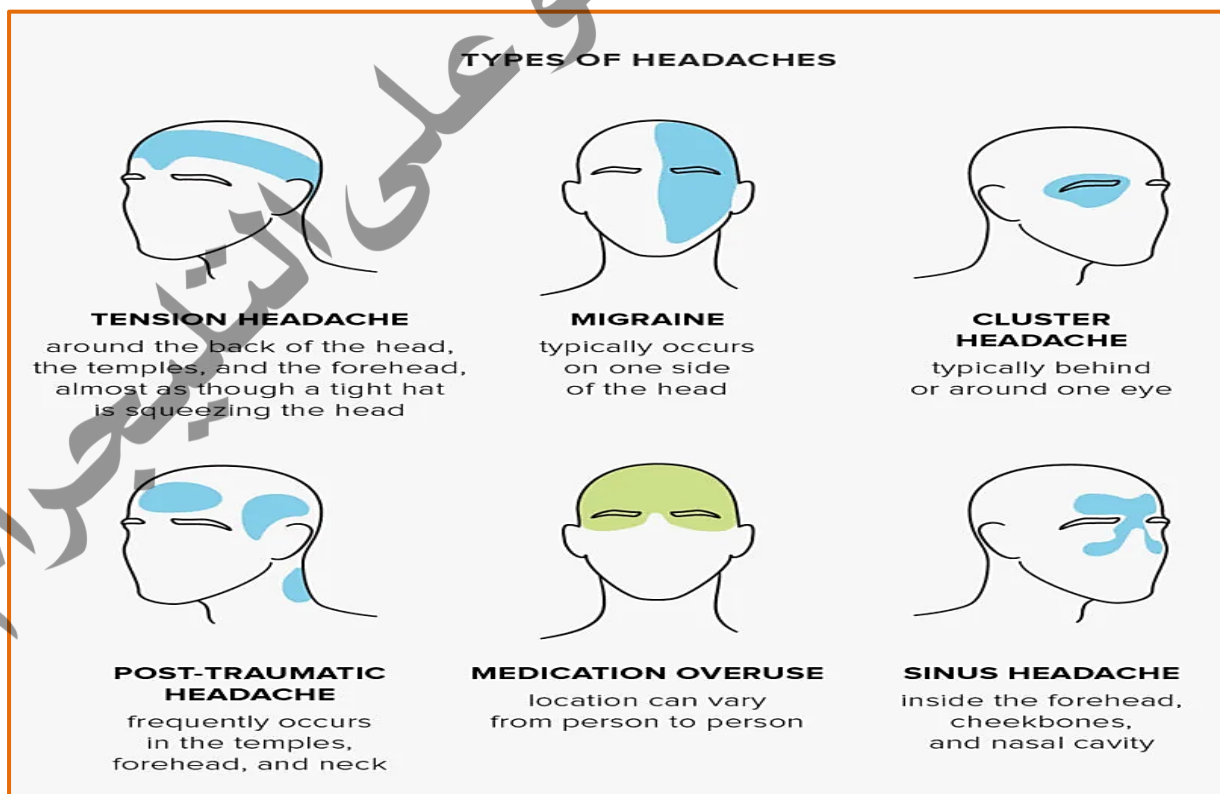
Pathophysiology of headache

The brain parenchyma does not have nociceptors. Thus, headache is typically the result of pain originating in surrounding structures, such as blood vessels, meninges, muscle fibers, facial structures, and cranial or spinal nerves. Stretching, dilatation, constriction, or any nociceptor stimulation within these structures can result in the perception of headache.



Classification of headache:

Primary headache	Secondary headache	Other
- Migraine - Tension-type headache - Cluster headache - Exertion headache	Headache attributed to.... - Head & neck trauma - Cranial or cervical vascular disorders - Nonvascular intracranial disorder - Substance use or its withdrawal - Infections	Disorders of cranial neuralgias & facial structures



Causes of headache

- **Illness** having a **infections, colds**, fevers, sinusitis, throat or ear disorders, and head injuries.
- Stress as Emotional tension and depression
- Drinking too much alcohol
- Bad posture
- Eyesight problems
- Not eating regular meals
- Not drinking enough fluids (dehydration)
- Taking too many painkillers
- Having your period or during menopause
- **Genetics:** Headaches, particularly migraines, tend to carry forward in families.
- **Migraines:** Too much physical activity can also activate migraines in adults.

Symptoms and signs of headache:

- Sensitivity to light, noise, and smell
- Nausea and vomiting, upset stomach, and abdominal pain
- Loss of appetite
- Feeling very warm or cold & Pale skin color
- Fever & feeling tired
- Dizziness and blurred vision
- Pain or tender scalp
- Stiff in neck and shoulder
- Behavioral changes as irritability or depression
- In severe cases unconsciousness.

Characteristics of Headache

- Pulsating or throbbing.
- A steady sensation of tightness or pressure.
- Sharp, lancinating (stabbing) pain.
- Icepick-like pain.

Conditions Relieving Headache

Migraine headaches are frequently relieved by:

- Darkness.
- Sleep.
- vomiting, or pressing on the temporal artery.
- relieved by recumbency.
- headaches caused by intracranial mass lesions may be less severe with the patient standing.

Conditions Exacerbating Headache

- Rapid changes in head position such as bending over or by events that transiently raise intracranial pressure, such as coughing and sneezing.
- Anger, excitement, or irritation.

Diagnostic evaluation

1. The patient's history and symptoms.
2. MRI, CT, skull x-ray.
3. Arteriogram.
4. EEG.
5. Cranial nerve testing.
6. Lumbar puncture to test CSF may be done to rule out other causes for the headaches.

Management of headache

Medical management

Treat underlying cause of headache.

Nursing management

- Assess the patient for location of headache, timing, characteristics, relieving and aggravating factors.
- Assist the patient to identify and reduce or eliminate aggravating factors.
- Encourage the patient to use alleviating techniques such as biofeedback or stress reduction.
- Provide a dark room and rest to reduce stimulation during a migraine headache.
- Give analgesics according to physician order.
- Teach the patient to use relaxation exercises and warm or cool moist compresses.
- Teach the patient about medications, appropriate dosage, expected action, side effects, and consequences of misuse.

Stroke

Learning objectives

- Define the stroke.
 - Discuss the pathophysiology of stroke.
 - Explain the causes and risk factors of stroke.
 - Differentiate the types of strokes.
 - Recognize signs and symptoms of stroke.
 - Discuss diagnostic evaluation of stroke.
 - Explain medical and nursing management of stroke.
 - Discuss the rehabilitation program for patient with stroke.
 - List the complication of stroke.
-

Introduction

Stroke is a neurological disorder characterized by blockage of blood vessels. Clots form in the brain and interrupt blood flow, clogging arteries and causing blood vessels to break, leading to bleeding. Rupture of the arteries leading to the brain during stroke results in the sudden death of brain cells owing to a lack of oxygen. A stroke can cause lasting brain damage, long-term disability, or even death.

Definition

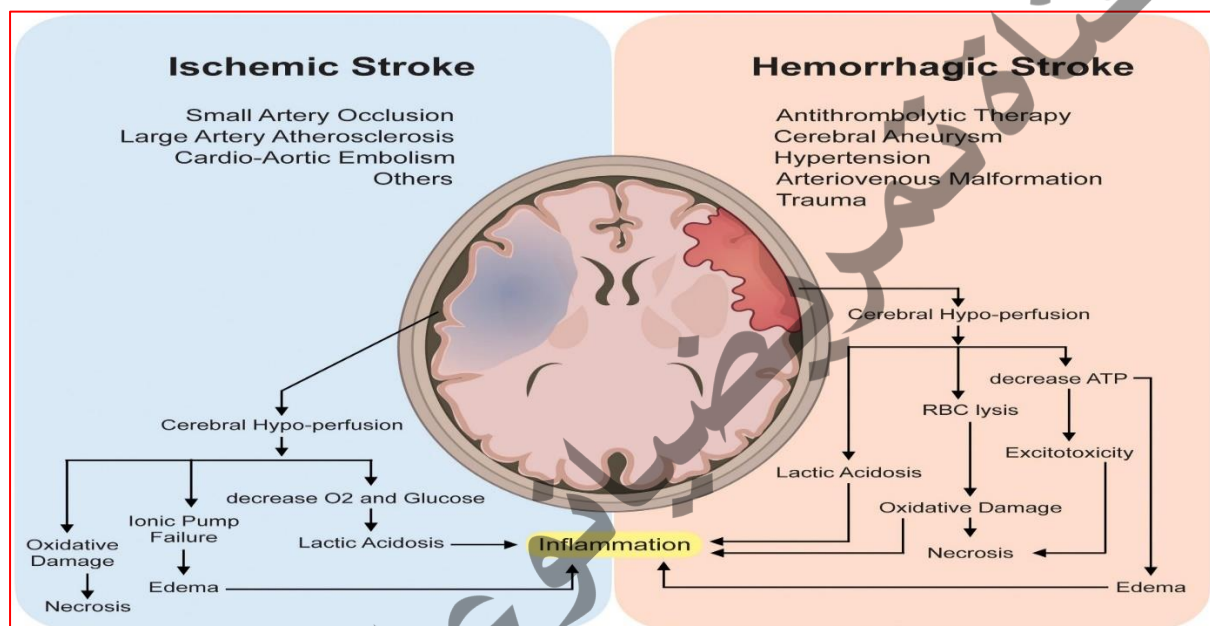
A stroke, sometimes called a brain attack **Stroke** is a descriptive term for the sudden onset of acute neurologic deficit persisting for more than 24 hours and caused by the interruption of blood flow to the brain.

Stroke is a syndrome characterized by four key features:

1. Sudden onset.
2. Focal involvement of the central nervous system.
3. Lack of rapid resolution.
4. Vascular cause.

Pathophysiology

The disruption in the blood flow initiates a complex series of cellular metabolic events. Decreased cerebral blood flow occurs when something blocks blood supply to part of the brain or when a blood vessel in the brain bursts. In either case, parts of the brain become damaged or die. A stroke can cause lasting brain damage, long-term disability, or even death.



Types of stroke:

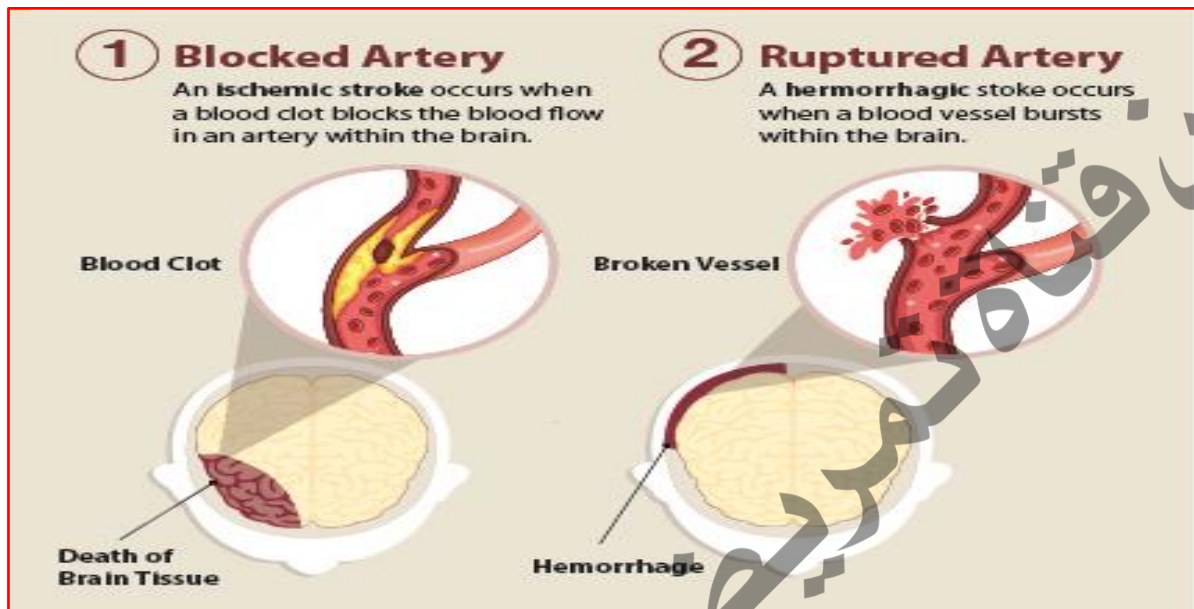
➤ Ischemic stroke:

- Most strokes are ischemic strokes. An ischemic stroke occurs when blood clots or other particles block the blood vessels to the brain.
- Fatty deposits called plaque can also cause blockages by building up in the blood vessels.

➤ Hemorrhagic stroke

Hemorrhagic stroke happens when an artery in the brain leaks blood or ruptures (breaks open). The leaked blood puts too much pressure on brain cells, which damages them.

- **A transient ischemic attack (TIA)** is sometimes called a “mini-stroke.” It is different from the major types of strokes, because blood flow to the brain is blocked for only a short time—usually no more than 5 minutes.



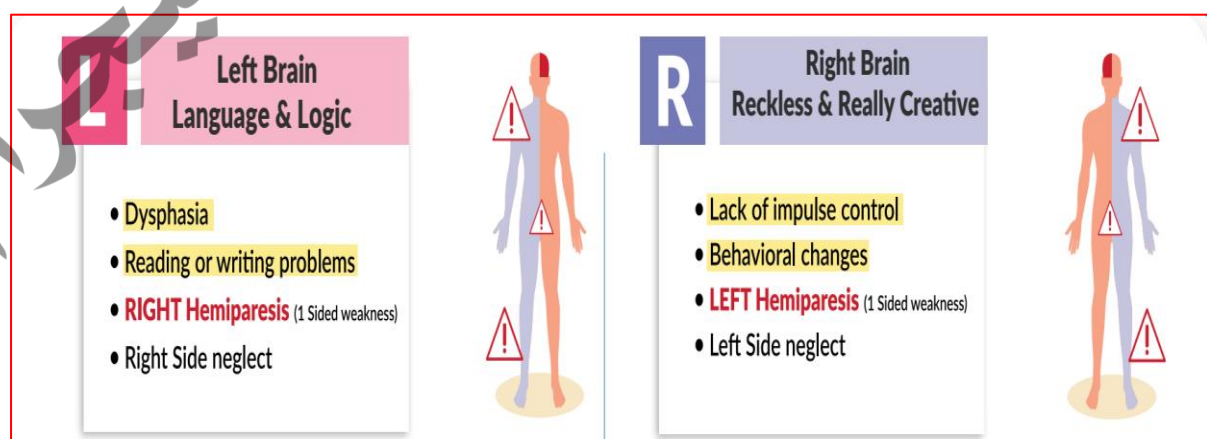
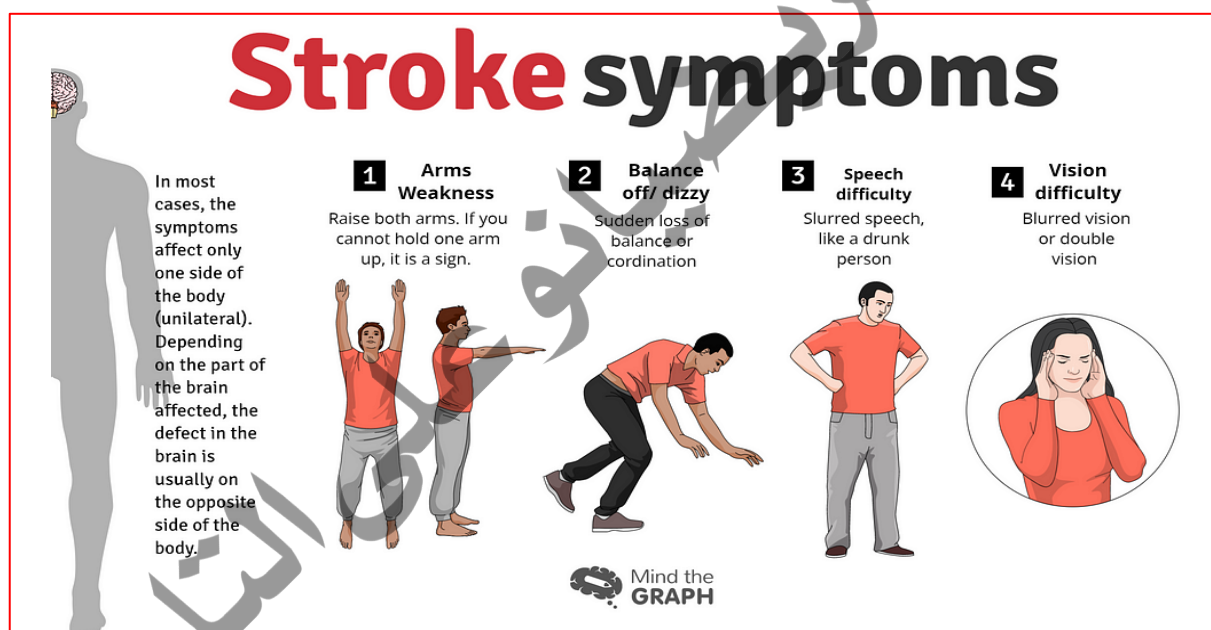
Risk factors of stroke

Nonmodifiable	Modifiable
• Advanced age (older than 55 years) • Gender (Male) • Race (African American)	• Hypertension • Atrial fibrillation • Hyperlipidemia • Obesity • Smoking • Diabetes • Asymptomatic carotid stenosis and valvular heart disease (eg, endocarditis, prosthetic heart valves)

Common signs and symptoms of a brain stroke include:

- **Numbness or weakness of the face.**
- **Change in mental status.** The patient experiences confusion.
- **Trouble speaking or understanding speech.**
- **Visual disturbances.**

- **Hemiparesis.** There is a **weakness** of the face, arm, and leg on the same side due to a lesion in the opposite hemisphere.
- **Hemiplegia.** **Paralysis** of the face, arm, and leg on the same side due to a lesion in the opposite hemisphere.
- **Ataxia.** Staggering, **unsteady gait** and inability to keep feet together.
- **Dysarthria.** This is the difficulty in forming words.
- **Dysphagia.** There is difficulty in swallowing.
- **Paresthesia.** There is numbness and tingling of extremities and difficulty with proprioception.
- **Aphasia.** a language disorder caused by damage in a specific area of the brain that controls language expression and comprehension.

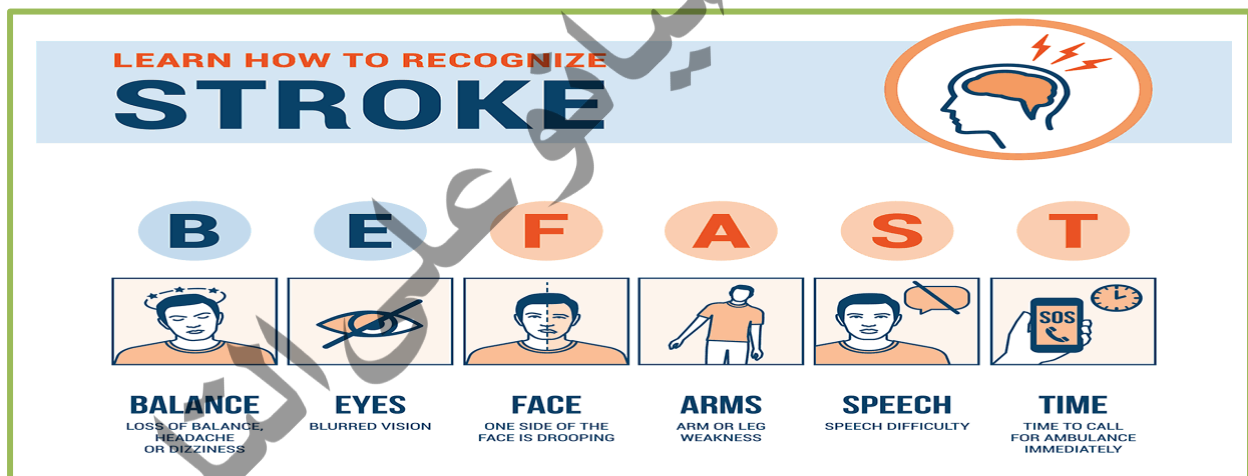


Warning signs of stroke indicate immediate call to emergency room

Men experience	Women often experience
• Facial numbness, weakness or drooping • Arm, face, or leg numbness or weakness, especially on one side of the body • Difficulty balancing • Speech difficulties or confusion. • Changes in eyesight or vision problems	• Loss of consciousness or fainting • General weakness • Shortness of breath • Sudden behavioral change • Hallucination • Nausea or vomiting, Hiccups • Seizures

Diagnosis:

1. Physical Examination



2- Computed tomographic (CT)

On arrival at the ED, a computed tomographic (CT) scan will be performed immediately to identify whether symptoms are caused by a hemorrhagic stroke OR Ischemic stroke changes will not be visible on a CT until several days after the event.

3- Electrocardiogram (ECG): to determine if atrial fibrillation is present.

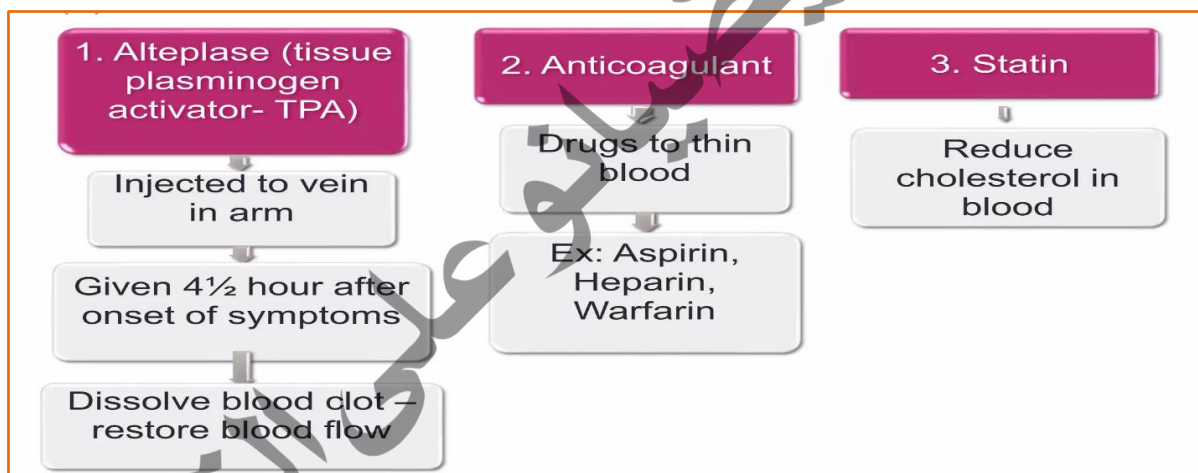
4- CBC, INR.

Prevention of stroke

- Controlling of Diabetes mellites
- Controlling of hypertension
- Life style modification
- Administer statin to control dyslipidemia
- Administer anticoagulant and antiplatelet for patient with atrial fibrillation
- Stop smoking.
- Perform regular exercise.

Management of stroke:

1- Pharmacological treatment:



2. Surgical treatment:



3. Nursing Management

Initial emergency care

- ABCs (airway, breathing, and circulation) are monitored.
- Oxygen is administered if oxygen saturation is less than 92% and the patient's level of consciousness (LOC) is reduced.
- monitoring vital signs.
- provide calm environment and dark room.
- elevate head of the bed to 35-45 degree.
- frequent neurological and hemodynamic assessment.
- monitoring for increasing headache and seizure and treat it.
- Swallow integrity must be assessed in all stroke patients, with the patient kept 'nil orally' (NPO) until the ability to safely manage oral intake.

After acute stage passed: The nurse assists the patient for:

- Improving Mobility.
- Preventing Joint deformity.
- Changing Position.
- Establishing Exercise Program.
- Improving Communication.
- Improving Thought Process.
- Improving Skin Integrity.
- Improving Family coping.

Rehabilitation for patients with stroke

1. Physical and occupational therapy are provided to maximize functioning and to progress the patient toward a return to baseline functioning.



2. Sensation changes may prevent the patient from being aware of pressure, temperature, or injuries on the affected side. Patients must be taught to be aware of these changes and protect the involved limbs.
3. Patient may find problems in communication Speech therapy can help the patient relearn to communicate.
4. Patient have poor judgement (they think they are able to perform activities that they done before getting stroke).
5. Emotional support from others is essential. Allowing them to talk about their frustration and anger may facilitate coping. Distracting the patient from inappropriate behavior may help.
6. The patient should never be reprimanded or punished because he or she no longer has the cognitive ability to control the behaviors.

Complication of stroke

- Venous thromboembolism (VTE).
- Pneumonia.
- Skin breakdown.
- Physical immobility.
- Mental confusion.
- Depression.

Student self-assessment

A 78-year-old client was admitted to the emergency department with numbness and weakness in the left arm and slurred speech. Which nursing intervention is a priority?

- A. Prepare to administer recombinant tissue plasminogen activator (rt-PA)
- B. Discuss the precipitating factors that caused the symptoms
- C. Schedule for a STAT computed tomography (CT) scan of the head
- D. Notify the speech pathologist for an emergency consult

The client, diagnosed with a right-sided cerebrovascular accident, is admitted to the rehabilitation unit. Which interventions should be included in the nursing care plan? Select all that apply.

- A. Position the client to prevent shoulder adduction
- B. Turn and reposition the client every shift
- C. Encourage the client to move the affected side
- D. Perform quadriceps exercises three (3) times a day
- E. Instruct the client to hold the fingers in a fist

The 85-year-old client diagnosed with a stroke is complaining of a severe headache. Which intervention should the nurse implement first?

- A. Administer a nonnarcotic analgesic
- B. Prepare for STAT magnetic resonance imaging (MRI)
- C. Start an intravenous line with D5W at 100 mL/hr
- D. Complete a neurological assessment

A nurse is interviewing a patient who is seeking relief for frequent headaches. Which description is consistent with symptoms of a migraine headache?

- A. Extreme tenseness in the area of the neck and shoulders.
- B. Tears flow from one eye and nasal drainage occurs with the headache.
- C. The pain of the headache wakes the patient from sleep.
- D. The pain throbs and is synchronous with the patient's pulse.

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Chapter eight

(Nursing care of musculoskeletal disease)

Learning objectives

At the end of this chapter student will be able to:

- Identify the basic parts of the musculoskeletal system
 - List the main functions of bones and muscles
 - Describe common musculoskeletal disorders
-

Musculoskeletal System

Introduction

The musculoskeletal system (also known as the locomotor system) is an organ system that enables an organism to move, support itself, and maintain stability during locomotion.

Basic anatomy and physiology of musculoskeletal system

Bones (Skeletal System)

- Store calcium and other minerals
- Make blood cells in the bone marrow
- Support and protect our body
- There are two types of bone marrow: yellow and red

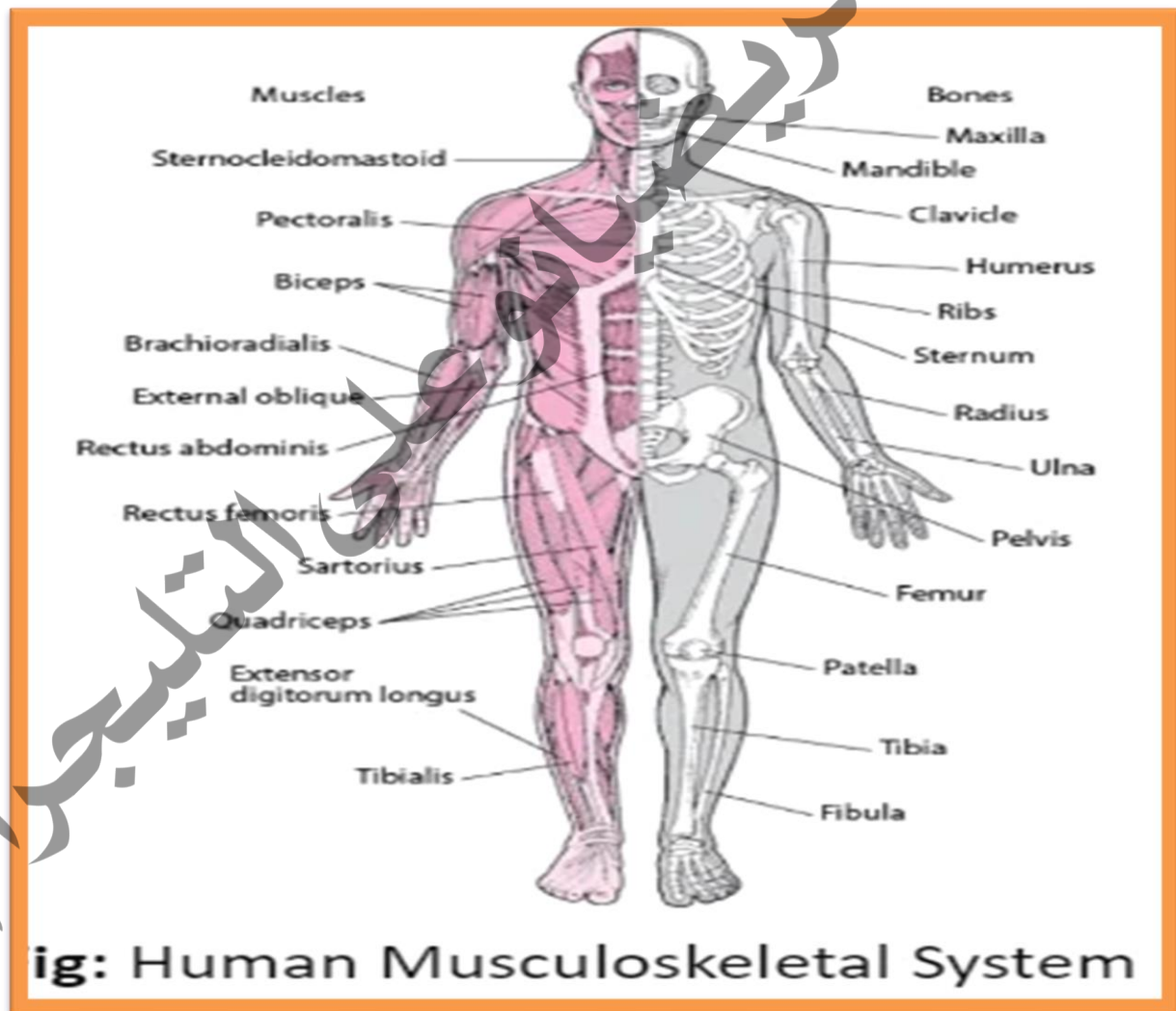
Muscles (Muscular System)

- Contract (get shorter) to move bones
- There are three types of muscles:

- **Skeletal muscles** - help us move (we can control these)
- **Smooth muscles** - found in organs like intestines (we cannot control these)
- **Heart muscle** - pumps blood (we cannot control this)

Other Important Parts

- **Tendons** - connect muscles to bones
- **Joints** - places where bones meet and can move
- **A ligament** is a fibrous elastic tissue connect the ends of bones together in order to form a joint.
- **A bursa** is a small fluid-filled sac made of white fibrous tissue and lined with synovial membrane.



Common musculoskeletal disorders

Fractures

Learning objectives:

At the end of the lesson the students will be able to:

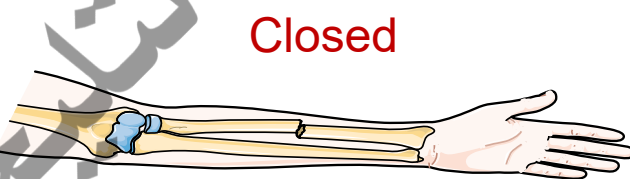
- State what a fracture is
- List the main types of fractures
- Identify causes and signs of fractures
- List basic ways to diagnose fractures
- Recognize basic fracture care steps

Definition

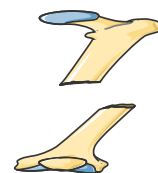
A fracture is a break of the continuity of the bone & is defined according to the type and extent.

➤ types of Fractures:

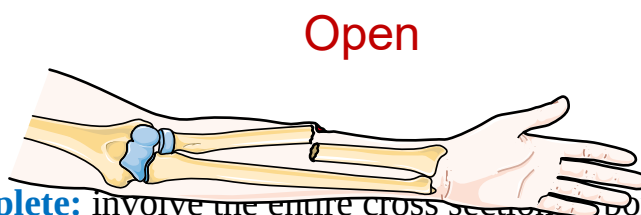
1. **Closed (simple fracture):** a bone is broken but there is no penetration extending from the fracture through the skin.



Complete



2. **Open (compound fracture):** there is a wound over the fracture site. And is more serious because the risk of contamination and infection.



3. **Complete:** involve the entire cross section of bone.

4. Incomplete (Greenstick): involve apportion of the cross section of bone.

5. Comminute: multiple pieces of bone.

6. Pathologic: through an area of diseased bone as osteoporosis.

➤ **Causes of fracture:**

- Bone stress greater than it can absorbed.
- A direct blow.
- Crushing force.
- Sudden twisting.
- Extreme muscle contraction

➤ **Signs & symptoms:**

1. Pain.
2. Swelling and discoloration.
3. Tenderness or discomfort.
4. Bone may break through the skin.
5. Deformity.

➤ **Diagnostic Evaluation:**

The diagnosis of fracture depends on:

- Assessment of the symptoms.
- Physical examination for the affected part.
- X-ray examination.
- Nerve conduction and electromyogram studies to detect nerve injury.

Management the fracture

Initial Management:

- Keep the injured area immobile.
- Immobilize with splints or bandages.
- Support the limb above and below the fracture site.

- Assess circulation and nerve function.
- Remove clothing from the uninjured side first.
- Move the patient as little as possible.

Specific Management for the fracture

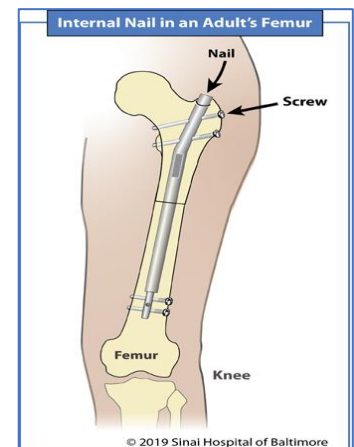
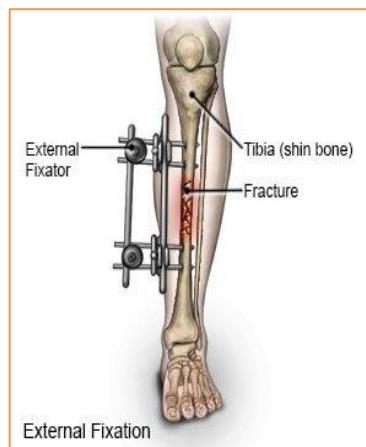
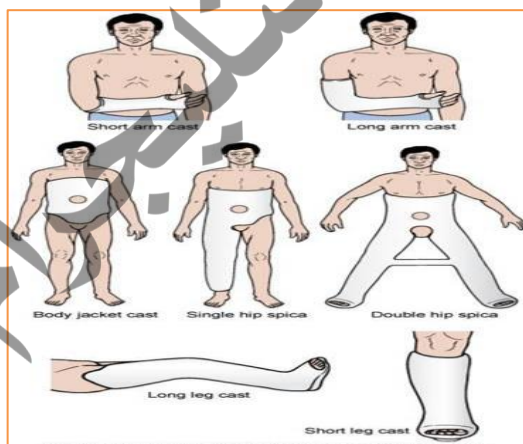
1. Reduction: Restoring bone fragments to normal alignment (closed or open methods).

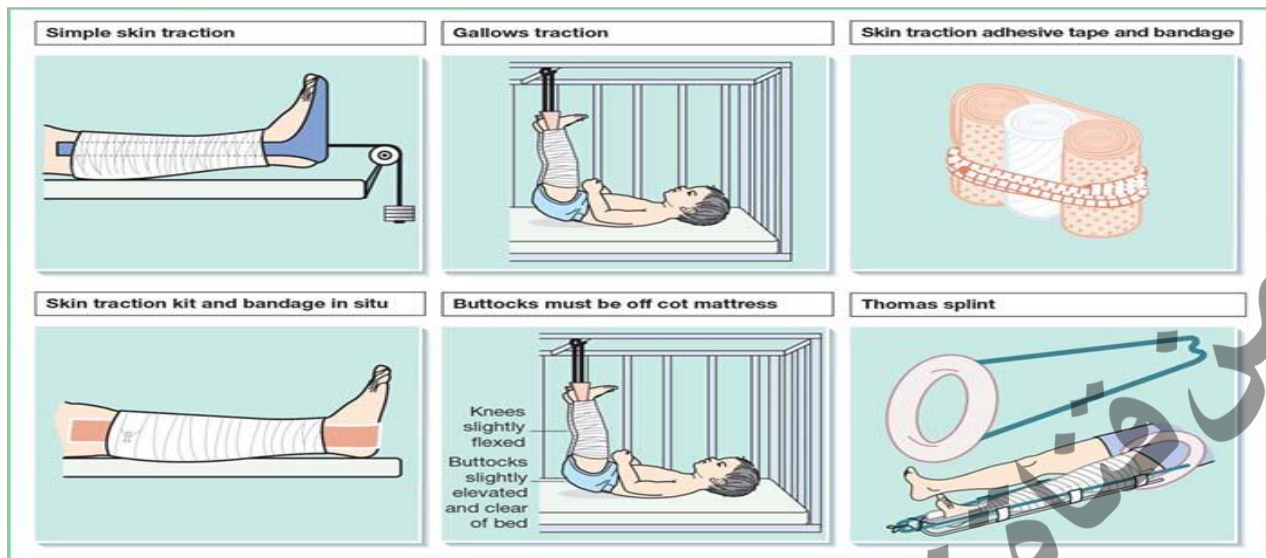
Nursing management:

Closed reduction	Open reduction.
Before procedure: • explain procedures & reassurance to the patient. • prepare patient for x- ray • prepare equipment During procedures: • Support the joints • handle the physician Post procedure care: • Prepare for x – ray	Preoperative care: • Perform neurovascular assessment • Skin preparation & Cover wound with sterile dressing Postoperative care: • Monitor patient for pain • Perform neurovascular assessment elevate the effected extremity to reduce swelling • prepare for x-ray as ordered • Check wound under aseptic technique

2. Immobilization

Definition: Using casts, splints, or traction to maintain alignment during healing.





Nursing management for patient in immobilization

1. For Patients with a CAST

1. Neurovascular Checks (The 5 Ps) - Every 8 Hours:

- **Pain:** Assess pain level and if it's worsening or unrelieved.
- **Pallor:** Check color of fingers/toes.
- **Paresthesia:** Ask about numbness or tingling.
- **Pulses:** Palpate distal pulses (rate, rhythm, strength).
- **Paralysis:** Check movement of fingers/toes.
- Assess capillary refill, temperature, and swelling.

2. Cast & Skin Assessment:

- Inspect skin around cast edges for irritation.
- Smell the cast for foul odor (infection sign).
- Report any **hot spots** on the cast.

3. Patient Education & Cast Care:

- **Keep cast clean & dry.**
- **DO NOT** stick anything inside the cast.
- **DO NOT** try to fix a broken cast.

- **Elevate** the casted extremity to reduce swelling.
- **Avoid weight-bearing** on the affected part as instructed.
- **Move fingers & toes regularly** to improve circulation.
- **Perform range of motion (ROM)** exercises on unaffected joints to prevent stiffness and muscle loss.
- Report any new or worsening pain, numbness, swelling, or changes in sensation immediately.

II. For Patients with internal/external fixation

A. Preoperative Care:

1. **Educate Patient:** Explain the procedure, expected pain, recovery time, appearance of fixation devices, and activity restrictions.
2. **Skin Preparation:** Prepare the surgical site as ordered.
3. **Perform Baseline Neurovascular Assessment.**
4. **Assure & Support Patient:** Address fears and concerns.

B. Postoperative & Ongoing Care:

1. **Monitor Vital Signs.**
2. **Assess & Manage Pain:** Provide analgesics as ordered.
3. **Frequent Neurovascular Checks**
4. **Elevate Affected Extremity** to reduce swelling.
5. **Wound Care:** Provide wound dressing for surgical sites under aseptic technique. Monitor for signs of infection (hotness, redness, changes in drainage color/odor).
6. **Encourage ROM Exercises:** For unaffected joints and muscles to prevent complications of immobility.
7. **Mobilization:** Assist with progressive ambulation using assistive devices (e.g., walker) as instructed by physician.

III. For Patients in traction

1. **Maintain Proper Alignment:**

- Ensure patient is on a **firm mattress**.
- Ensure the **pull is in line** with the bone's long axis.
- **Do not allow weights to touch the floor.**

2. **Neurovascular Checks - Every 8 Hours & PRN:** Assess pain, numbness, and color (e.g., blueness).

3. **Skin Inspection:** Frequently check skin under straps/tape for irritation or breakdown.

4. **Pin Site Care:** Clean pin tracts with sterile applicators as needed. Inspect for infection signs.

5. **Patient Education:**

- Teach patient allowed movement.
- Emphasize **not to interrupt the traction pull.**

6. **Encourage Participation:** Support patient involvement in daily activities as tolerated within traction limits.

Key Nursing Actions in Rehabilitation:

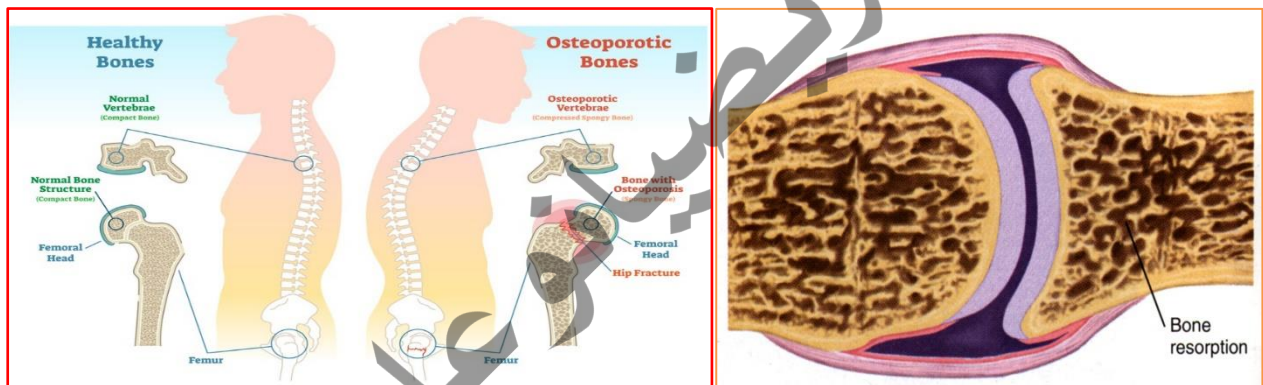
- **Collaborate:** Understand and support physical therapy goals.
- **Encourage Movement:** Promote ROM exercises for unaffected joints to reduce immobility risks.
- **Teach Safe Use:** Instruct on how to use assistive devices (walkers, crutches).
- **Provide Emotional Support:** Listen to the patient, encourage them to express feelings.
- **Prepare for Discharge:** Provide clear instructions for home care and importance of follow-up visits.

Osteoporosis

Learning objectives:

At the end of the lesson the student will be able to:

- Define osteoporosis
- Describe pathophysiology of the osteoporosis
- Enumerate causes and typed of osteoporosis
- List the clinical manifestation of osteoporosis
- Mention diagnostic studies of osteoporosis
- Describe the management of osteoporosis



Definition:

It is irreversible metabolic bone disease, in which the rate of bone resorption increases over the rate of bone formation causing a loss of bone mass and weakening of the bone making them more susceptible

Path physiology

- The rate of bone resorption increases over the rate of bone formation causing loss of bone mass
- Calcium & phosphate salts are lost creating brittle & fragile bones.
- The bones become weakened from the demands of weight bearing.
- Fractures may occur with little force.

- The vertebral column's overall mass is diminished leading to kyphosis & loss of weight.

➤ Causes & types

According to etiology of osteoporosis it can be classified into primary & secondary types.

Primary osteoporosis	Secondary osteoporosis
• The exact cause unknown • It includes two subtypes due to aging process A. Postmenopausal osteoporosis: Due to estrogen deficiency (leads to calcium deficiency).	• It is developed secondary to one of the following causes: • Long therapy of steroids (e.g., corticosteroids) • long therapy of heparin • Long therapy of anticoagulants • Hyperthyroidism • Malabsorption

Common risk factors for osteoporosis:

- Thin, small framed body
- Prolonged immobility (e.g., hemiplegia, paralysis)
- Calcium deficient diet.
- Heavy cigarette smoking
- Vita mine D deficiency
- Sedentary life style

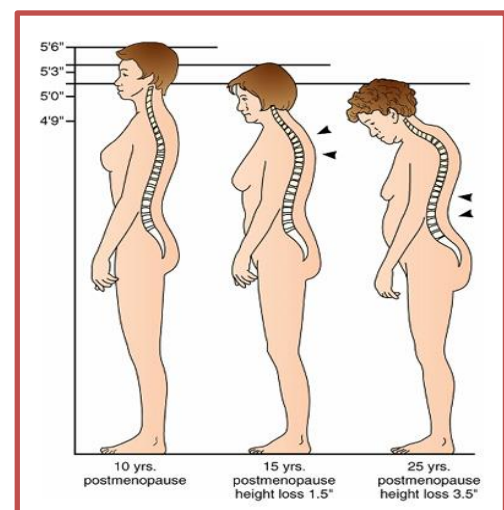
Clinical manifestation

This disease is asymptomatic until later stages, for this reason, it is called the silent killer.

The symptoms include:

- Loss of height due to collapse of bone
- Marked kyphosis of the thoracic spine
- Back pain radiates around the trunk
- Tenderness and restriction of motion
- Possibility of fractures

Diagnostic studies



1. Lab investigation:

- Serum calcium & phosphorus within normal
- Serum alkaline phosphates low level
- Urine calcium increase then returns to normal

2. Radiographic measures

3. Bone x-ray

4. Bone density measures as

- Ultrasonic heel density studies

Management of osteoporosis

• General management

1. Nutritional supplements of dietary calcium & vitamin D to support normal bone metabolism
2. Adequate exposure to sunlight especially for Childs
3. Weakened vertebral should be supported with a back brace
4. Physical therapy including gentle exercise to held calcium in bones
5. Pain management using prescribed types of analgesics
6. Mobility in hazard free environment

• Pharmacological therapy

Drug therapy	Action
Calcium supplements with vitamin D.	Decrease bone resorption Increase bone mass
Calcitonin	Increase vertebral bone mass Inhibit bone resorption
Selective estrogen receptor modulators	Increase bone mass Decrease bone resorption
Bisphosphonates.	Increase bone mass and decrease bone loss.

• Surgical treatment

- Joint replacement.

- Closed or open reduction with internal fixation. Open reduction, internal fixation.

• **Nursing management**

1. Educate Patient (Understanding & Treatment):

- Explain osteoporosis: what it means and what causes it.
- Teach how to slow or stop its progression.
- Discuss the treatment plan and medication adherence.

2. Relieve Pain:

- Encourage rest in bed several times a day (supine on back or side-lying on a firm mattress).
- Advise knee flexion for comfort.
- Apply intermittent local heat and provide back rubs for muscle relaxation.
- Promote good posture and teach proper body mechanics.

3. Improve Bowel Movement:

- Start a high-fiber diet early.
- Encourage increased fluid intake.
- Administer prescribed stool softeners as needed.

4. Prevent Injury:

- Encourage regular walking and daily outdoor weight-bearing activity (to help with Vitamin D).
- Reinforce good body mechanics and posture.
- Advise on fall prevention strategies (e.g., removing rugs, good lighting, assistive devices).

Complications:

- Fractures
- Bone deformities
- Trauma

Student self-assessment

The nurse is instructing a 65-year-old female client diagnosed with osteoporosis. The most important instruction regarding exercise would be to:

- A. Exercise doing weight bearing activities.
- B. Exercise to reduce weight.
- C. Avoid exercise activities that increase the risk of fracture.
- D. Exercise to strengthen muscles and thereby protect bones.

A client fell off a ladder and the healthcare provider suspect a fracture of the right wrist. Which manifestation should the nurse anticipate observing in the client? (Select all that apply)

- A. Crepitus
- B. Visible deformity
- C. Pain
- D. Cyanosis of nail beds
- E. Absence of radial pulse

The nurse is providing discharge teaching to the family of an older adult client who was treated for a fracture after a fall. Which recommendation should the nurse include in the teaching?

- A. Always wear socks when ambulating.
- B. Use a step stool when possible.
- C. Start a mild exercise program.
- D. Remove the rubber mat from tub.

A client sustained a radial fracture and a cast was just applied. The client states that there is unrelieved pain and numbness in the fingers on the affected side. Which intervention should be a priority?

- A. Notifying the healthcare provider for cast removal
- B. Elevating the extremity
- C. Preparing for fasciotomy
- D. Performing frequent neurovascular checks

A client had a cast applied to a fractured limb, and the healthcare provider has ordered frequent neurovascular checks. Which assessment should the nurse perform? (Select all that apply.)

- A. Paresthesia
- B. Pain
- C. Position
- D. Color
- E. Temperature

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Chapter nine

(Nursing care endocrine of disease)

Learning objectives

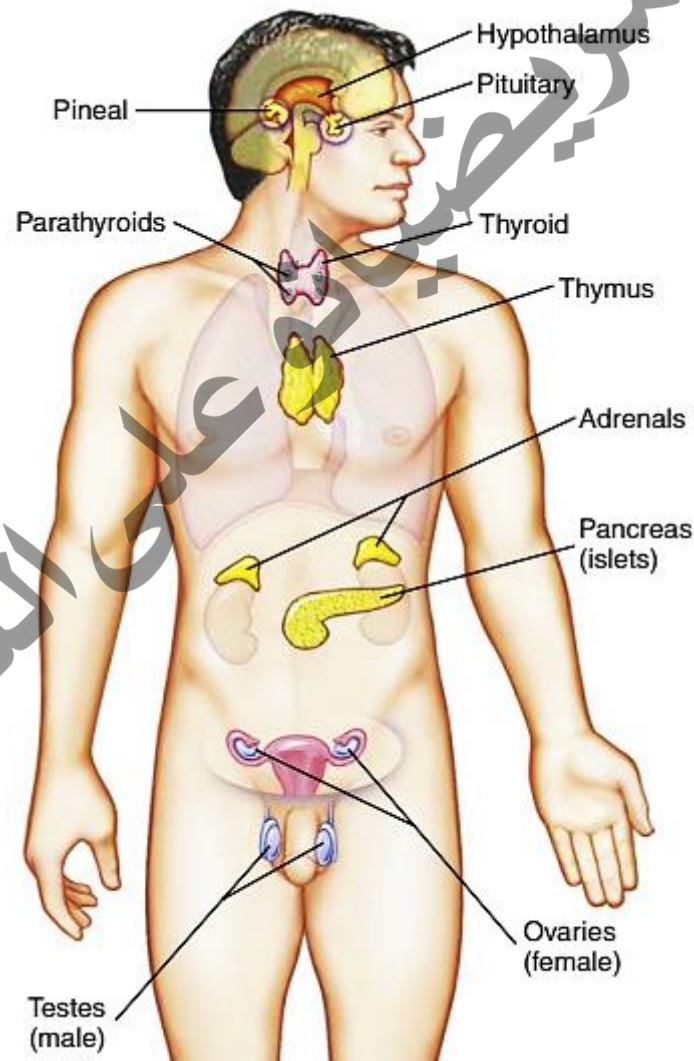
At the end of this chapter the students will be able to:

- Describe anatomy of endocrine system.
- Identify physiology of endocrine system.

Introduction

Homeostasis refers to the process of maintaining a stable internal environment. There are two major systems in the body for maintaining homeostasis: the nervous system and the endocrine system. The endocrine system is often responsible for the regulation of longer-term processes.

Anatomy of endocrine system



Endocrine glands – organs whose only function is the production and release of hormones. These include:

- ✓ Pituitary gland.
- ✓ Thyroid gland.
- ✓ Parathyroid gland.
- ✓ Adrenal gland.

Organs that are **not** pure glands (as they have other functions as well as the production of hormones) but contain relatively large areas of hormone-producing tissue. These include:

- ✓ Hypothalamus.
- ✓ Pancreas.

physiology of endocrine system

- **Hypothalamus:**
 - ✓ Regulates hunger, thirst, sleep and wakefulness plus most of your involuntary mechanisms including body temperature.
- **Thyroid glands:**
 - ✓ Regulates energy and metabolism.
- **Pituitary gland:**
 - ✓ Controls all other endocrine glands; influences growth, metabolism and regeneration.
- **Parathyroid gland:**
 - ✓ Secretes the hormones necessary for calcium absorption.
- **Pancreas:**
 - ✓ Aids in the digestion of protein, fats and carbohydrates. Produces insulin which controls blood sugar level.
- **Adrenal gland:**
 - ✓ Secretes hundreds of compounds including cortisone & adrenal which helps you react to emergencies. Regulates metabolic processes in the cells, water balance, and blood pressure.

Diabetes Mellitus

Learning objectives

At the end of this lesson the students will be able to:

- Define diabetes mellitus.
- Understand the pathophysiology of diabetes mellitus.
- List the causes of diabetes mellitus.
- Identify the types of diabetes mellitus.
- Recognize the signs and symptoms of diabetes mellitus.
- List the diagnostic evaluation of diabetes mellitus.
- Explain nursing interventions for diabetes mellitus.

Introduction:

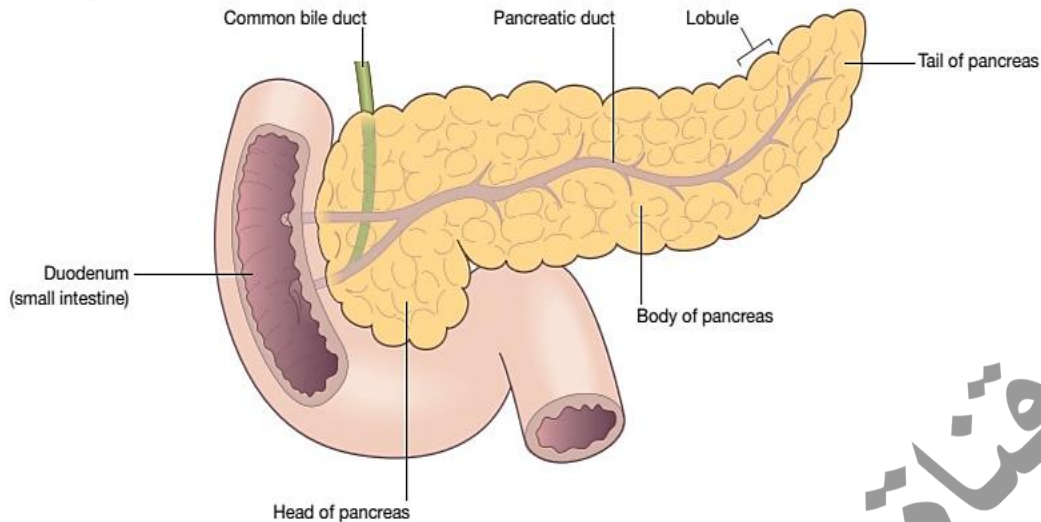
Diabetes is a chronic condition that can lead to serious complications like blindness, kidney failure, heart attacks, and strokes. However, with proper education and self-care, people with diabetes can prevent or delay these problems and live healthy, active lives.

Definition of Diabetes Mellitus:

- Diabetes mellitus is a condition caused by a lack or reduced effectiveness of insulin in the body.
- It is a disease that affects multiple systems due to abnormal insulin production, poor insulin use, or both.

Pathophysiology of Diabetes Mellitus:

- The pancreas produces insulin from beta cells in the islets of Langerhans.
- After eating, blood glucose rises. In healthy individuals, insulin helps return glucose to normal levels within 2 hours.
- Insulin helps store excess glucose as glycogen in the liver.
- Insulin and glucagon work together to keep blood glucose balanced.
- In diabetes, insulin is either not produced enough or doesn't work properly, so glucose stays high in the blood.
- The body starts using fat for energy, leading to abnormal fat metabolism and cholesterol buildup in arteries.
- Burning fat instead of glucose causes ketone production, which may lead to ketoacidosis.
- High glucose levels overwhelm the kidneys, causing glucose to pass into urine and leading to polyuria (frequent urination).



Causes of Diabetes Mellitus:

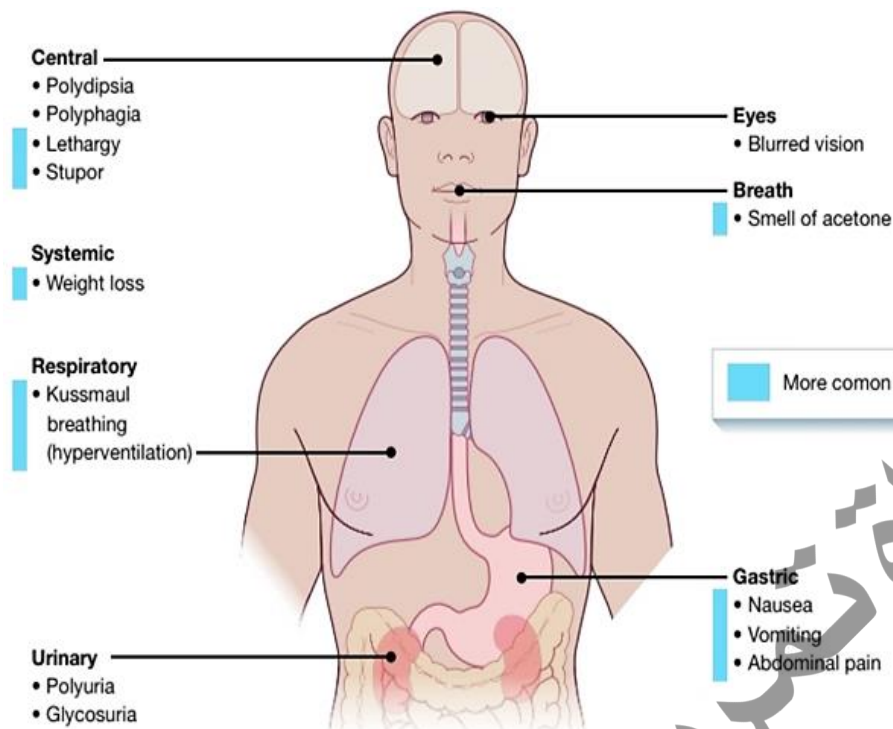
- Genetic factors
- Autoimmune disorders
- Viral infections
- Environmental factors (e.g., obesity, lack of exercise, stress)

Types of Diabetes Mellitus:

- **Type 1 Diabetes Mellitus (IDDM):** The body cannot produce insulin.
- **Type 2 Diabetes Mellitus (NIDDM):** The body resists insulin, even if insulin levels are normal or high.
- **Gestational Diabetes:** Occurs during pregnancy in women who didn't have diabetes before.
- **Secondary Diabetes:** Caused by another medical condition or treatment that affects blood glucose.

Signs and Symptoms:

- The "Three Ps":
 - **Polydipsia** – excessive thirst
 - **Polyuria** – frequent urination
 - **Polyphagia** – increased hunger
- Nocturia (urinating at night)
- Dehydration
- Fatigue
- Blurred vision
- Abdominal pain
- Headache
- Ketones in blood or urine (especially in type 1 or advanced type 2)



Diagnostic Evaluation:

Diabetes is diagnosed using:

- Medical history and physical exam
- Blood tests:
 - Fasting blood glucose
 - Post-meal glucose
 - HbA1c (average glucose over 3 months)
- Lipid profile
- Kidney function tests (creatinine, microalbuminuria)
- Electrolytes
- Urinalysis
- Acetone (if needed)
- Dental check-up
- ECG (if needed)
- Eye exam (fundoscopy)
- Foot exam
- Neurological exam
- Weight monitoring

Nursing Interventions:

non-Pharmacological:

- Assess patient's knowledge about diabetes care.
- Help patient set target glucose levels and actions for high/low readings.
- Teach glucose monitoring before meals and at bedtime.

- For NPO or tube-fed patients, check glucose every 4–6 hours; every 1–2 hours if on insulin drip.
- Teach insulin or medication administration based on glucose levels.
- Ensure meals match medication timing to prevent hypoglycemia.
- Observe and teach signs of hypo- and hyperglycemia, including causes and treatment.
- Refer to dietitian for nutrition education.
- Teach patients on insulin to eat extra complex carbs before exercise if glucose <100 mg/dL.

Diabetes Education:

Diet:

- Eat balanced meals at regular times (no more than 6 hours apart).
- Limit sugars and sweets (e.g., soda, desserts, candy, jam).
- Macronutrient goals:
 - Protein: 15–20%
 - Fat: <35%
 - Carbohydrates: 45–60% (from whole grains, fruits, vegetables, low-fat milk)
 - Fiber: 25–50 g/day

Fluid Intake:

- Drink water when thirsty.

Physical Activity:

- Assess readiness and any health restrictions.
- Set realistic goals and identify barriers.
- Explain how exercise improves glucose control and heart health.
- Moderate aerobic exercise: 150 minutes/week over 3+ days.
- Resistance training improves insulin sensitivity.
- Check glucose before exercise; carry fast-acting glucose.
- Avoid exercise if ketones are present (risk of worsening hyperglycemia).

Medications:

- Rotate insulin injection sites to avoid tissue damage.
- Use one area for a week before switching.
- Take oral medications before meals.
- Adjust insulin dose based on glucose and carb intake.

Foot Care:

- Wear proper shoes at all times.
- Report any sores immediately.
- Daily foot care tips:

- Wash and dry feet with warm water
- Use alcohol-free lotion (not between toes)
- Inspect feet daily
- Wear leather shoes and light cotton socks
- Never walk barefoot
- Avoid tight socks and crossing legs
- Trim nails straight across
- Visit podiatrist regularly (3–4 times/year)

Pharmacological:

- **Type 1 DM:** Requires insulin from diagnosis.
- **Type 2 DM:** May need oral hypoglycemic agents (OHAs) to stimulate insulin or improve sensitivity.
- **Aspirin therapy:** 75–162 mg/day for cardiovascular risk.
- **Statins:** For patients with heart disease or age >40 with risk factors.

Thyroid Disorders

Learning objectives

At the end of this chapter the students will be able to:

- Define hypothyroidism
- Understand the pathophysiology of hypothyroidism
- List the causes of hypothyroidism
- Identify signs and symptoms of hypothyroidism
- List diagnostic evaluation of hypothyroidism
- Discuss nursing interventions for hypothyroidism
- Define hyperthyroidism
- Understand the pathophysiology of hyperthyroidism
- List the causes of hyperthyroidism
- Identify signs and symptoms of hyperthyroidism
- List diagnostic evaluation of hyperthyroidism
- Discuss nursing interventions for hyperthyroidism

Introduction:

The thyroid gland has two lobes connected by a small bridge called the isthmus. It produces three main hormones:

- **Triiodothyronine (T3)**
- **Thyroxine (T4)**
- **Calcitonin**

These hormones help regulate metabolism, growth, and calcium levels in the body.

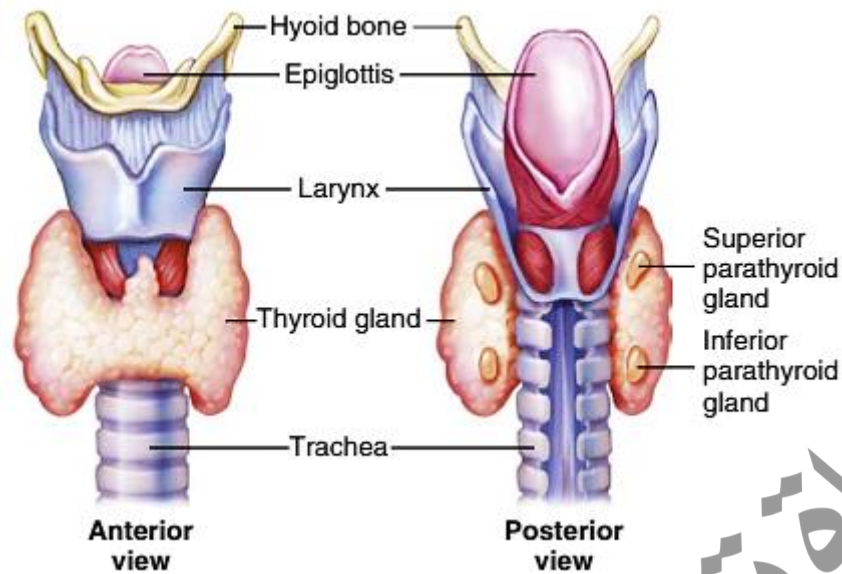
Hypothyroidism

Definition:

Hypothyroidism is a condition where the thyroid gland does not produce enough thyroid hormones.

Pathophysiology:

- Most common in women over 50 years old.
- In infants, it can cause serious growth and developmental problems.
- Often caused by autoimmune destruction of the thyroid (e.g., Hashimoto's thyroiditis).
- Low thyroid hormone levels slow metabolism, increase cholesterol, and cause weight gain, fatigue, and cold intolerance.
- Muscle activity becomes sluggish, and the patient may feel sleepy and weak.



Causes:

Primary Hypothyroidism:

- Congenital defects
- Low thyroid hormone production
- Thyroid inflammation
- Iodine deficiency
- Hashimoto's thyroiditis (autoimmune)

Secondary/Tertiary Hypothyroidism:

- Pituitary or hypothalamic damage
- Postpartum pituitary necrosis
- Treatment of hyperthyroidism (medications or surgery)
- Resistance to thyroid hormone at the tissue level

Signs & Symptoms:

- Fatigue
- Weight gain
- Slow heart rate (bradycardia)
- Constipation
- Mental dullness
- Feeling cold
- Shortness of breath
- Dry skin and hair
- Decreased sweating
- Heart failure (due to weak heart pumping)
- High cholesterol (hyperlipidemia)
- Myxedema (non-pitting swelling of face, hands, feet)
- Slow digestion

- Hand pain or numbness (carpal tunnel)
- Poor memory

Diagnostic Evaluation:

- History and physical exam
- Serum T3 and T4 (if ordered)
- TSH and free T4 levels
- Thyroid peroxidase antibodies (autoimmune marker)
- TRH stimulation test

Nursing Interventions

Non-Pharmacological:

1. For Activity Intolerance:

- Assess fatigue level
- Assist with daily tasks
- Allow rest between activities
- Gradually increase activity as medication works

2. For Constipation:

- Monitor bowel movements
- Encourage normal routine (e.g., morning coffee)
- Increase fluids (8 glasses/day if heart is stable)
- Add fiber: fruits, vegetables, bran
- Encourage walking
- Use toilet instead of bedpan
- Request stool softener if needed
- Avoid enemas unless necessary

3. For Nutrition Imbalance (Weight Gain):

- Weigh weekly
- Refer to dietitian
- Encourage exercise within limits
- Reassure patient that weight will improve with treatment

Pharmacological:

- Treated with oral thyroid hormone (usually levothyroxine).
- Start with low doses and increase gradually to avoid side effects or heart problems.

Patient Education:

- Explain the need for lifelong medication and regular follow-up.
- Emphasize warmth and comfort due to cold intolerance.
- Teach skin care to prevent breakdown.
- Warn older adults about sedatives—use lowest dose if needed.

- Teach constipation prevention: fiber, activity, stool softeners, regular toilet routine.

Hyperthyroidism

Definition:

Hyperthyroidism is a condition where the thyroid gland is overactive and produces too much thyroid hormone.

Pathophysiology:

- More common in women, especially ages 20–40.
- Also called thyrotoxicosis.
- Excess thyroid hormone increases metabolism and sensitivity to stress hormones (catecholamines).
- Leads to overactive sympathetic nervous system and changes in fat and carbohydrate metabolism.

Causes:

- **Graves' disease** (autoimmune—most common cause)
- Overactive thyroid nodules (toxic adenoma, multinodular goiter)
- Thyroid inflammation (thyroiditis)
- Thyroid or pituitary tumors
- Overuse of thyroid medication
- Radiation exposure
- Genetic factors
- Smoking (especially in women)

Signs & Symptoms:

- Nervousness
- Excessive sweating
- Heat intolerance
- Palpitations
- Fast heart rate (tachycardia)
- Fatigue
- Weakness
- Weight loss
- Tremors
- Increased appetite
- Warm, moist skin
- Possible bulging eyes (exophthalmos)
- Enlarged thyroid (goiter)
- Frequent bowel movements

- Insomnia

Diagnostic Evaluation:

- History and physical exam
- ECG (electrocardiogram)
- Blood tests: free T4, TSH
- TRH stimulation test
- Eye exam
- Radioactive iodine uptake (RAIU) test

Nursing Interventions for Hyperthyroidism

Non-Pharmacological Care

1. For Hyperthermia (High Body Temperature)

- Monitor temperature regularly
- Give acetaminophen as prescribed (avoid aspirin—it can worsen symptoms)
- Use a cooling blanket if ordered
 - Set it 1–2°C below current body temperature
 - Wrap arms and legs with towels to prevent shivering
- Offer fluids to replace those lost through sweating

2. For Diarrhea

- Provide a low-fiber diet
- Serve small, bland meals (e.g., bananas, rice)
- Monitor for signs of dehydration
- Keep skin clean and dry
- Use barrier cream to protect skin from irritation

3. For Imbalanced Nutrition (Weight Loss)

- Determine healthy weight based on height
- Weigh weekly to track progress
- Refer to a dietitian for high-calorie meals and supplements

4. For Disturbed Sleep

- Create a quiet, restful environment
- Offer earplugs or soft music to reduce noise

Pharmacological Care

Drug Therapy

Anti-Thyroid Medications:

- **Iodine:**
 - Used short-term before surgery or during thyrotoxic crisis
 - Blocks production and release of T3 and T4
 - Reduces blood flow to the thyroid, making surgery safer

- Maximum effect in 1–2 weeks; not suitable for long-term use
- **Beta-Blockers (e.g., Propranolol):**
 - Help control fast heart rate and other symptoms

Radioactive Iodine Therapy (RAI)

- Most common treatment for non-pregnant adults
- Destroys thyroid tissue to reduce hormone production
- May cause dry mouth or throat (thyroiditis/parotiditis)
 - Use water, ice chips, or baking soda gargles (10 mL in 250 mL water)
- Effects take 2–3 months
- Anti-thyroid drugs and beta-blockers are used during this period
- High chance of developing hypothyroidism afterward → lifelong hormone replacement may be needed

Surgical Therapy: Thyroidectomy

Indications:

- Large goiter pressing on the trachea
- No response to medications
- Thyroid cancer
- Not suitable for RAI therapy

Types:

- **Subtotal Thyroidectomy:** Removes ~90% of the gland
- **Total Thyroidectomy:** Requires lifelong hormone replacement

Preoperative Teaching:

- Teach coughing, deep breathing, and leg exercises
- Practice neck movements and head support techniques
- Explain post-op care and temporary difficulty speaking
- Administer anti-thyroid drugs, iodine, and beta-blockers to stabilize hormone levels

Postoperative Care:

- Monitor thyroid hormone levels regularly
- Expect temporary hypothyroidism
- Avoid giving thyroid hormone immediately (to allow natural recovery)
- Reduce calorie intake to prevent weight gain
- Ensure adequate iodine intake (not excessive)
- Encourage regular exercise
- Avoid hot environments
- Schedule regular follow-up visits
- If total thyroidectomy: teach lifelong hormone replacement

Nutritional Therapy

- High-calorie diet: 4000–5000 kcal/day
- Six meals + snacks rich in:
 - Protein
 - Carbohydrates
 - Vitamins (A, B1, B6, C)
 - Minerals
- Avoid spicy and high-fiber foods (can worsen diarrhea)
- Consult dietitian for personalized meal plans

Patient Education

- Teach signs of both hyperthyroidism and hypothyroidism
- Explain how and when to take medications
- Stress the importance of regular lab tests and follow-up visits
- Encourage reporting of new or worsening symptoms

Student self-assessment

1. A nurse is assessing a client newly diagnosed with Type 2 diabetes mellitus. Which of the following findings is most consistent with this diagnosis?

- A. Sudden onset of weight gain and bradycardia
- B. Polyuria, polydipsia, and blurred vision
- C. Cold intolerance and constipation
- D. Increased appetite with low blood glucose

2. A client with Type 1 diabetes reports feeling shaky, sweaty, and confused. What is the nurse's priority action?

- A. Administer insulin
- B. Check blood glucose level
- C. Call the healthcare provider
- D. Give 1 liter of IV normal saline

3. Which laboratory value indicates poor glycemic control in a diabetic client?

- A. HbA1c 5.8%
- B. Fasting glucose 95 mg/dL
- C. HbA1c 8.5%
- D. Random glucose 120 mg/dL

4. A nurse is teaching foot care to a diabetic client. Which statement indicates understanding?

- A. "I will soak my feet daily in warm water."
- B. "I will walk barefoot at home to strengthen my feet."
- C. "I will inspect my feet daily for cuts or blisters."
- D. "I will use heating pads to warm my feet in winter."

5. A client with diabetic ketoacidosis (DKA) is admitted. Which intervention should the nurse perform first?

- A. Administer IV insulin
- B. Start potassium replacement
- C. Begin fluid resuscitation
- D. Obtain urine ketone levels

6. A client with hypothyroidism is prescribed levothyroxine. Which instruction should the nurse include?

- A. Take the medication with meals
- B. Avoid dairy products while on this medication
- C. Take the medication in the morning on an empty stomach
- D. Discontinue the drug if heart rate increases

7. Which clinical manifestation is most associated with hypothyroidism?

- A. Tachycardia and weight loss
- B. Heat intolerance and diarrhea
- C. Fatigue and dry skin
- D. Exophthalmos and tremors

8. A nurse is reviewing lab results for a client with suspected hypothyroidism. Which pattern confirms the diagnosis?

- A. ↓ TSH, ↑ T3/T4
- B. ↑ TSH, ↓ T3/T4
- C. ↓ TSH, ↓ T3/T4
- D. ↑ TSH, ↑ T3/T4

9. A client with myxedema coma is admitted to the ICU. What is the nurse's priority assessment?

- A. Blood glucose
- B. Pupillary response
- C. Airway and respiratory status
- D. Urine output

10. A nurse is caring for a client with hyperthyroidism. Which symptom should the nurse expect?

- A. Bradycardia
- B. Cold intolerance
- C. Weight gain
- D. Heat intolerance

11. Which medication is commonly used to treat hyperthyroidism and requires liver function monitoring?

- A. Levothyroxine
- B. Methimazole
- C. Insulin glargine
- D. Prednisone

12. A client receiving radioactive iodine therapy for Graves' disease should be taught to:

- A. Avoid all dairy products
- B. Increase iodine-rich foods
- C. Use separate utensils and avoid close contact for several days
- D. Take the medication with food

13. A nurse is monitoring a client post-thyroidectomy. Which finding requires immediate intervention?

- A. Hoarseness

- B. Mild neck swelling
- C. Positive Chvostek's sign
- D. Pain at the incision site

14. A client is experiencing thyroid storm. Which of the following is the nurse's priority?

- A. Administer oral levothyroxine
- B. Provide cooling measures and beta blockers
- C. Encourage ambulation
- D. Start high-dose corticosteroids

15. Which of the following clients should the nurse assess first?

- A. A diabetic client with blood glucose of 250 mg/dL and no symptoms
- B. A hypothyroid client with dry skin and fatigue
- C. A hyperthyroid client with heart rate of 130 bpm and anxiety
- D. A diabetic client with HbA1c of 6.5%

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Chapter ten (Nursing care of cancer)

Learning objectives

At the end of this chapter the students will be able to:

- Define the cancer.
- Describe the Pathophysiology of the cancer.
- State the Characteristics of cancer cells.
- Identify etiology of the cancer
- Mention risk factors of the cancer
- Recognize the types of cancer
- Enumerate the signs and symptoms of cancer
- Describe the prevention of the cancer
- Differentiate between the types of cancer Management
- Identify and understand a common cancer disorder (Colon Cancer).

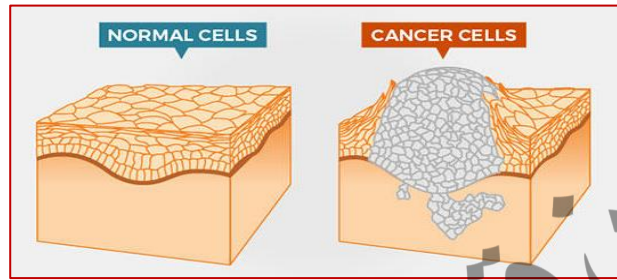
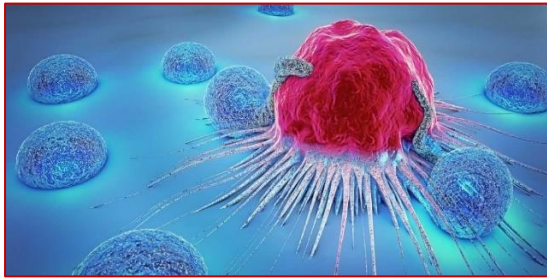
Introduction

Cancer causes cells to divide uncontrollably. This can result in tumors, damage to the immune system, and other impairment that can be fatal. Cancer is a broad term. It describes the disease that results when cellular changes cause the uncontrolled growth and division of cells.

Terminology

- **Neoplasm:** Abnormal tissue growth (tumor).
- **Benign:** Non-cancerous, encapsulated, doesn't spread.
- **Malignant:** Cancerous, invades tissue, can spread.
- **Metastasis:** Cancer spread to other body parts (via lymph/blood).
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Cancer

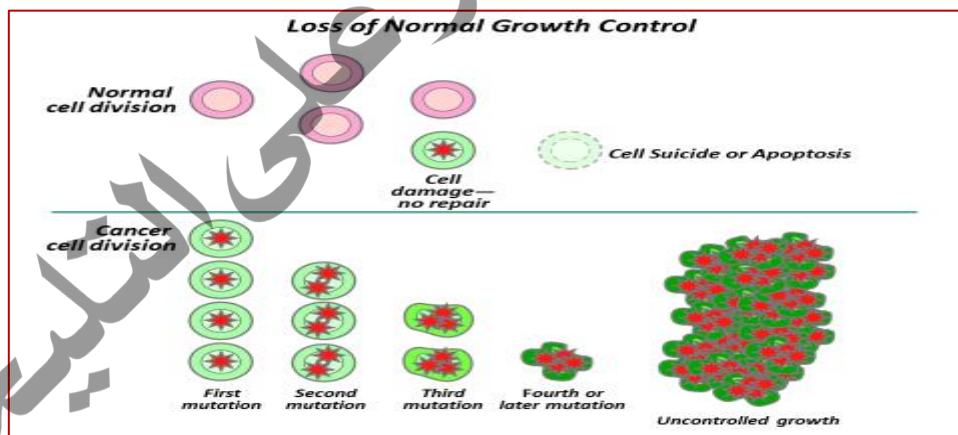


Definition of cancer

Abnormal cell growth with ability to invade or spread. Begins when a cell's DNA mutates, leading to uncontrolled transformation.

Pathophysiology

- Abnormal cell forms a "clone," ignoring normal growth signals.
- Cells become invasive, infiltrate tissues, and spread via lymph/blood vessels.



Etiology (Causes):

- **Physical Agents:** Sunlight, radiation, chronic irritation/inflammation.
- **Chemical Agents:** Tobacco, pesticides, formaldehyde, asbestos.
- **Genetic/Family Factors:** Runs in families (5-10% due to genetics, shared environment/lifestyle).

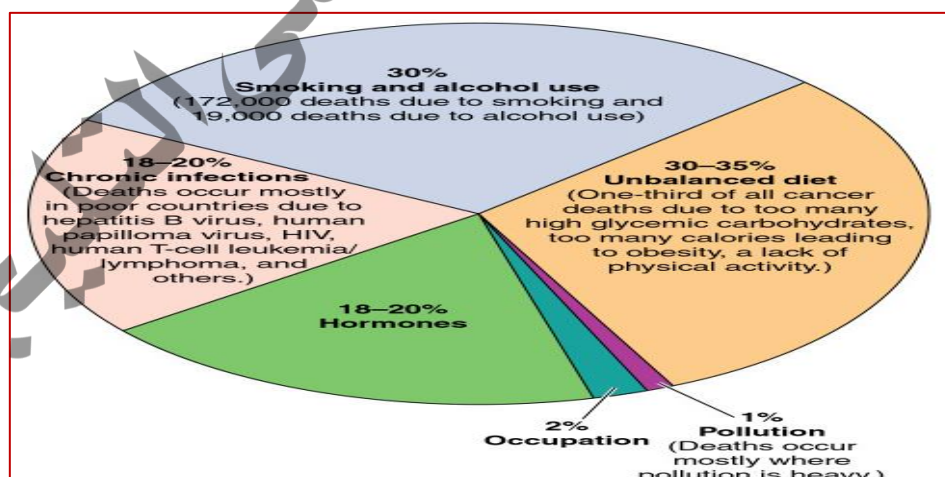
- **Dietary Factors:** High fat, alcohol, salt-cured/smoked meats, nitrates/nitrites, high calories, obesity.

Risk factors:

The exact cause of cancer is **unknown** but there are many risk factors that lead to cancer.

Controllable risk factor	Uncontrollable risk factor
1. Smoking. 2. Obesity and fatty diet. 3. Alcohol and drug up use, 4. Ultraviolet radiation, AIDS, HBV, HCV 5. Certain chemical, air pollution, pesticides. 6. Stress.	1. Age: the risk of many cancers increases with age. 2. Family history of cancer. 3. Genetic condition. 4. Previous history of cancer. 5. Previous chemotherapy and radiation therapy.

Factors Believed to Contribute to Global Causes of Cancer



➤ Cancer Characteristics & Stages

A. Tumor Staging (TNM System):

- **T:** Size/extent of Primary Tumor.

- **N:** Lymph Node involvement.
- **M:** Extent of Metastasis.

B. Benign vs. Malignant Tumors:

Characteristics	benign	Malignant
Cell characteristics	Not Resemble normal cells	Resemble normal cells
Mode of growth	Grow by expansion not penetrate normal tissue	Invade normal cells and destroy it
Rate of growth	Slow	Rapid
Metastasis	Doesn't spread by metastasis	Spread by metastasis
General effects	Doesn't cause general effect unless interfere with vital function	Often cause general effect
Tissue destruction	Doesn't cause tissue damage	Extensive tissue damage
Ability to cause death	Not cause death	Usually cause death

C. Types of Cancer:

- **Carcinomas:** Develop on cells covering body surfaces (most common).
- **Sarcomas:** Develop on supporting tissues (bone, cartilage, fat, muscle).
- **Lymphomas:** Develop on lymph nodes/immune system tissues.
- **Leukemias:** Develop on immature blood cells in bone marrow.

Warning Signs of Cancer (Caution Sig):

1. Change in bowel or bladder habits
2. A sore that does not heal
3. Unusual bleeding or discharge
4. Thickening or lump in the breast, testicles, or elsewhere
5. Indigestion or difficulty swallowing

6. **Obvious change in the size, color, shape, or thickness of a wart, mole, or mouth sore**
7. **Nagging cough or hoarseness**

Clinical manifestation

1. Mechanical:

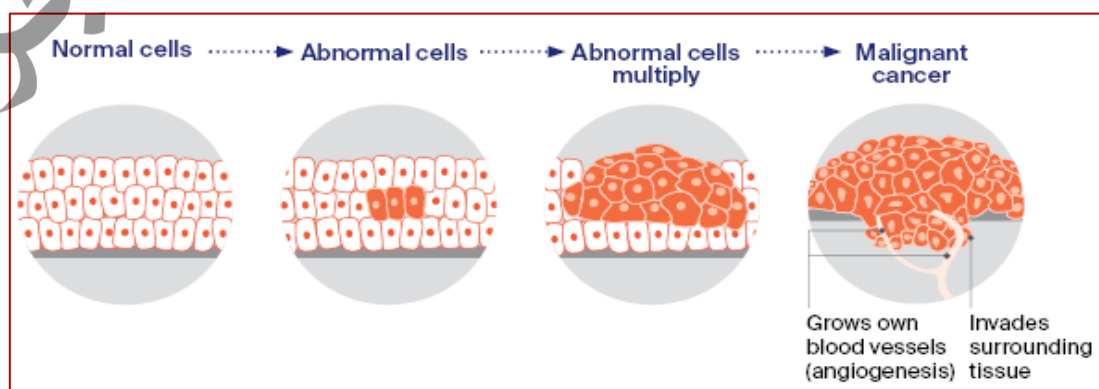
- a. Pressure upon surrounding tissue led to pain.
- b. Obstruction
- c. Interruption of blood supply led to necrosis and ulceration.
- d. Disturbance of surrounding tissue.

2. Systemic effect:

- a. Anorexia, weakness & weight loss.
- b. Disturbance of body metabolism, fluid and electrolyte imbalance, interfere with organ function.

Stages of Cancer:

- **Stage I:** Localized to one body part; usually curable.
- **Stage II:** Locally advanced.
- **Stage III:** Locally advanced with lymph node involvement.
- **Stage IV:** Metastasized (spread to other organs/body).



Detection and prevention

- **Detection:** Screening (self-exams, mammography, blood tests, rectal/pelvic exams).
- **Prevention:**
 - Healthy diet (vegetables, fruit and high fiber diet).
 - Increase vitamin A&C intake.
 - Maintain ideal body weight.
 - Reduce salt, smoked food and limit intake of red meat.
 - Stop smoking and alcohol intake.
 - Low sugar and high in complex carbohydrate.
 - Avoid exposure to sun light.

Diagnosis of cancer

1. Biopsy: obtain tissue sample for analysis of cells suspected to be malignant
2. X- Ray & Ultrasound.
3. Endoscopy.
4. Blood test & stool.
5. Computerized axial tomography
6. Magnetic resonance imaging.
7. Diagnostic surgery.

Management of cancer (Treatment modalities):

Cancer Management Goals:

- **Cure:** Eradicate disease.
- **Control:** Prolong survival.
- **Palliation:** Relieve symptoms.

The treatment of cancer depends on:

1. Types of tumors.
2. Location and size of tumor.
3. Patient condition.
4. Presence of metastasis.

Treatment Modalities & Nursing Management:

➤ **Surgical Management**

Surgery: is local treatment to remove the tumor. Tissue around the tumor and nearby lymph nodes may also be removed during the operation.

Surgery Goals:

1. **Curative:** removal of primary site of tumor when early detection.
2. **Prophylactic:** removal of lesion which, if left in the body are to develop into cancer.
3. **Diagnostic:** take biopsy
4. **Palliative:** to relieve complication (obstruction) or to relieve pain
5. **Reconstructive/Rehabilitative.**

Nursing management

- Give pre & post operation care.
- Give general care for cancer patient.
- Care for nutritional deficits and disorder of coagulation
- Strict aseptic technique to prevent infection.
- Rehabilitation to promote body function.
- Importance of follow up.
- Medication & diet

➤ **Radiation treatment**

Definition: use highest energy rays that can kill any cell or tissue. Approximately 60% of cancer patient receive radiotherapy.

Uses:

- Prophylactics
- Control of cancer cells
- Curative: destroyed tumor cells
- Palliative: to relieve pain

Radiation Safety Precautions:

- **Distance:** Stay as far as necessary.
- **Time:** Spend minimal time in radiation areas.
- **Shielding:** Use protective materials.

Nursing management

Instruct the patient about:

1. Avoid washing marked skin with soap (water only).
2. No perfume/medication on treated site.
3. Wear soft, cotton clothes over treated area.
4. Protect skin from sun exposure.
5. Avoid sharp equipment near site.
6. Plenty of rest and balanced diet.
7. Report any skin damage/changes.

➤ Chemotherapy treatment

Definition: Cytotoxic drugs destroy cancer cells or prevent growth.

Administration: Hospital, outpatient, home; routes (topical, oral, Intramuscular, Intravenous, Subcutaneous).

▪ Nursing management:

- Assess fluid/electrolyte balance (anorexia, nausea vomiting, altered taste, mucositis, and, diarrhea).
- Modify risks for infection/bleeding (bone marrow suppression).
- Administer safely, monitor for extravasation.
- Protect caregivers: use PPE, follow policies for handling/disposal.

➤ **Biotherapy (immunotherapy):**

- Treatment that alters the body's biological response to fight infection and disease.
- Alters the immune system with either stimulatory or suppressive effect
- Produce anti-tumor activities.

➤ **Hormonal therapy:**

- Used against hormonally sensitive tumors like breast and prostate.
- Creates unfavorable growth environment.

▪ **Side effects of cancer treatment**

1. Bone marrow suppression led to the patient high risk for infection, bleeding and anemia.
2. Gastrointestinal effects lead to anorexia, loss of appetite, nausea and vomiting.
3. Hair loss
4. Lung and Kidney damage
5. Low white blood count
6. Liver damage
7. Heart damage
8. Damage to the veins
9. Tumor lysis syndrome
10. fatigue and pain

▪ **General Nursing management of Patients with Cancer**

- **Maintain Tissue Integrity:** Skin care, comfort, pain relief, prevent trauma/infection.
- **Promote Nutrition:** Manage anorexia, offer small, frequent meals.
- **Relieve Pain:** Assess pain source/factors, provide interventions.

- **Decrease Fatigue:** Promote comfort, reduce stressors.
- **Improve Body Image/Self-Esteem:** Help patient cope with changes.
- **Monitor & Manage Complications:** Lab studies, vital signs, neurological status, fluid/electrolyte balance, Arterial Blood Gases, pulse oximetry.
- **Promote Home & Community-Based Care:** Teach self-care, facilitate follow-up, provide reassurance.
- **Assist in Grieving Process:** Provide emotional support.

Colon cancer

Learning objectives:

At the end of the lesson the students will be able to:

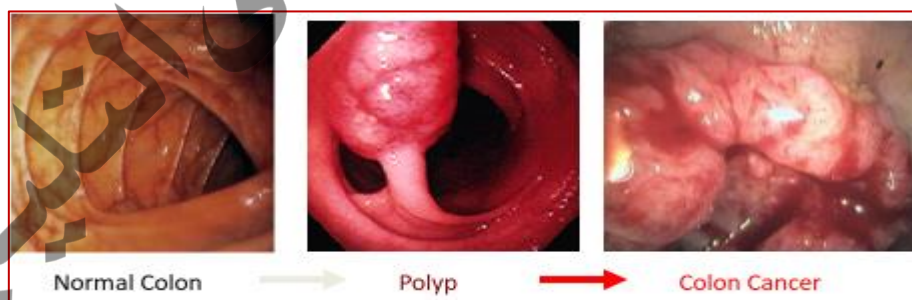
- Define the colon cancer.
- Describe pathophysiology of the colon cancer.
- List signs and symptoms of colon cancer.
- Mention diagnostic procedure used for colon cancer.
- Explain management of the colon cancer.

Definition:

Malignant growth of tumor that begins from the inner wall of the colon or rectum. Can also involve the anal canal.

Pathophysiology:

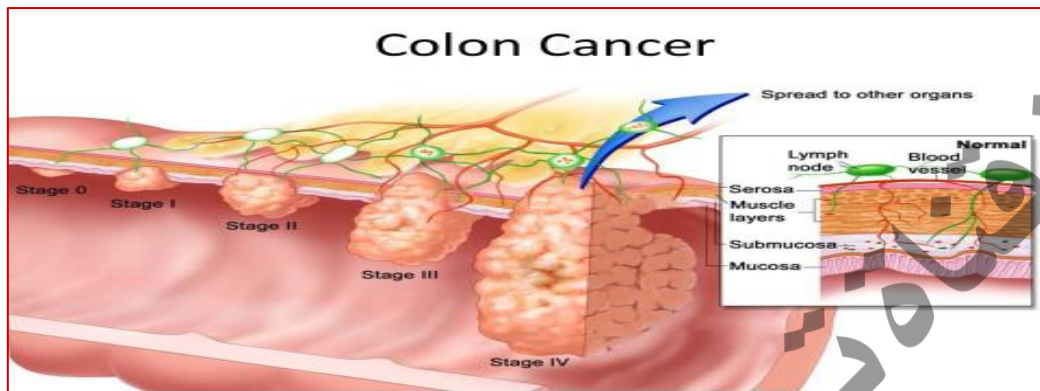
The formation of colorectal is a multistep process. Begins with an abnormal growth of tissue known as a polyp (precursors to the disease) originating from the innermost wall of the colon.



Stage of colon cancer

- **Stage 0:** Mucosal. Take out with colonoscopy
- **Stage I and 2:** Submucosal and into the muscle but not through the wall.
Surgery can cure these cancers

- **Stage 3:** Lymph nodes are involved and chemotherapy and surgery are needed
- **Stage 4:** The cancer has metastasized.



Etiology:

1. Inflammatory bowel disease
2. genetic mutations
3. polyps

Risk factors:

- **Non-modifiable:** Age, family history
- **Modifiable:** Physical inactivity, obesity, high consumption of red or processed meats, low consumption of fruits and vegetables, alcohol use, tobacco use

Signs and symptoms:

Right side	Left side	Rectum
1. Abdominal pain 2. bleeding 3. weakness 4. Fatigue 5. Palpable abdominal mass 6. Malena 7. Bowel obstruction	1. Colicky pain 2. bleeding 3. Obstruction 4. weakness and fatigue 5. ribbon like stools 6. nausea and vomiting	1. Pain 2. Rectal bleeding & Bloody stools 3. Altered bowel pattern 4. Perineal and buttock pain

Diagnosis:

- Entire Large bowel evaluation: Colonoscopy - Double-Contrast Barium Enema - CT Colonography
- Flexible sigmoidoscopy (FSIG): examine lower half of the bowel to the splenic flexure

Colon management

1. Chemotherapy,
2. Biologic/Targeted Therapy.
3. Radiation Therapy.
4. **Surgical:**
 - Colostomy (surgical opening from bowel to outside, creating a stoma).
 - Colectomy (partial or total removal of colon).

Nursing management

- Assess for pain and other abnormalities.
- Administer chemotherapy/radiation and provide associated care.
- Provide care for patients undergoing bowel surgery (pre/post-operative care).

Discharge and Home Healthcare Guidelines

- **Patient Teaching:**
 - Wound care (when to notify physician for signs of infection),
 - Avoid heavy activities,
 - Gentle skin care, avoid lotions/perfumes on treated areas,
 - Wear soft clothing,
 - Medication teaching (purpose, side effects).
- **Follow-up:** Stress regular follow-up visits.

- **Prevention:**

- Take an occasional aspirin
- Don't smoke
- Avoid red meat
- Get enough calcium and vitamin D
- Exercise.
- Genetic counselling

Complication:

- 1- Liver cancer
- 2- Lung cancer
- 3- Intestinal obstruction
- 4- Intestinal perforation

Student self-assessment

Which of the following is a characteristic of benign tumors?

- A. Invasive growth
- B. Immature, poorly differentiated tissue
- C. Presence of metastasis
- D. Fully differentiated tissue

There are two types of tumors: benign and malignant. A difference between the two types is:

1. Malignant nuclei are small and regular; benign are large and irregular.
2. Malignant cells multiply uncontrollably; benign cells have controlled growth.
3. Malignant cells don't invade; benign cells sometimes do.
4. Malignant cells are exact copies; benign cells become disorganized.

Your patient will begin chemotherapy next week. Based on the understanding of the effects of chemotherapy, what patient education would be appropriate?

1. Use effective birth control due to teratogenic effects.
2. Expect continuous muscle aches from long-life span cell toxicity.
3. Anticipate significant sickness and learn nausea prevention.
4. Take stool precautions as most drugs excrete focally.

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Chapter eleven

(Nursing care with emergency)

Learning objectives

At the end of this lesson the students will be able to:

- Define key terms.
 - Describe how burns are classified.
 - List the complications that are common with burn victims.
 - Describe the usual treatment for burns.
 - Assess the nursing care needs of a client with burns.
 - Plan and implement effective nursing care for clients with a burn injury.
-

Introduction

Burns are among the most devastating injuries an individual can experience. Burns can be painful and disfiguring and require long hospitalizations. Severe burns can be life threatening or fatal. Most burns occur in the home and are preventable.

Definition

A burn is an injury to the tissues of the body caused by heat, chemicals, electric current, or radiation. The resulting effects are influenced by the temperature of the burning agent, duration of contact time, and type of tissue that is injured.

Types of Burn Injury

- **Thermal Burns:** These are the most common and happen due to heat exposure such as flames, hot liquids (scalds), or contact with heated objects. Severity depends on temperature and duration of exposure.
- **Chemical Burns:** Chemical burns are the result of contact with acids, alkalis, and organic compounds. In addition to tissue damage, eyes can be injured if they are splashed with the chemical.

- **Smoke and Inhalation Injury:** Smoke and inhalation injuries from breathing hot air or noxious chemicals can cause damage to the respiratory tract.
- **Electrical Burns:** Electrical burns result from intense heat generated from an electric current. Direct damage to nerves and vessels, causing tissue anoxia and death, can also occur.

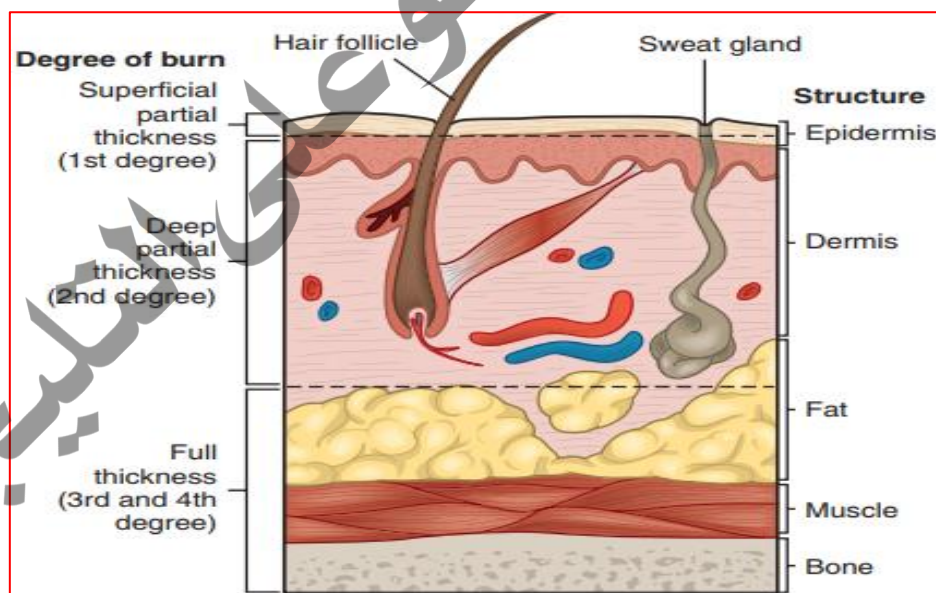
Classification of Burn Injury

The treatment of burns is related to the severity of the injury. Severity is determined by:

- (1) depth of burn.
- (2) extent of burn calculated in percent of total body surface area (TBSA).
- (3) location of burn.
- (4) patient risk factors (e.g., age, past medical history).

Depth of Burn

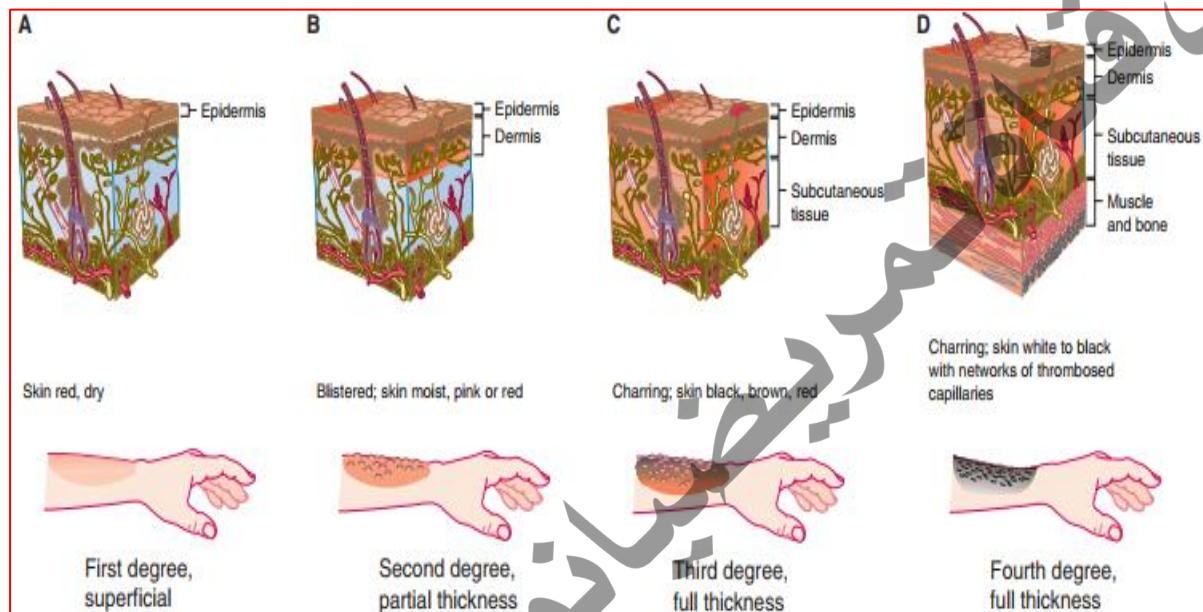
Burn injury involves the destruction of the integumentary system. the skin is divided into three layers: epidermis, dermis, and subcutaneous tissue.



The Epidermis, The outermost layer of the skin. It's nonvascular (contains no blood vessels) and functions as a protective barrier. It is about as thin as a sheet of paper.

The Dermis, which lies below the epidermis, is approximately 30 to 45 times thicker than the epidermis. the dermis contains connective tissues with blood vessels, hair follicles, nerve endings, sweat glands, and sebaceous glands.

The Subcutaneous Tissue, which contains major vascular networks, fat, nerves, and lymphatics. The subcutaneous tissue acts as a heat insulator for underlying structures, including muscles, tendons, bones, and internal organs.



Classification of Burn Injury Depth			
Classifications	Appearance	Possible Cause	Structures Involved
Partial-Thickness Skin Destruction			
Superficial (first degree) burn	Erythema, blanching on pressure, pain and mild swelling, no vesicles or blisters (although after 24 hr skin may blister and peel).	Superficial sunburn Quick heat flash	Superficial epidermal damage with hyperemia. Tactile and pain sensation intact.

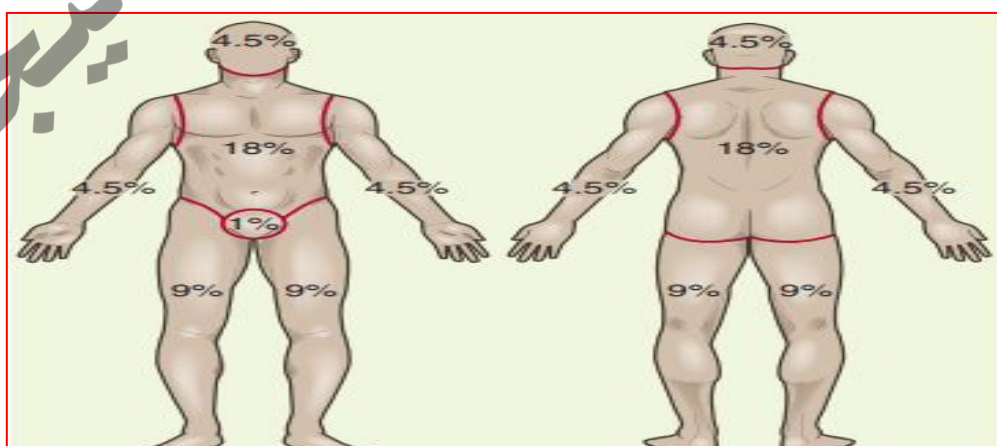
Deep (second degree) burn	Fluid-filled vesicles that are red, shiny, wet (if vesicles have ruptured). Severe pain caused by nerve injury. Mild to moderate edema.	Flame, Flash, Scald Contact burns Chemical, Tar Electric current	Epidermis and dermis involved to varying depths. Skin elements, from which epithelial regeneration occurs, remain viable.
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Full-Thickness Skin Destruction

Third- and fourth degree burns	Dry, waxy white, leathery, or hard skin; visible thrombosed vessels. Insensitivity to pain because of nerve destruction. Possible involvement of muscles, tendons, and bones.	Flame, Scald Chemical, Tar Electric current	All skin elements and local nerve endings destroyed. Coagulation necrosis present. Surgical intervention required for healing
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Extent of Burn

- Two commonly used guides for determining the TBSA affected or the extent of a burn wound are the Lund-Browder chart and the Rule of Nines.
- the Rule of Nines is often used for initial assessment of a burn patient because it is easy to remember.
- For irregular- or odd-shaped burns, the patient's hand (including the fingers) is approximately 1% TBSA.



Nursing Management:

➤ Assessment

- Assess circumstances surrounding the injury: time of injury, mechanism of burn.
- Monitor vital signs frequently and body temperature; monitor respiratory status closely.
- Check peripheral pulses on burned extremities hourly.
- Monitor fluid intake (IV fluids) and output (urinary catheter) and measure hourly.
- Assess neurologic status: consciousness, psychological status, pain and anxiety levels, and behavior.

➤ Interventions

1. **Promoting Gas Exchange and Airway Clearance:** Provide humidified oxygen, and monitor arterial blood gases (ABGs), pulse oximetry, and carboxyhemoglobin levels.
2. **Restoring Fluid and Electrolyte Balance:** Monitor vital signs and urinary output (hourly), central venous pressure (CVP), and Maintain IV lines and regular fluids at appropriate rates, as prescribed.
3. **Maintaining Normal Body Temperature:** Provide warm environment: use heat shield, space blanket, heat lights, or blankets.
4. **Minimizing Pain and Anxiety:** Use a pain scale to assess pain level (i.e., 1 to 10); differentiate between restlessness due to pain and restlessness due to hypoxia and Administer IV opioid analgesics as prescribed.
5. **Preventing Infection:** Provide a clean and safe environment; protect patient from sources of cross contamination (e.g., visitors, other patients, staff, equipment).
6. **Maintaining Adequate Nutrition:** Initiate oral fluids slowly when bowel sounds resume; record tolerance if vomiting and distention do

not occur, fluids may be increased gradually and the patient may be advanced to a normal diet or to tube feedings.

7. Promoting Skin Integrity.

8. Promoting Physical Mobility

9. Supporting Patient and Family Processes

10. Monitoring and Managing Potential Complications

Trauma

Learning objectives

At the end of this lesson the students will be able to:

- Identify trauma injury.
 - List etiologic and risk factor for trauma injuries
 - Enumerate clinical diagnosis and laboratory test for the trauma patient.
 - Discuss the assessment and management of patients with trauma
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Definition

Trauma is injury due to an often-sudden external force that affects one or many aspects of the body. Injury can be minor or it can be fatal.

Etiology

- Blunt injury comprises about 75% of all trauma.
- Approximately 70% of traumas are unintentional.
- Traumatic brain injury (TBI)
- Motor vehicle crashes are the leading mechanism of traumatic death.
- Falls are responsible for the majority of all deaths in people older than 65.

Predictive/risk factors

- Lower socioeconomic status is associated with a higher proportion of overall trauma
- Men are nearly three times as likely as women to become victims of trauma
- Elderly has the highest rates of suicide.
- Children are most likely to drown.

Clinical diagnosis

➤ History

- The examiner should first inquire about circumstances of the injury to set a context, often provided at time of presentation, as mechanism of injury can provide clues **AMPLE** questions should be ascertained:
- **Allergies.**
- **Medication**
- **Past medical history** including tetanus status.
- **Last meal.**
- **Events** leading to injury.

➤ Physical examination

- The examination should begin with the **patient's airway, breathing, and circulation ('ABCs')**, while the patient's clothing is removed to provide complete exposure.
- Ask the patient to speak, visualizing the work of **breathing**, and listening for breath sounds to quickly evaluate **airway patency** and respiratory compromise.
- Heart rate and blood pressure in conjunction with a pulse examination provide information on **the circulatory system**.
- Disability or the patient's **neurologic status** should also be assessed by speaking to the patient and assessing responsiveness.
- If ABCs are intact, a **secondary survey should proceed with a thorough head-to-toe examination.**
- This should include inspection and palpation and a neurologic assessment.

Laboratory diagnosis

➤ List of diagnostic tests

- Typical initial tests include:
- Complete blood count (CBC)

- kidney function, electrolyte abnormalities, and blood glucose level.
- Lactate level as an indicator of hypoperfusion.
- Alcohol and urine drug screen.
- Arterial blood gas (ABG)
- Prothrombin time (PT) or international normalized ratio (INR) in a patient taking warfarin.
- Blood type and cross in a patient likely to need blood product transfusions.
- Urinalysis or urine dipstick if possible genitourinary trauma

List of imaging techniques

- (ECG) electrocardiogram.
- Chest X-rays are often routine.
- Pelvic X-rays are used to evaluate blunt injuries.
- Focused assessment using sonography for trauma (FAST).
- CT head: penetrating trauma to the head, or blunt trauma to the head
- CT spine: spine tenderness or new motor or sensory neurologic defects.
- MRI spine: new motor or sensory deficits or persistent pain without bone disruption.
- CT abdomen: abdominal tenderness or stable patients with blunt trauma.
- Bronchoscopy/esophagography/esophagogram: penetrating neck injuries.

Treatment

- placement of two preferably large bore IVs.
- initiation of oxygen by nasal cannula or facemask.
- cervical collar should be placed for cervical spine immobilization until the extent of injury is determined.
- As with triage and diagnosis, treatment should be individualized.
- Initial management is guided by ABCs, which will sequentially identify the earliest and most threatening potential sources of mortality.

- focus should shift to causes of circulatory compromise, as hemorrhage
- Initiation of warmed isotonic IV fluids should begin on any patient with a systolic blood pressure <110 mmHg and a heart rate >100 beats/min with a rapid search to identify the origin of blood loss
- Initiation of blood product transfusion should be based on clinical judgment.
- Blood products should ideally be limited to maintain hemoglobin above 7 g/dl.

Prevention/management of complications

- MODS is a complex process due to inflammatory or immune dysregulation with deleterious systemic effects.
- Management includes rapid identification of the cause, appropriate use of antibiotics when indicated, aggressive nutritional support, prevention of hypoxemia and hypotension, avoidance of nephrotoxic agents, use of lung protective ventilation strategies, limitation of blood product transfusions.

Coma and Brain Death

Learning objectives

- Define coma and brain death.
 - Identify etiology of coma.
 - Explain pathophysiology of coma
 - Discuss Clinical diagnosis of brain death.
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Introduction

Disorders of consciousness are conditions that affect a person's ability to be alert and aware of one's environment. Operationally, levels of consciousness can be thought of as being on a spectrum. A person who is fully conscious is alert and can fully perceive themselves and their environment.

Coma: patients with severe cerebral dysfunction who are unresponsive are said to be in a coma. This may result from several life-threatening conditions including structural injuries, vascular pathology, toxicity from drugs, other metabolic disarrangement, or epileptic abnormality. Coma may be potentially reversible.

Brain death: a person who has been determined to be 'brain dead' has suffered irreversible cessation of all brain function. Brain death is a concept not known prior to the advent of ventilators. The determination of brain death is based on accepted medical practices and local law.

Disease classification

The Glasgow Coma Scale (GCS) is often used to describe a person's level of consciousness after injury or medical event. The score is divided into three sections (eye, verbal, and motor). Each section is given a numerical value based on the patient's best response.

The sum of all three values provides the score. The scale is graded from 3 to 15 and individuals who score 8 or less are considered to be in a coma.

Eye response	Verbal response	Motor response
4. Spontaneous eye opening 3. Eye opening on verbal command 2. Eye opening to pain 1. No eye opening	5. Oriented 4. Confused 3. Inappropriate words 2. Incomprehensible sounds 1. No verbal response T. Endotracheal tube placement	6. Following commands 5. Localizes to pain 4. Withdrawal from pain 3. Flexion in response to pain 2. Extension in response to pain 1. No motor response

Etiology of coma

- Trauma: contusions, diffuse axonal injury, hemorrhage.
- Toxins: lead, carbon monoxide, cyanide.
- Drugs: sedative medications including benzodiazepines, barbiturates, opiates.
- Metabolic derangements: hypoxia, hypotension, hypothermia, hyperthermia, hyponatremia, hypernatremia, hypothyroidism, hepatic encephalopathy, uremia, diabetic ketoacidosis, hyperosmolar non-ketotic coma.
- Infections: bacterial meningitis, brain abscesses, epidural abscesses, viral encephalitis.
- Intracranial hemorrhages: subarachnoid hemorrhage, epidural hemorrhage.
- Ischemic infarctions: large hemispheric, brainstem, pons.
- Seizures: status epilepticus, epileptic encephalopathy.
- Inflammatory: encephalitis, vasculitis.

Clinical diagnosis of brain death

➤ Physical examination

Coma or unresponsiveness:

- No cerebral response to pain in all extremities (nail bed pressure, supraorbital pressure).

Pupils:

- No response to bright light.

Ocular movement:

- No oculoccephalic reflex: eyes appear to be painted on with head movement.
- No oculovestibular reflex: no deviation of the eyes to irrigation in each ear of 60 mL of cold water.

Facial sensation and motor response:

- No corneal reflex.
- No jaw reflex.
- No grimacing to deep pressure on nail bed, supraorbital ridge, or temporomandibular joint.

Pharyngeal and tracheal reflexes:

- No response after stimulation of posterior pharynx.
- No cough response to endotracheal suctioning.

➤ Apnea test

Prerequisites:

- Core temperature $>36.5^{\circ}\text{C}$.
- Systolic blood pressure ≥ 100 mmHg.
- Normal PCO₂ (PaCO₂ 40 mmHg).
- Normal PO₂ (preoxygenate to a PO₂ >200 mmHg).

Procedure:

- Make sure patient is connected to a pulse oximeter. Disconnect from the ventilator. Deliver 100% oxygen into the trachea. This may be done via a cut nasal cannula tubing. Monitor for respiratory movements.
- Obtain arterial blood gas at around 7–8 minutes and reconnect to the ventilator.
- If respiratory movements are noted at any time, the test is not consistent with brain death.

- If SBP is <90 mmHg or patient develops arrhythmia or significant desaturations, obtain arterial blood sample and abort the test.
- Test is consistent with brain death if PCO₂ rises to >60 mmHg or rises 20 mmHg from baseline and no respiratory movements are observed.

من فضاء نهج ريشيانو على التليجرام

Student self-assessment

A full-thickness burn is:

- A. Classified by the appearance of blisters.
- B. Identified by the destruction of the dermis and epidermis.
- C. Not associated with edema formation.
- D. Usually very painful because of exposed nerve endings.

Identify Classification of Burn Injury Depth?

List the Etiology of coma

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على تطبيق التليجرام



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